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VARIABILITY AND CONTROL IN SPRINGBOARD DIVING

by

MOHSEN SAYYAH

A Doctoral Thesis

**Submitted in partial fulfilment of the requirements for the award of
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ABSTRACT

VARIABILITY AND CONTROL IN SPRINGBOARD DIVING

Mohsen Sayyah, Loughborough University, 2017

Elite springboard divers typically make very precise and reproducible movements when they perform the same dive many times. However, variability is always present in both technique and outcome. While it is desirable to have low outcome variability this may necessitate real-time adjustments which result in increased technique variability from trial to trial. The aim of the present research was to determine whether feedback control adjustment is used during (a) the hurdle takeoff, (b) the dive takeoff, and (c) the dive flight phase. 15 forward pike dives and 15 forward 2½ somersault pike dives, performed by an international diver, were video recorded at 250 Hz and manually digitised followed by DLT reconstruction of joint centre locations. Orientation angle and joint angles were calculated and fitted with quintic splines to give angular velocities. Foot placements, mass centre location and velocity were determined along with angular momentum about the mass centre. In the hurdle takeoff no adjustment was made to reduce the variability in the foot location at hurdle landing. In the dive takeoff phase an angle-driven simulation model was used to determine the expected variation in mass centre velocity and angular momentum at the instant of takeoff arising from the variation in velocity and angular momentum at touchdown. The simulated variation at the instant of takeoff was greater than the variation in the recorded performances indicating that some adjustment had been made during the takeoff phase. In the flight phase an angle-driven simulation model was used to determine the expected variation in orientation angle at water entry arising from the variation in velocity and angular momentum at takeoff. The variation in the orientation angle at entry obtained from the simulations was greater than the variability in the actual performances, indicating that the diver had used feedback control adjustments in the flight phase to reduce his performance outcome variability. The variation in the angular momentum at takeoff was reflected in the average hip angle in flight, indicating that the hip angle was adjusted to be larger to compensate when the initial angular momentum was greater. The use of feedback control adjustments found in this study demonstrated that variability has a functional role in human movement.

PUBLICATIONS

Conference presentations:

M, Sayyah, Yeadon, M. R., Hiley, M. J. and King, M. A. (2016). Factors influencing variation in dive height in 1-m springboard diving. *In the BASIS Biomechanics Interest Group conference.*

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DEDICATION

To my family:

My parents, Jamshid and Monir

&

My siblings, Maral and Nima

Table of Contents

ABSTRACT	I
PUBLICATIONS.....	II
ACKNOWLEDGMENTS.....	III
DEDICATION	IV
LIST OF FIGURES	VIII
LIST OF TABLES	X
CHAPTER 1.....	1
INTRODUCTION	1
1. INTRODUCTION.....	1
2. STATEMENT OF PURPOSE	7
3. RESEARCH QUESTIONS	7
4. CHAPTER ORGANISATION	9
CHAPTER 2.....	11
LITERATURE REVIEW.....	11
2.1. SPRINGBOARD DIVING.....	11
2.2. MOVEMENT CONTROL.....	11
2.2.1. Motor programme	13
2.2.2. Feedback control	16
2.2.3 Feedforward control.....	17
2.3. MOVEMENT CORRECTION	19
2.4. APPLICATION OF SIMULATION MODELS	20
2.5. SUMMARY.....	22
CHAPTER 3.....	23
DATA COLLECTION AND ANALYSIS	23
3.1. DATA COLLECTION	23
3.1.1. Camera set-up	23
3.1.2. Performances.....	24
3.1.3. Fulcrum	24
3.1.4. Camera calibration.....	25
3.1.5. Dynamic loading of the springboard.....	31
3.1.6. Anthropometric measurements	32
3.1.7. DLT image reconstruction.....	32
3.2. DIGITISATION	33
3.2.1. Board neutral position.....	34
3.2.2. Foot placement.....	35
3.2.3. Flight phase.....	36
3.2.4. Takeoff phase	38
3.2.5. Order and frequency of digitising	38
3.2.6. The accuracy of digitising.....	39
3.3. KINEMATIC VARIABLES	41
3.3.1. Joint angles and mass centre locations	41
3.3.2. Body orientation	42
3.3.3. Velocity, momentum and rotation potential.....	43
3.4. SUMMARY.....	43
CHAPTER 4.....	44
PARAMETER DETERMINATION.....	44
4.1. INTRODUCTION.....	44
4.1.1. Angle-driven model of a diver and springboard	44
4.1.1.1. The springboard model.....	44
4.1.1.2. Foot-springboard interface.....	45
4.1.1.3. The diver model	45

4.1.2. Angle-driven simulation model of aerial movement	46
4.2. SEGMENTAL INERTIA PARAMETERS.....	47
4.2.1. Wobbling mass parameters	51
4.3. SPRINGBOARD PARAMETERS.....	52
4.3.1. Vertical stiffness and effective board mass	52
4.3.1.1. Dynamic loading of the springboard	53
4.3.2. Plotting T^2 against m_L	57
4.3.3. Equation of the board stiffness	58
4.3.4. Board deflection	59
4.3.5. Vertical deflection and board rotation relationship	61
4.3.6. Vertical and horizontal deflection relationship	64
4.3.7. Moment of inertia	66
4.4. FOOT-SPRINGBOARD INTERFACE PARAMETERS	66
4.5. SUMMARY.....	67
CHAPTER 5.....	68
PERFORMANCE ANALYSIS	68
5.1. INTRODUCTION.....	68
5.2. METHOD	70
5.3. RESULTS	70
5.3.1. Peak dive height.....	70
5.3.2. Angle and time variability	71
5.3.3. Angle time history.....	73
5.3.3.1. Takeoff phase.....	73
5.3.3.2. Flight phase.....	75
5.3.4. Comparison between 101B AND 105B	77
5.3.4.1. Takeoff phase.....	77
5.3.4.2. Flight phase.....	77
5.3.5. Orientation angle.....	77
5.3.6. Factors contributing to dive height	79
5.3.6.1. Correlation	79
5.3.6.2. Stepwise multiple regression	81
5.3.7. Factors contributing to entry angle	82
5.3.7.1. Correlation	82
5.3.7.2. Stepwise multiple regression	83
5.4. DISCUSSION AND CONCLUSION	83
5.5. SUMMARY.....	87
CHAPTER 6.....	88
VARIABILITY IN FOOT PLACEMENT	88
6.1. INTRODUCTION.....	88
6.3. RESULTS	91
6.3.1. Correlation	94
6.3.2. Stepwise multiple regression analysis	95
6.3.2.1. Hurdle landing.....	95
6.3.2.2. Hurdle length	96
6.4. DISCUSSION	98
6.5. SUMMARY.....	100
CHAPTER 7.....	101
VARIABILITY AND CONTROL IN TAKEOFF	101
7.1. INTRODUCTION.....	101
7.2. METHOD	103
7.2.1. Model input	105
7.2.2. Matching simulation	106
7.2.3. Objective score	107
7.2.4. Accuracy of the simulation	108

7.3. PERTURBATION	109
7.3.1. Individual perturbation	110
7.3.2. Combined perturbation	114
7.3.3. Lowest point perturbation	116
7.4. ACTUAL PERFORMANCE	117
7.5. DISCUSSION	120
7.6. SUMMARY	125
CHAPTER 8.....	126
VARIABILITY AND CONTROL IN FLIGHT PHASE	126
8.1. INTRODUCTION	126
8.1.1. Motor control	126
8.2. METHOD	128
8.2.1. Normalizing the angular momentum	129
8.2.2. Tuning angular momentum	131
8.2.3. Rotation potential.....	131
8.2.4. Matching performances	133
8.2.5. Accuracy of simulation.....	136
8.3. SIMULATION	137
8.3.1. First Analysis	137
8.3.2. Second Analysis	137
8.3.3. Net change	138
8.4. RESULTS	138
8.4.1. Actual performance	138
8.4.2. Simulation outcome.....	141
8.4.2.1. Analysis 1	141
8.4.2.2. Analysis 2	145
8.4.2.3. Net change & RMS differences	149
8.4.3. Actual performance and simulation outcome	151
8.4.4. Regressions	153
8.5. DISCUSSION	158
8.6. SUMMARY	165
CHAPTER 9.....	166
SUMMARY AND CONCLUSION	166
9.1. INTRODUCTION.....	166
9.2. SUMMARY OF MAIN FINDING	166
9.2.1. Kinematic variability	166
9.2.2. Adjustment in foot placement.....	167
9.2.3. Movement control in takeoff	167
9.2.4. Movement control in flight.....	168
9.3. RESEARCH QUESTIONS	170
9.4. CONCLUSION	172
REFERENCES	173
APPENDIX 1	183
INFORMED CONSENT FORM	183
APPENDIX 2	185
ANTHROPOMETRIC MEASUREMENTS	185
APPENDIX 3	187
INERTIA PARAMETERS	187
APPENDIX 4	189
SUBJECT PROFILE	189

LIST OF FIGURES

Figure 1: 1-m springboard forward pike dive. Positions: A=hurdle takeoff, B=touchdown, C= maximum board depression, D= takeoff, E= entry, (A to B)= hurdle flight, (B to C)= board depression, (C to D)= recoil phase.....	1
Figure 2.1. Schmidt's (1975) notion of schema.....	14
Figure 2.2. Feedback loop system.....	16
Figure 3.1. Camera field of view while the diver standing on the end of the board.	23
Figure 3.2. 1-m springboard, Ponds Forge, Sheffield, United Kingdom.....	25
Figure 3.3. The calibration pole with the seven balls in vertical position.	26
Figure 3.4. A two-dimensional plan of the pool and the measures of x-y coordinates of P1 to P6.	27
Figure 3.5. Generating pseudo-3D coordinates using anthropometric measurements with the assumption of symmetrical body movement.	31
Figure 3.6. Measurements of 1-m springboard.....	33
Figure 3.7. Calibration image.	34
Figure 3.8. The diver position at hurdle touchdown (left), hurdle takeoff (middle) and hurdle landing (right) while the foot is flat on the board.	35
Figure 3.9. The diver position at hurdle takeoff (left) and hurdle landing (right) when the board is in neutral position.	36
Figure 3.10. Dive takeoff position at board neutral (left) and entry position (right).	37
Figure 3.11. Diver position at board neutral position at landing from hurdle and dive takeoff.....	38
Figure 3.12. Low accuracy and poor precision (left), low accuracy and good precision (right), accuracy according to BIPM and ISO 5725.....	39
Figure 3.13. Body orientation angle as the angle between the vertical and a line from the midpoint of the knee centres to a fixed point on trunk and vertical.	42
Figure 4.1. The springboard modelled as a rod with three degrees of freedom.	44
Figure 4.2. Model of the foot-springboard.	45
Figure 4.3. Model of a diver and springboard.	46
Figure 4.4. Model of human airborne.	47
Figure 4.5. The two segments model of foot: a triangle and a rod segment.	48
Figure 4.6. Dynamic loading of the springboard on the board tip (left) and one step back from the board tip (right).	54
Figure 4.7. Determination of k and m_e at one step back from the board tip.....	56
Figure 4.8. Determination of k and m_e at board tip.	56
Figure 4.9. Plotting board stiffness against toe distance from the board tip.....	58
Figure 4.10. The board angle as the angle between a line fitted with the ball, toe and the board tip and the board at neutral position.	60
Figure 4.11. Linear regression of the board angle against the board vertical deflection using performances of 10 forward pike dives.....	61
Figure 4.12. Linear regression of the board angle against the board vertical deflection using performances of 9 forward $2\frac{1}{2}$ somersaults pike dives.....	62
Figure 4.13. Regression of horizontal and vertical deflection of the board tip of 15 forward pike dives.	64
Figure 4.14. Regression of horizontal and vertical deflection of the board tip of 15 forward $2\frac{1}{2}$ somersaults pike dives.....	65

Figure 5.1. 1-m springboard forward pike dive. Positions: A= hurdle takeoff, B= touchdown, C= maximum board depression, D= takeoff, E= Entry, (B to C)= board depression, (C to D)= board recoil phase.	68
Figure 5.2. Centre of mass movement pattern in hurdle flight (A to B), board contact phase (B to C) and aerial (C to D) phases of 15 forward pike (101B) and 15 forward 2 ½ somersault pike dives (105B).	71
Figure 5.3. Average standard deviation and time histories of joint angles during dive takeoff phase of 15 forward and 15 forward 2½ somersault pike dives.	74
Figure 5.4. Average and time histories of joint angles during dive flight phase of 15 forward and 15 forward 2½ somersault pike dives.	76
Figure 5.5. Mean and time histories of standard deviation of body orientation angle during dive contact and flight phases of 15 forward pike (left) and 15 forward 2½ somersault pike (right) dives.	78
Figure 6.1. Forward dive, A= hurdle touchdown, B= hurdle takeoff, C= landing from hurdle, (B to C) = hurdle length.	88
Figure 6.2. Linear regression of foot placement at takeoff of forward pike dives (morning session), the first 7 dives (squared points), the last 8 dives (crossed points).	91
Figure 6.3. Linear regression of foot placement at takeoff of forward 2½ somersault pike dives (afternoon session), the first 8 dives (squared points), the last 7 dives (crossed points).	92
Figure 6.4 Linear regression of foot placement at takeoff of the last 8 dives in the morning (squared points) and the first 8 dives in the afternoon (circle points).....	93
Figure 6.5. Mean and standard deviations of toe distance from the board tip at hurdle contact and dive takeoff of the last 8 dives in the morning (101B) and the first 8 dives in the afternoon (105B).....	94
Figure 7.1. 1-m springboard forward pike dives. Board contact phase (A to C), Hurdle landing (A), Maximum board depression (B), Takeoff (C), Board depression phase (A to B), Board recoil phase (B to C).	101
Figure 7.2. Standard deviation time history of orientation, hip, knee and arm angles during the takeoff phase of 15 forward and 15 forward 2½ somersault pike dives. The dash-line is the average time at maximum board depression.	119
Figure 8.1. 1-m springboard forward pike (solid line) and forward 2½ somersault pike (dotted line) dives. Phases: hurdle contact (A-B) and hurdle flight (B-C), contact (C-D), flight (D-E). Φ_t = initial body orientation angle at takeoff at board neutral position, Φ_e = entry orientation angle.	126
Figure 8.2. Matching simulation (solid line) and performance (dashed line) orientation angles (°) during the aerial phase of 15 forward pike dives.	134
Figure 8.3. Matching simulation (solid line) and performance (dashed line) orientation angles (°) during the aerial phase of 15 forward 2½ somersault pike dives.	135
Figure 8.4. The angle envelope generated from 15 forward pike dives.	140
Figure 8.5. The angle envelope generated from 15 forward 2½ somersault pike dives.	140

LIST OF TABLES

Table 3.1. Number of dives performed during data collection.....	24
Table 3.2. Comparison between the measured and manufacturing reference.....	25
Table 3.3. Horizontal distance, lean and extra height factors	28
Table 3.4. Horizontal deviation of the markers	28
Table 3.5. The amount of leaning in each pole position	29
Table 3.6. Space coordinates of the 26 control points for camera calibration (m).....	30
Table 3.7. Board neutral coordinates calculated from 10 frames of the calibration data.....	35
Table 3.8. The sequence of the body landmarks.....	37
Table 3.9. RMS differences across all time points between trial 1 and re-digitised trial 1, trial 15 and re-digitised trial 15.....	41
Table 4.1. Body segmental inertias calculated from anthropometric measurement.....	48
Table 4.2. The mass and inertia values of the diver's right foot.....	49
Table 4.3. Measured and calculated dimensions for two-segment foot	50
Table 4.4. A comparison of mass segments obtained in Kong (2005) and this study	50
Table 4.5. A comparison of segments moment of inertias obtained in Kong (2005)	50
Table 4.6. Length of segments in Kong (2005) and this study.....	51
Table 4.7. Segmental inertia parameters for the fixed and wobbling component.....	52
Table 4.8. The oscillation period of different body weights and free load at board tip.....	55
Table 4.9. The oscillation period of different body weights and free load one step back from the board tip.....	55
Table 4.10. Comparison of the board stiffness and effective board mass.....	57
Table 4.11. The board digitised coordinates at board neutral position at touchdown of 15 forward and 15 forward 2½ somersaults pike dives	59
Table 4.12. Mean toe distance to the board tip during contact phase of forward pike and forward 2½ somersaults pike dives	63
Table 5.1. Mean and standard deviation of the Kinematic variables of 15 forward pike and 15 forward 2½ somersault pike dives.....	72
Table 5.2. Mean and standard deviation of vertical and horizontal velocity of 15 forward pike dives (101B) and 15 forward 2½ somersault pike dives (105B).....	72
Table 5.3. Mean and standard deviations of time in different phases.	73
Table 5.4. A list of the kinematic variables used in SPSS to perform regression analysis	79
Table 5.5. Pearson Correlation for all significant variables against peak dive height of 15 forward pike dives (101B)	80
Table 5.6. Pearson Correlation for all significant variables against peak dive height of 15 forward 2½ somersault pike dives (105B)	80
Table 5.7. Model summary of a stepwise multiple regression analysis of kinematic variables against peak dive height in forward pike and forward 2½ somersault pike dives	81
Table 5.8. Stepwise multiple regression of takeoff variables and dive height against entry orientation angle in forward pike and forward 2½ somersault pike dives	83
Table 5.9. Pearson correlation of all significant variables with entry orientation angle in forward pike and forward 2½ somersault pike dives	83
Table 6.1. Correlations between variables of the last 8 dives in 101B and the first 8 dives in 105B	95
Table 6.2. Stepwise multiple regression analysis of hurdle landing distance against dive number, hurdle touchdown distance and hurdle flight CM horizontal velocity in 101B and 105B	96

Table 6.3. Model summary of the multiple regression of hurdle length against dive number, hurdle touchdown distance and hurdle flight CM horizontal velocity of the last 8 trials in the morning (101B) and the first 8 trials in the afternoon (105B) 97

Table 7.1. Initial touchdown variables used in the simulation model in forward dives	105
Table 7.2. Initial touchdown variables used in the simulation model in forward 2½ somersault pike dives.....	105
Table 7.3. Lower and upper bounds of the parameters in the matching process.....	107
Table 7.4. Root mean square differences of the six scores between simulation and video data (S _{ANG} score) in forward pike and forward 2½ somersault pike dives.....	108
Table 7.5. Initial touchdown conditions of CM horizontal and vertical velocities and angular velocity of 13 forward and 14 forward 2½ somersault pike dives.....	109
Table 7.6. Rotation potential at takeoff obtained in simulation outcome by varying the initial condition using ±SD of the initial CM horizontal, vertical and angular velocities in forward pike dives	111
Table 7.7. Rotation potential at takeoff obtained in simulation outcome by varying the initial condition using ±SD of the initial CM horizontal, vertical and angular velocities in forward 2½ somersault pike dives	111
Table 7.8. Root mean square difference between rotation potential obtained in each individual perturbation and the original matching simulation of 13 forward pike dives	113
Table 7.9. Root mean square difference between rotation potential obtained in each individual perturbation and the original matching simulation of 14 forward 2½ pike dives	113
Table 7.10. Root mean square difference between rotation potential obtained in each combined perturbation of initial touchdown conditions and the original matching simulation of 13 forward pike dives	114
Table 7.11. Root mean square difference between rotation potential obtained in each combined perturbation of initial touchdown conditions and the original matching simulation of 13 forward pike dives	115
Table 7.12. Root mean square difference between rotation potential obtained in each combined perturbation of initial lowest point and the original matching simulation of 13 forward pike dives.....	116
Table 7.13. Root mean square difference between rotation potential obtained in each combined perturbation of initial lowest point and the original matching simulation of 14 forward 2½ pike dives.....	117
Table 7.14. Mean and standard deviation of the angles at touchdown and takeoff, flight time and CM velocity in contact phase of 15 forward and 15 forward 2½ somersault pike dives	118

Table 8.1. Entry orientation angle, rotation potential, normalised and SI values of the angular momentum and flight time of 101B and 105B	130
Table 8.2. Mean and standard deviation of the rotation potential in straight somersault in 101B and 105B	132
Table 8.3. Mean and minimum hip and arm angles in 101B and 105B.....	132
Table 8.4. Root mean square differences of the orientation angles between simulation and video data (S ₂ score).....	136
Table 8.5. Mean and standard deviation of the angles at takeoff and entry, CM velocities, flight time, angular momentum and rotation potential during flight of 15 forward and 15 forward 2½ somersault dives.....	139
Table 8.6. Entry orientation angle in 101B obtained from Analysis 1.....	141
Table 8.7. Entry orientation angle in 105B obtained in Analysis 1	142
Table 8.8. Simulated entry orientation angle obtained from the simulations of 15 trials 101B while retaining joint angle time history the same as actual performance	143

Table 8.9. Simulated entry orientation angle obtained from the simulations of 15 trials 105B while retaining joint angle time history the same as actual performance	144
Table 8.10. The outcome of simulation 1 using joint angle time histories of 15 forward pike dives while retaining the flight time and angular momentum of dive 1	145
Table 8.11. The outcome of simulation 1 using the joint angle time histories of 15 forward 2½ somersault pike dives while retaining the flight time and angular momentum of dive 1	146
Table 8.12. Simulated entry orientation angle obtained from the simulations of 15 trials of 101B while retaining the initial conditions the same as actual performance.....	147
Table 8.13. Simulated entry orientation angle obtained from the simulations of 15 trials of 105B while retaining the initial conditions the same as actual performance.....	148
Table 8.14. Differences between each individual simulated and mean entry angle from 15 simulations and the net changes in analysis 1 and analysis 2 of 101B.....	149
Table 8.15. Comparison between simulated orientation angle and actual entry angle and the net change between analysis 1 and analysis 2 of 105B	150
Table 8.16. Mean and standard deviation of orientation angle at initial takeoff conditions and entry in actual performance and simulation of 101B and 105B.....	151
Table 8.17. Stepwise multiple regression analysis of rotation potential against average hip angle, minimum hip angle, average arm angle and minimum arm angle during the flight phase of 15 forward and 15 forward 2½ somersault pike dives	154
Table 8.18. Linear regression analysis of (a) arm angle at maximum arm variation against orientation angle at maximum hip variation and (b) hip angle at entry against orientation angle at maximum arm variation of 15 forward 2½ somersault pike dives	157
Table 8.19. Linear regression analysis of hip angle at maximum hip variation against rotation potential of 15 forward pike dives and 15 forward 2½ somersault pike dive	157

CHAPTER 1

INTRODUCTION

1. INTRODUCTION

A diving performance can be divided into three main phases: a) hurdle b) contact c) aerial. Figure 1 shows the hurdle contact, hurdle flight, dive contact and flight phases. The diver starts the hurdle with approaching steps and jumps from an active leg towards the end of the diving board to achieve hurdle flight. At the end of the hurdle flight, the diver contacts the board using both feet to begin the contact phase and depress the board. The contact phase can be subdivided into board depression and board recoil phases. Greater hurdle height results in a greater vertical landing velocity at touchdown and thus more energy is available to depress the springboard. To initiate rotation, the angular momentum is generated during the recoil phase by hip flexion in forward dives. However, losing energy by performing extra flexion may lead to a loss in dive height, shorter flight time and short of rotation at entry. The optimal flexion in hip and knees at touchdown greatly depends on the diver's skill and muscular strength in which extra flexion following the touchdown would be minimised to avoid energy absorption (Sanders & Wilson, 1988; Sanders & Gibson, 2000).

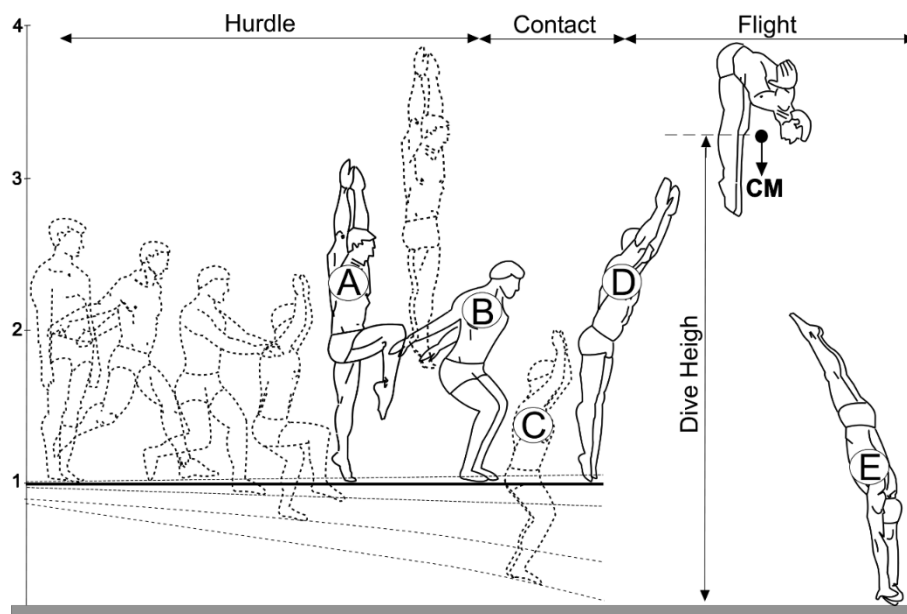


Figure 1: 1-m springboard forward pike dive. Positions: A=hurdle takeoff, B=touchdown, C= maximum board depression, D= takeoff, E= entry, (A to B)= hurdle flight, (B to C)= board depression, (C to D)= recoil phase.

Generating sufficient angular momentum, obtaining maximum dive height and therefore flight time, travelling safely with a reasonable board clearance distance and accuracy in entry orientation angle are the main objectives for competitive springboard diving (Miller and Munro, 1985). Although the dive height, board clearance distance and angular momentum required for executing somersaults are all determined during the contact phase, the angular momentum is constant during the flight phase and the balance between the moment of inertia and angular velocity is responsible for any change or adjustment in orientation in this phase. Since the diver must finish the dive before entering the water, the body should straighten with arms extended overhead. Thus, a correct entry requires not only physical strength but also correct timing in each individual phase (Wilson and Cramer, 1974).

The diver's motor outcome often seems very repeatable and stereotyped but, there is always variability in their performances. Although Hiley et al. (2013) investigated the effect of skill level and task difficulty on movement variability in gymnastics and Hiley & Yeadon (2003b, 2005) considered the margin error of releasing window in giant circle in gymnastics, the study of variability has been mainly focused on basic movements such as jumping (James et al., 2000; Rodano & Squadrone, 2002; Wilson et al., 2008), sprint start (Bradshaw et al., 2007), running (Bates et al., 1983; Devita & Bates, 1988; Lees & Bouracier, 1994; Diss, 2001; Ferber et al., 2002; Queen et al., 2006), race walking (Preatoni, 2007; Preatoni et al., 2010) and baseball pitching (Fleisig et al., 2009).

Movement variability has been traditionally viewed as noise that should be minimised or eliminated; however, recent studies reassessed its role and reported that variability may have a functional role in human movement (Preatoni et al., 2013; Bartlett, Wheat, & Robins, 2007; Hamill, van Emmerik, Heiderscheit, & Li, 1999). Although variability has been reported to be due to planning and execution errors (Cohen & Sternad, 2009; Newell & Corcos, 1993; van Beers et al., 2004), movement variability due to adjustment has been viewed to be functional since feedback correction strategies to control the movement have been demonstrated in gymnastics such as in a hand balance (Yeadon & Trewartha, 2003), twisting somersaults (Yeadon & Hiley, 2014; Yeadon & Mikulcik, 1996), giant circles on high bar (Hiley and Yeadon, 2016). The variability of the diver's

movement may be identified as an unwanted source of error which is to be minimised to help technique proficiency. However, the diver's adjustment during the performance may appear as a deliberate source of variation to get an optimal output. The effect of variability on consistency of performance in springboard diving can be investigated to understand the difference between execution variability and adjustment.

Consistency and accuracy in the whole body angular orientation at takeoff and entry in springboard diving determines the quality of a dive (Golden, 1981; Miller, 1974; O'brien, 1992; Slobounov et al.,1997). Variability in the generation of angular momentum can significantly affect the performance outcome as too much angular momentum at takeoff may result in over rotation at entry or lack of enough angular momentum may end up with less rotation at entry. Adjustments are made in performances of such movements to compensate for too much angular momentum by configurational movement. Although the variability in the diver's joint angle time histories may be viewed as an unwanted error which needs to be minimised to improve accuracy, a diver's adjustment of body configuration during flight may be a deliberate compensation for variations in takeoff, leading to increased joint angle variability and decreased entry angle variability.

While there are a limited number of investigations regarding the treatment of variability in complex sports movements such as the giant circle on high bar (Hiley & Yeadon, 2003b, 2005) and the Tkatchev (Hiley and Yeadon, 2012), there has been no investigation regarding the treatment of variability in technique in springboard diving. The study of movement correction is also limited to a few studies. For the control of twist in a simulated target trampoline movement, a single symmetrical arm adduction movement using delayed feedback control could correct a perturbation (Yeadon & Hiley, 2014). Variation in technique and increasing strength in gymnastics could improve the consistency of performance and success in a limiting movement (Hiley & Yeadon, 2012).

In springboard diving, the relationship between gaining height, rotation and distance travelled has been recognised and optimal takeoff techniques in terms of generating both linear and angular momentum have been developed (Miller and Munro 1984; McCormick, Subbiah, & Arnold, 1982; Jones & Miller, 1996; Kooi and Kuipers, 1994;

Sanders and Wilson 1988; Walker, 2010; Sanders & Gibson, 2000). However, little is known about the variability in different stages of dives. Future studies into the characteristics of technique proficiency and a realistic level of variability in whole body coordinated movements in springboard diving may wish to investigate the extent to which the precision of repetition could affect the consistency and success in a diver's performance. The treatment of variability to understand control systems in sporting movements would not be possible by establishing the magnitude of timing and angle variability through observing performances whereas using computer simulation model makes it feasible to investigate variability and control in complex sport movements.

Regardless of the variation that appears through repetitions, the determination of the movement variability and the interpretation of biomechanical data could determine the diver's control strategy to reduce the variation by following on multiple levels such as recording the diver changes over time, modelling optimal performance through a theoretical approach and simulation. In fact, in order to understand the control system, it is essential to identify the cause of the movement variability. It may not be practical to determine the effect of changing initial conditions of a sport movement on performance outcome through observation or video analysis. Simulation modelling enables one variable to be changed at a time. The variability in the recorded performance outcome represents the variation arising from initial conditions and joint torque time histories; application of simulation models make it feasible to separate execution variability and adjustment.

Development of the simulation model, determination of subject-specific parameters, evaluation of the model, and application of the model are a four-stage methodology required with each model to answer specific research questions (King and Yeadon, 2015). Simulation models have been used to investigate control in sports movement. The simulation model of aerial movement was applied to demonstrate that non-twisting straight somersaults are inherently unstable and require continual proprioceptive feedback control (Yeadon and Mikulcik, 1996). The model provided a means for understanding how athletes must make movements in the aerial phase of twisting somersaults to initiate, control, and stop the twist, demonstrating that movements of arms and hips control the twist and the tilt (Yeadon, 1997). A tumbling model and the

aerial movement model were used to demonstrate that feedforward control might be used during takeoff to compensate for the variability in approach characteristics (King and Yeadon, 2003).

The principle of sensory correction allows humans to assume control over skeletal-muscular moving systems (Bernstein, 1996), sending signals ahead of the movement in prediction of error as feedforward control and/or using feedback control as a self-regulating system for error detection and correction (Holst, 1954; Adams, 1971; Schmidt and Lee, 1999). The control in discrete fast movement such as in the throwing of a dart is feedforward and feedback control is used in movement with longer duration (longer than reaction time) such as coordination of the action in a tennis rally (Klapp, 1975).

The interaction between the diver and springboard highlights one aspect of the complexity of the event since divers need to know how to make best use of a variable springboard and how to cope with any perturbations in order to have a good performance at takeoff, in the air and an accurate performance outcome. Even though these events have been learned by repeated attempts, the coordinated movements are not identical from trial to trial and as a consequence there is still coordination variability due to errors in the selection of motor commands, sensory information and execution (van Beers et al., 2004; Bartlett et al., 2007; Cohen and Sternad, 2009). Feedback control increases kinematic variability in certain aspects of repeated trials of the same skill (Yeadon and Mikulcik, 1996), each feedback correction needs to be performed with accuracy to minimise variability in the performance outcome. Both feedforward and feedback control strategies are used in aerial movement, such that takeoff and the early part of flight phase use solely feedforward control whereas feedback control is used to modify the planned feedforward movement in the later stages of the aerial phase in order to make configuration changes to control rotational movement during 1.4 s flight time of a twisting somersault (Yeadon and Hiley, 2014). Although, this might suggest that feedback from initial touchdown conditions is used to control the board contact phase and feedback from initial takeoff conditions is used to control the aerial movement, no evidence has been reported on whether there is any compensation for variation at takeoff and what kind of information processing

(feedforward or feedback) is used to control board contact phase and the aerial movement in springboard diving.

The correct orientation angle at entry into the water is often considered one of the primary performance outcomes of the dive, since it is the last part of the movement the judges see, with incorrect orientation at entry accompanied by greater splash of the water. Success in competitive diving requires consistency in achieving appropriate somersault orientation at water entry. For elite divers, it is to be expected that they will be very consistent at takeoff and entry so that the initial conditions of the contact and aerial phases will have low variability. The variability at takeoff is dependent on the initial touchdown conditions and body configuration during the board contact phase and entry will be dependent upon the variability of the initial takeoff conditions and the variability of the configuration changes during flight phase arising from planning and execution errors. Additionally, there is the possibility of making adjustments to body configuration during the board contact and flight phases in order to achieve correct takeoff and entry orientation. There will be limits to the level of adjustment that can be made in the contact and aerial phases. The aim of this research is to investigate the extent to which configuration changes (technique) during the contact and flight phases control the takeoff conditions and entry angle in 1-m springboard forward dives.

2. STATEMENT OF PURPOSE

The focus of the present research is to investigate the variability of a springboard diver's movement and understand control systems in diving together with the extent to which the variability in the early stages of a dive can affect the final phase and entry and consequently the whole performance. Following this, the execution variability of a diver was quantified, and biomechanical data was analysed to understand the core strategy of the diver that governs the movement. A simulation model of the contact phase of springboard diving was applied to investigate the effect of variation at touchdown and maximum board depression on the takeoff conditions and a simulation model of aerial movement was used to investigate the extent to which configuration changes (technique) during the flight phase control the performance outcome (entry angle) in 1-m springboard forward dives.

3. RESEARCH QUESTIONS

Q1. How consistent is the diver from dive to dive? i.e. what is the variation in the durations of phases, mass centre velocities and the joint angles at start and end of each phase in 1-m springboard forward dives?

As kinematic data were recorded (Chapter 3), joint angle time histories of a diver in 15 forward pike and 15 forward 2½ somersault pike dives were calculated. The mean and standard deviations of the time of phases, mass centre velocities and the joint angles at touchdown, takeoff and entry were calculated. The movement pattern of the mass centre location in the takeoff and flight phases was determined (Chapter 5).

Q2. Is there any compensation for variability in foot placement of the hurdle?

Since landing near the end of the board decreases the effective linear vertical board stiffness, foot placement has a key role in changing the board stiffness. The diver starts the hurdle by jumping from an active leg toward the end of the diving board and there is inevitably an amount of variability in foot placement from trial to trial. It might be expected that there could be an adjustment to compensate for the variability of foot placement in the hurdle (Chapter 6).

Q3. To what extent can variation at touchdown and at maximum board depression affect the takeoff conditions?

The role of the board contact phase is fundamental for the flight phase and the whole dive because the dive height, board clearance distance and angular momentum are all determined during this mechanically important phase. A diver can perform the same dive with a slightly different technique and make corrections to compensate for any perturbation during the takeoff, flight, and/or entry. It was hypothesised that the divers make adjustment during the contact phase to cope with variation of the initial touchdown conditions leading to consistency at takeoff. A subject-specific angle-driven simulation model of a diver and springboard (Kong, 2005) was used to investigate variability and control during the contact phase. The kinematic data (Chapter 3) and the model parameters (Chapter 4) were used based on data obtained in this study to customise the simulation model. The variation of the actual performance was used to vary the model initial conditions to investigate its effect on takeoff conditions. The time and magnitude of variation in actual performances and simulation outcome determined the difference between execution variability and adjustment (Chapter 7).

Q4. Is there any adjustment in the flight phase of 1-m springboard forward dives?

Although there are many different types of mathematical or physical models of tracking movement behaviour, the most important aspect of a system is how it uses feedback to detect the error and how it initiates a correction. If a simulation model is used to model the human movement, the non-living outcome of the model can be compared with the human counterparts to understand how the human behaves (Schmidt and Lee, 1999). A computer simulation model of human airborne movement of Yeadon et al. (1990b) was used to investigate whether there is any adjustment during the flight phase. The kinematic data (Chapter 3) and the model parameters (Chapter 4) were used to customise the simulation model based on experimental data obtained in the present study. The model was applied to examine the variability in performance outcome arising from variation at takeoff and body configuration during the flight phase. It was hypothesised that there is feedback correction during the flight phase if the variation in the simulation outcome is greater than the variability in the actual performance (Chapter 8).

4. CHAPTER ORGANISATION

Chapter 2 consists of a review of the literature on diving, springboard characteristics, movement variability, motor control and application of simulation models. The pros and cons of different research techniques are addressed and limitations of previous studies are discussed.

Chapter 3 describes the method applied to collect experimental data directly from an international male springboard diver and the subsequent data processing and analysis. The protocol of digitising the recorded videos is also explained.

Chapter 4 explains the procedure to calculate subject-specific model parameters required as inputs to both the angle driven simulation models of board contact and flight phases. An explanation of the method used to determine model parameters including body segmental inertia, strength, wobbling mass, springboard and visco-elastic parameters.

Chapter 5 analyses forward pike and 2½ somersaults pike dives performed by the diver of the present study. The variability and consistency in different aspects of the movements are discussed. The relationships between variables in different phases of dives along with factors influencing dive height and entry angle are identified. Answers to the first research question addressed in Chapter 1 is also given.

Chapter 6 analyses the variability of foot position on the springboard to investigate whether the diver makes any correction for variability in hurdle. Answers to the second research question addressed in Chapter 1 is also given.

Chapter 7 applies the angle-driven model of a diver and springboard to investigate the movement variability and control in 1-m springboard forward dives. Answers to the third research question addressed in Chapter 1 are also provided.

Chapter 8 provides the procedure of applying the angle-driven simulation model of aerial movement to investigate the movement variability and control during the flight phase. Answers to the fourth research question addressed in Chapter 1 is also provided.

Chapter 9 a summary of the main finding of the study is provided. This also includes the answers to all research questions posed in Chapter 1. The limitations of the present study with suggestions for future research are also presented.

CHAPTER 2

LITERATURE REVIEW

2.1. SPRINGBOARD DIVING

Looking at the history, the diving action can be traced from a Greek wall painting created about 470 B.C. (Holloway, 2006) to the first modern diving competition in England in the 1880s (Wilson, 1883). The diving competitions were introduced at the third Olympic games in 1904 and the first platform and elastic diving was presented in the Olympic games in 1908 (IOC, 2011). The modern era of springboard diving arguably began when it underwent major changes not only by better facilities and equipment but also by changes in the rules and structure of the competitions. The construction of the diving board evolved from wood covered by coconut matting to aluminium alloy, the length increased from 3.66 m to 4.88 m, the slope changed from 0.020 m elevation to completely level and a rigid pivot replaced with flexible boards (Miller, 1985; Rewt, 1993).

In the early existence of competitive diving, divers used to run toward the end of the board to obtain sufficient horizontal momentum to clear the board due to the rigid wooden plank (Miller and Munro, 1985). In later years when the springboards with easily adjustable fulcrum emerged, a hurdle pre-flight approach was substituted for the traditional walking and running approaches to lower the difficulty of generating momentum and to increase the duration of the hurdle and dive flight phases (Miller et al., 1998; Miller, Zecevic & Taylor, 2002). During the past few decades, there were only a few professional springboard divers capable of executing a forward 1½ somersault dive with a double twist from a 3-m board (Sprigings et al., 1990). Nowadays dives such as 2½ somersaults have been performed even on 1-m board almost routinely. It is obvious that this improvement in performance is a result of the significant role of better coaching methods and advances in springboard technology.

2.2. MOVEMENT CONTROL

The study of the movement of a patient with a gunshot wound in the back by Lashley (1917) is the first important documentation that the movement has been centrally

controlled. Helmholtz (1925) and Holst (1954) argued that the central nervous system should have information about the motor commands sent to the muscles, which are responsible for moving the eyes. Without this information, there would not be an accurate visual perception. Bernstein (1967) reported that there is a motor command centre that has the responsibility of what the response sequence need to be. A comparator receives the feedback from the response to test it against the ideal one in the command centre, and an error signal is resulted for a correction. Motor program has been called different names such as Victrola record (Hunter, 1930) or a score (Weiss, 1950). Adams (1971) believed that there is a motor program and defined it as central organisation for the movement.

Bernstein (1996) observed that dexterity appears in the interaction of motor act with the changing environment not in the motor act itself. One of the important features of dexterity, as an ability that defines the relation of the central nervous system to motor skills, is the flexibility with respect to the changing environment. Although it is important to classify the actions and motor tasks, it makes too difficult to understand the laws of motor behaviour not only because various terms are used to explain the movements in the research literature but also it is dependent on the kinds of performances under consideration (Schmidt and Lee, 1999).

For instance, the environmental and situational characteristics of open skills can change as the performer plans his response. When a wrestler tries to takedown, the action, could be defined as an open skill (Poulton, 1957). The performers with open skills have to plan the movement in a way that they can control the environmental uncertainty whereas it is predictable in closed skills (Schmidt, 1975). The goal and environmental conditions in closed skills are relatively constant. A predictable environment might happen when the situation is variable but the changes have been learned as a result of practice. Closed skills require very consistent and stable performances in a predictable environment such as springboard diving (Schmidt and Lee, 1999).

2.2.1. Motor programme

Although there are a number of studies that tried to understand the motor performances (Honzik, 1936; Chase et al. 1961; Laszlo, 1967; Laszlo & Manning, 1970), Wilson's (1961) study on the sensory control of the locust's wings was the most powerful as he found that there is a central motor program governing the wing movement.

With the advent of electronic computer, Henry's (1960) tried to use the notion of the memory drum element in the early computers as an analogy to human system and proposed the memory-drum theory. His idea was that the programs applied by the machines can be used to understand the human movement. One of the most popular ways to understand human behaviour in the environment is to study the fundamental concept of human information processing. The motor program has been defined by Keele (1968) as a sequence of stored commands that is formed before the initiation of the movement and allows the entire arrangement to be run without any influence by peripheral feedback. The information from environment acts as an input through the sensory organs begins the process of information to consider what the input causes to happen inside the system.

The interval between the presentation of a stimulus and the beginning of the response is one of the most important measure of the human behaviour. This time interval is called reaction time which is the time from the arrival of a suddenly presented and unanticipated signal to the beginning of the response to it. The stimulus identification, response selection and response programming are defined as three stages in reaction time. After sensing the stimulus, the subject identifies the stimulus information as part of a pattern, decides what response to make, selects the appropriate action, and finally, the action will be initiated. The last period, which is the preparation of the motor control and the initiation of the action, is called the response programming or the organisation and initiation of the action and is processed and translated into a set of muscular actions (Schmidt and Lee, 1999).

Schmidt's (1975) found that if there is any error in the initiation of the movement, it takes from 0.12 s to 0.2 s (one reaction time) to correct the movement. The process

involving the generation of sensory error information, perceiving it, and initiating corrections in response to the errors requires about one reaction time. Thus, the decisions about when to stop the movement must have been made before the movement begins. Schmidt and Gordon (1977) experienced that the typical minimum simple reaction times range from 0.13 s to 0.2 s with the average 0.144 s. Schmidt and Lee (1999) reported that it takes 0.04 s to 0.08 s from the presentation of the stimulus to the activation of the muscle. They concluded that the slowness of the information process stage, the evidence of planning of the movement in advance as well as the evidence of executing the movement without afferent information are three reasons that movements are controlled by programs.

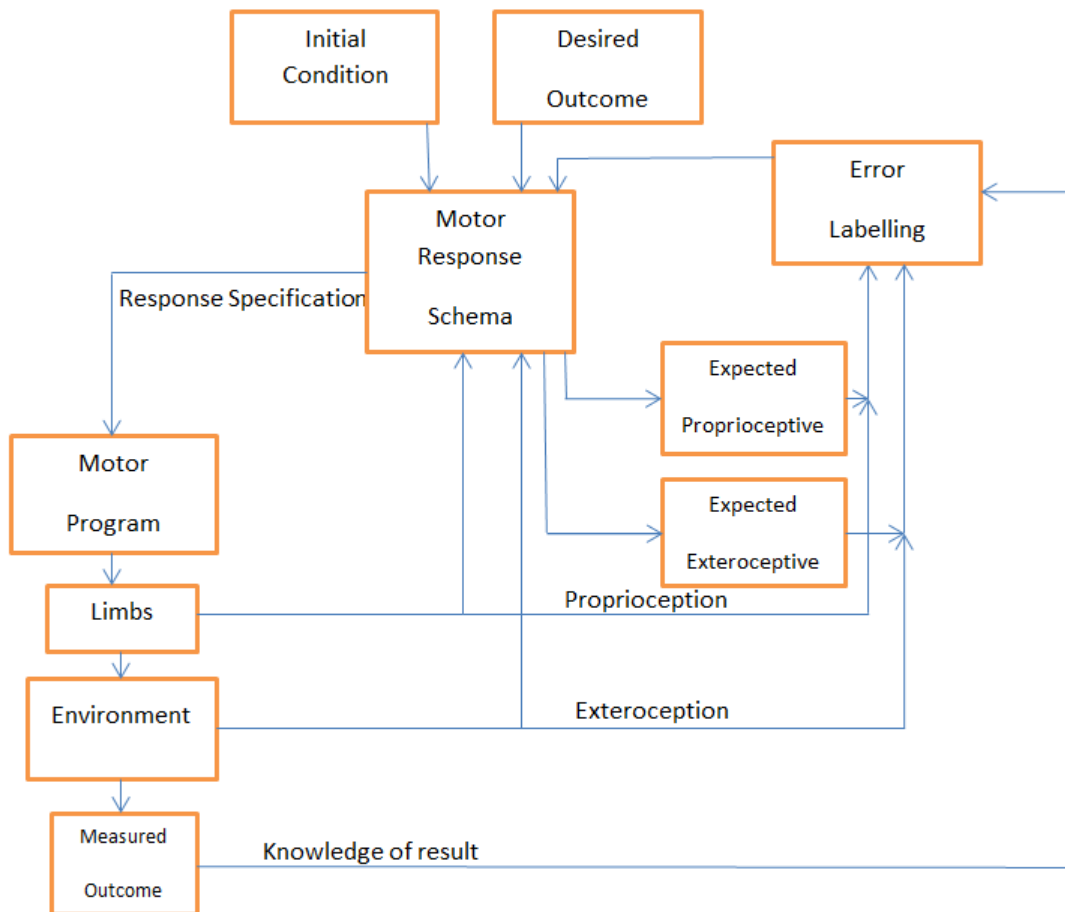


Figure 2.1. Schmidt's (1975) notion of schema.

According to the Schmidt's (1975) notion of schema (Figure 2.1), when subjects are planning to execute a movement, they select a desired outcome, note the initial

conditions, and then begin the process of the motor program. In order to determine what set of specifications is required to achieve the desired outcome, they work on the relationship between the past actual outcomes and past response specifications. At the same time, from the relationship between the actual outcome and the sensory consequences, the selection of the expected proprioceptive feedback and exteroceptive feedback are selected. This allows the generation of the response specifications which are the best prediction of the desired outcome. The muscle contractions produce the movement of the limbs and the body makes changes in the environment. Then, the reference correctness can compare the feedback from outcome of the movement and the feedback that expects to receive. The implication is that the movement is correct if the two feedbacks are the same and there is no need for adjustment but, if they are not the same an error is signalled and a correction is needed (Schmidt and Lee, 1999).

Immediately after running the motor program, efferent information starts flowing out to the muscular system with all of the details during the first 0.2 s of the movement. This means when the performer executes the movement, he/she cannot change it for at least 0.2 s. When a movement is executed the knowledge of the result (KR) is stored in the motor program. The expected sensory consequences are dependent on the development of the relationship between the response outcome resulted from previous trials and the actual sensory feedback received (Schmidt, 1975). The feedback from early stages of the sequence could be involved to provide adjustments for the later stages.

Knowledge of result is not fed back only to the motor program for updating it but also to the error labelling system to improve the accuracy of this system in identifying the future error signal arising from the deviations of the actual and expected proprioceptive and exteroceptive feedbacks. Since all sources of information of movement are used for updating the motor program, they provide revised estimates of the expected sensory information and response specifications for the next trial of the movement (Schmidt, 1975).

2.2.2. Feedback control

Sensory control of action can be considered in ways similar to mechanical systems control. Closed loop system can control itself for a long period of time. This system is called closed loop because the loop of control from the environment to decisions to action and back to the environment again is completed or closed (Figure 2.2). The system receives and processes the feedback coming from stimulus identification stage and then send the program instructions to the muscles to reduce the error in the response programming stage. The various sensory information is sent back to the reference correctness for analysis, and decisions about the future actions (Schmidt and Lee, 1999). The sensory information which is used in the closed loop control is often called feedback or movement-produced feedback as it is the result of performed actions.

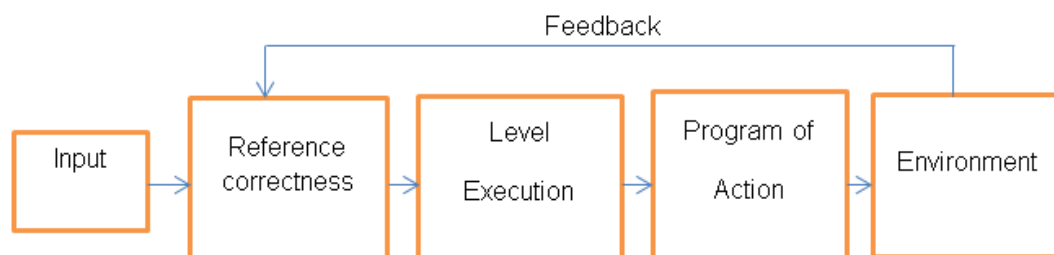


Figure 2.2. Feedback loop system.

There are many different forms of sensory information associated with the performed actions and are often considered as sensation and perception. There are three groups of sensory information (or receptors) available during movements; interoceptors information about the state of internal organs, exteroceptors that have information about the movement of objects in the environment and proprioceptors provide information about the movement itself. Vestibular, muscles, joints, golgi tendons and cutaneous receptors are the set of sensors that provide proprioceptors information (or kinaesthetic) for the movement of the body. These receptors play fundamental roles for feedback control (Abbs & Winstein, 1990). The processing of feedback against a reference of correctness in the closed loop system causes the detection of the error and subsequently results in correction.

While a movement is in process, the reference correctness changes constantly and the feedback from the moving limbs and the feedback from expecting movement are matched to detect the error in the trajectory of the movement and correct it. In this way, the subject learns a set of references of correctness that the closed loop system is tracked during the movement. Klapp (1975) found that by eliminating the feedback the long movements were totally disrupted but the short movement were maintained with reasonable accuracy and concluded that feedback is more crucial for the longer movement (longer than one reaction time). On the other hand, there would not be enough time in the closed loop system for rapid movement to generate error, detect it, determine the correction and initiate and correct the action. This process requires about 0.15 s to 0.2 s to complete the process but, the time of movement itself in rapid movement might be less than 50 ms. Thus, the movement might finish before the correction begins (Schmidt and Lee, 1999).

Before the initiation of actions, sensory organs provide information about the initial conditions of the motor system. If the environment is not perfectly predictable feedback during the movement as a monitoring function acts to control the action. If the reference of correctness is set to determine the end location of the movement, the feedback could be involved and set the position. In other words, sensory information perhaps provides information resulting from the error made on the previous trial to make an adjustment and reduce the error of the movement on the subsequent trial (Schmidt and Lee, 1999).

2.2.3 Feedforward control

In the 19th century William James (1890) proposed the first open loop theory for motor control which is the theory of the reflex chaining. The idea was that a movement starts by sending a signal to contract a muscle. The feedback is generated from the muscle contraction or the movement caused by the contraction or both of them. This sensory information or stimulus acts as a trigger for the next contraction. The second contraction generates its own feedback to trigger the third contraction. Such a procedure continues until all the contractions are completed. Under this hypothesis, when skilled actions are generated, there is no conscious control of the movement. In

other words, attention is needed only to initiate the first action and the rest of the movement is run off automatically (Schmidt and Lee, 1999).

There is no mechanism or feedback for error control in open loop system. The input goes through the system and applies its influence, the system acts on the input, and an output is produced. (Adam, 1971). Keele (1968) and Schmidt (1972) claimed that there is no involvement of the peripheral feedback in the movements with duration less than 0.2 s.

Although the open loop control depends on the type of response, it could run itself for nearly 0.4 s before it could be corrected. As long as the time of movement is constant, the amount of possible feedback involvement in a movement longer than 200 ms is dependent mainly on the movement time. The performer becomes a lot more independent of feedback for execution due to the lags in processing the feedback. The problem in the motor skills for the performers is to develop these open loop programs to make themselves free from feedback involvement (Schmidt, 1975).

Holst (1954) proposed that a copy of efferent information is fed forward to correct the incoming visual sensation and found that an image of the motor commands has been fed forward to the eyes to compensate for the eye's movement. It has been reported that the visual perception system has already been informed about the upcoming movement of the eye so that the pattern of changed input can be estimated correctly. Feedforward control is provided in human motor system, sending signals ahead of the movement to prepare the system for upcoming motor command. Therefore, it plays a key role in error detection and correction as it occurs often in prediction of the error (Schmidt and Lee, 1999).

According to Fitts' (1954) model of human movement, the movement time is dependent on both distance to the target and the width of the target. Klapp (1975) showed that Fitts's law may be associated with the long movements feedback control but no longer holds for very short movements because they are programmed in advance. Lee (1976) proposed the theory of time to collision to explain motor behaviour of a driver to control braking based on visual information and concluded that the driver uses information

about time to collision for controlling braking rather than information about distance, speed, acceleration or deceleration.

2.3. MOVEMENT CORRECTION

Bernstein (1996) believed that the corrections to imaginable changes in the environment should be programmed in advance or even run in prediction of the perturbation. The rapid movements are structured before the initiation of the movement, and when it is initiated no modification will occur until the program runs its course for the next few milliseconds. Although the details of control are determined centrally, low level reflexive activities will play the role of regulation in rapid movement (Schmidt and Lee, 1999).

The executions of correct movements are not dependent on the precision of the signals from central nervous system to the muscles due to the excessive degree of freedom. Bernstein (1996) noted that the forces of interaction between the body's segments and reactive forces are not reflection of the central nervous information. Motor skill cannot be characterized as a formula of signals to effectors as the information from the brain to muscles is different for repetitions of the same task.

The study of movement correction is limited to a few studies. Increasing strength and variation in technique in gymnastics could improve the consistency of performance and success in a limiting movement (Hiley & Yeadon, 2012). A single symmetrical arm adduction using delayed feedback control could correct a perturbation in a simulated target trampoline movement to control the twist (Yeadon & Hiley, 2014). The movement variability in a gymnast's technique in a backward giant circle has the characteristic of improving performance (Hiley & Yeadon, 2016).

Dynamical systems theory has been used as a framework to provide individualised plots of motor performance and interpret the variability and stability demonstrated by individuals in supporting physical performance. Although it has been applied for studying motor behaviour, application of simulation models can be considered a parallel approach as both provide a theoretical framework to explain movement variability in supporting physical performance.

2.4. APPLICATION OF SIMULATION MODELS

Development of the simulation model, determination of subject-specific parameters, evaluation of the model, and application of the model are a four stage methodology required with each model to answer specific research questions (King and Yeadon, 2015).

Experimental studies in sports biomechanics have been often problematic: not only may they produce an artificial environment but also unwanted changes in addition to the designed change (Yeadon, 1997). In the last 20 years a variety of computer simulation models have been developed to answer a range of research questions in sports biomechanics. These simulation models have provided insight into the mechanics of sports movements, contributions to performance, optimisation of sports technique and control of sports movements. The main benefit of using such simulation models to address research question is that it is possible to change just one variable at a time while the ideal theoretical experiments are carried out (King and Yeadon, 2015).

Alexander (1990) applied a simple model to predict optimum takeoff techniques in high and long jumpers. He found that long jumpers run at a near maximal approach speed whereas high jumpers run at moderate speed. Yeadon et al. (1990b) developed an 11-segments angle-driven simulation model of aerial movement. The model was applied in various studies in sports biomechanics such as the 1-m and 3-m springboard diving (Yeadon, 1993), gymnastics apparatus dismounts (Yeadon et al., 1990a; Yeadon, 1994), gymnastics floor exercise (Yeadon and Kerwin, 1999), control of non-twisting straight somersaults (Yeadon and Mikulcik, 1996), rotations of the human body during free flight (Yeadon, 1997), high bar (Yeadon and Hiley, 2000; Hiley and Yeadon, 2003a; 2003b; 2005; 2008; 2012; 2016), tumbling (Yeadon and King, 2002; King and Yeadon, 2003; 2004). The application of the simulation model provided a mean for understanding how athletes must make movements in the aerial phase to control the action.

A simulation model of triple jump was applied to investigate the takeoff technique in each phase of the triple jump. It was found that the symmetrical arm movement during

the takeoff phase provides a substantial increase in phase distance (Allen, King and Yeadon, 2010). Yeadon, Jackson and Hiley (2014) applied a computer simulation model to investigate the extent to which the conditions at touchdown and the technique during table contact affected peak post-flight height in a straight handspring somersault vault. They found that optimising either the touchdown conditions or the contact technique increased post-flight height by 0.1 m whereas optimising both together increased post-flight height by 0.4 m above that of a simulation matching the recorded performance. They concluded that changes in initial conditions and contact technique have a similar influence on post-flight height.

Wilson, Yeadon and King (2007) applied a torque-driven model of the foot contact phase and found that the constraints such as angular momentum at takeoff, joint angle limits, and robustness to perturbations should be taken into account for maximising jump height. Kong (2005) used a simulation model of a diver and a springboard and concluded that a diver can perform the same dive with slightly different techniques and make corrections to compensate for any mistakes during the takeoff, flight, and/or entry.

Yeadon (2013) investigated various asymmetrical movements to produce twist during three somersaults in freestyle skiing using a computer simulation model of aerial movement and found that more twists could be performed than are currently seen in competitions and concluded that reduction in equipment mass might make the twists easier to achieve. Hiley & Yeadon (2016) determined an optimisation criterion best characterised the technique in backward giant circle prior to release adapted by a gymnast. They found that optimising based on maximising the likelihood of success in the presence of noise provided a better characterisation of the technique. They concluded that the movement variability in the gymnast's technique has the characteristic of improving performance (maximising success).

2.5. Summary

This chapter provided a review of the literature on springboard diving, feedforward and feedback control and application of computer simulation models. The information obtained from the literature was used to further investigate the movement variability and control in springboard diving through the application of simulation models.

CHAPTER 3

DATA COLLECTION AND ANALYSIS

3.1. DATA COLLECTION

3.1.1. Camera set-up

A Fastec TS3 high speed camera (Fastec Imaging Corporation) was chosen to record a number of diving performances from 1-meter springboard. The camera was positioned on the deck as far as possible from the springboard. A 30 mm (1:1.4) lens was connected to the camera and it was set-up with the field of view about 5 metres wide covering the whole sequence from approach to the entry of a dive. The frequency, exposure time and resolution of the camera were selected 250 Hz, 4 ms and 1280 x 1024 pixels respectively. A series of numbers were positioned at the left corner of the field of view to identify the number of each recorded trial (Figure 3.1). The number was changed after each trial to the number of the next performance. The diver started warming up and performed a number of practice dives. The camera's field of view was checked and set-up during warm up. The camera was post triggered and cropped to an average of 5 seconds and the video data were stored on the camera's memory card. In order to observe the fulcrum in the field of view, a Panasonic DMC-FZ200 as a backup camera was positioned next to the main camera to record the whole sequence from fulcrum to the entry of the dive.



Figure 3.1. Camera field of view while the diver standing on the end of the board.

3.1.2. Performances

An elite male diver competing at international level (mass 69.7 kg, height 1.79 m) participated in the study. In addition to informed consent approved by University of Loughborough (Appendix 1) and anthropometric measurements (Appendix 2), a subject profile of competition and training background were obtained (Appendix 4). After explaining the purpose and details of the study, the diver performed 15 forward pike dives (101B) and 15 forward 2½ somersaults pike dives (105B) from the 1-m springboard (Table 3.1). In the dive code the first number indicates the dive group (1=forward), the second gives the takeoff position (0=from the feet), the third indicates the number of half somersaults (5=2½ somersaults), and the final letter defines the body configuration in flight (B=pike). The easier dives (101B) were performed in the morning and more difficult dives (105B) in the afternoon. The diver was asked to perform his best dives as in a real competition. No additional instructions, specific comments or corrections were provided to the diver. In order to avoid the effect of fatigue there was a rest period of two hours between the morning and the afternoon sessions. In addition, during data collection, the diving coach was asked to score each trial of the diver's performances.

Table 3.1. Number of dives performed during data collection

	Forward pike	2½ somersaults forward pike
Number of trials	15	15

3.1.3. Fulcrum

Since the diving board moves violently and might move the fulcrum, the position of the fulcrum was checked after each dive to ensure it was the same for all dives (Figure 3.2).

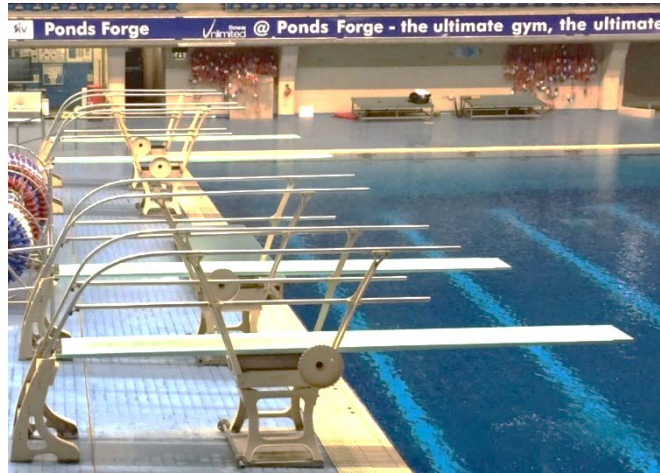


Figure 3.2. 1-m springboard, Ponds Forge, Sheffield, United Kingdom

3.1.4. Camera calibration

The first camera calibration was performed after collecting the first data in the morning and the second calibration was carried out after the second session in the afternoon in case that the camera was moved between the sessions. A calibration pole with seven balls attached was used to calibrate the camera (Figure 3.3). The numbers of the balls were 0 to 6 from bottom to the top of the calibration pole and the distance of each ball was known as it was measured prior to the data collection. The calibration pole was placed horizontally on the floor and distance of each ball to the bottom of the pole was measured. Table 3.2 shows the comparison between the measured and the manufacturing reference values.

Table 3.2. Comparison between the measured and manufacturing reference

Distance of the balls to the bottom of the calibration pole		
Ball number	Measured distance (mm)	Manufacturing reference (mm)
0	95	99
1	1181	1183
2	2164	2165
3	3166	3166
4	4151	4149
5	5171	5168
6	6101	6100

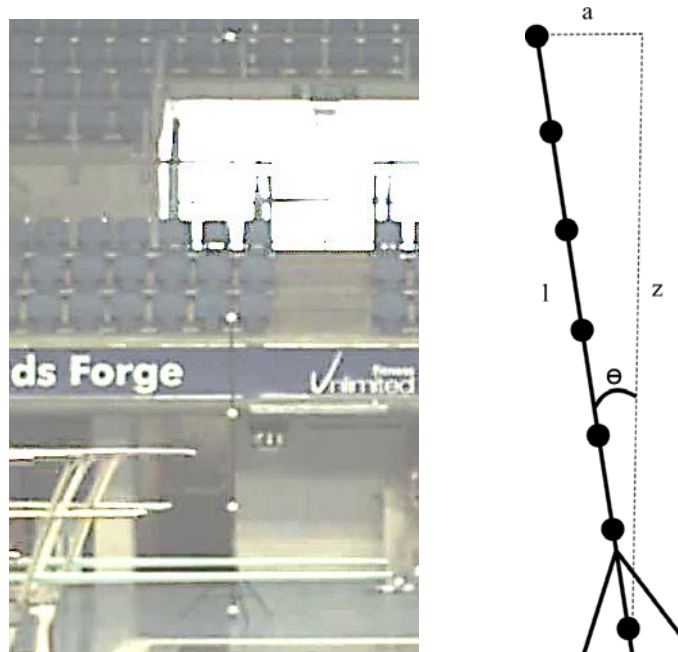


Figure 3.3. The calibration pole with the seven balls in vertical position.

The calibration pole was positioned in six different known locations, three at the near side (P1 - P3) and three at the far side (P4 – P6) of the pool deck. After recording a short period in each position, the horizontal location of the bottom of the pole was measured. It was assumed that the right-hand side of the body (RHS) ($X = 0$) was -0.1 m away from the middle of the board (Figures 3.4 and 3.5).

During placing the pole vertically on the pool deck for camera calibration, the pole leaned towards the water as the pool deck was not level (Figure 3.3). The slopes of leaning at the near side (P1, P2 and P3) and far side (P4, P5 and P6) of the pool were 0.05 and 0.06 respectively. In order to evaluate the markers height above the water level, an extra height was added to the original markers height as follows:

Near side of the pool: Extra height = $d_n \times 0.05$ m

Far side of the pool: Extra height = $d_f \times 0.06$ m

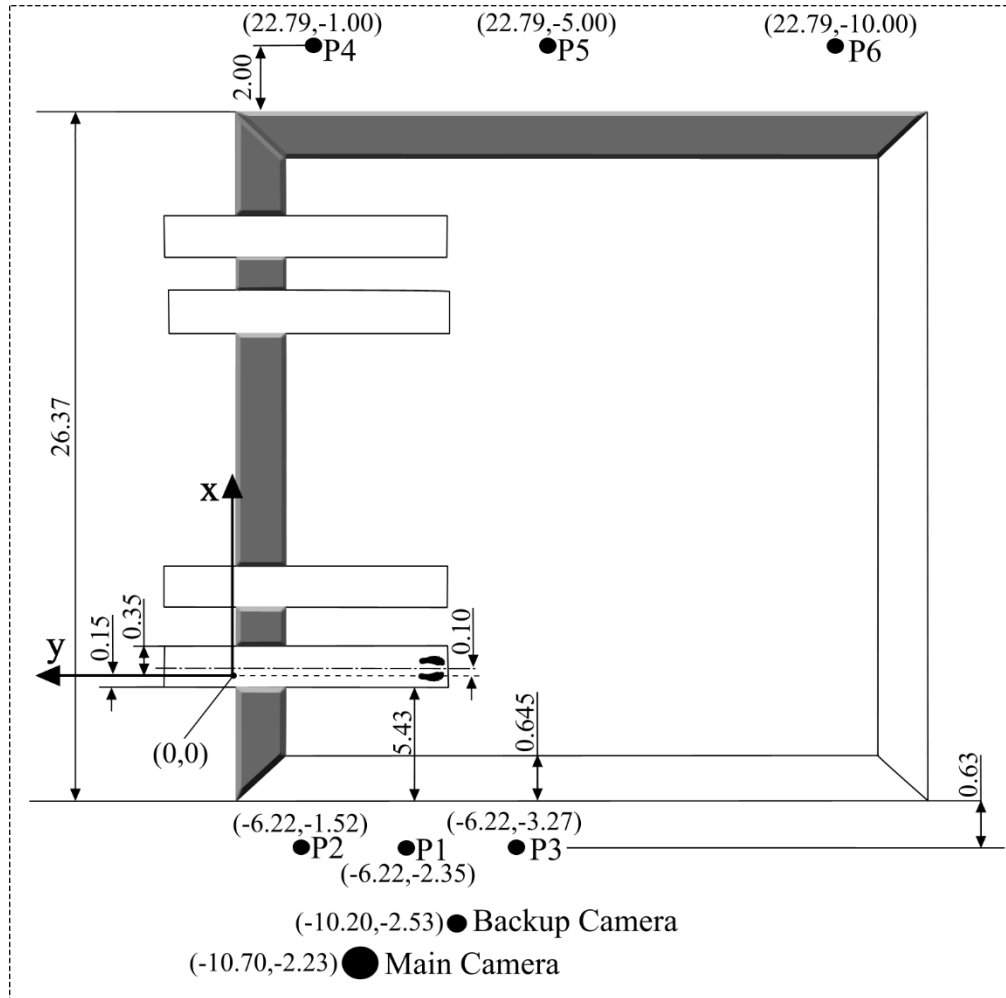


Figure 3.4. A two-dimensional plan of the pool and the measures of x-y coordinates of P1 to P6.

Where 0.05 and 0.06 are the amount of leaning at near and far side of the pool, respectively, d_n is the horizontal distance of the pole positions (P1, P2, P3) to the edge of the water at the near side of the pool and d_f is the horizontal distance of the pole positions (P4, P5, P6) to the edge of the water at the far side of the pool. The horizontal distance to the edge of the water (d_n and d_f), lean and extra height were calculated for the camera, pole and all balls positions. The amounts of extra height were 0.064 m and 0.159 m for the positions at near side (P1, P2 and P3) and far side (P4, P5 and P6) of the pool, respectively (Table 3.3).

Table 3.3. Horizontal distance, lean and extra height factors

Facility	Positions	horizontal distance to the edge of the water (m)	Lean slope	Extra height (m)*
Pole	Near side (P1, P2, P3)	1.275	0.05	0.064
	Far side (P4, P5, P6)	2.645	0.06	0.159
Camera	Near side	5.765	0.05	0.288

* extra height = horizontal distance x lean slope

The x-coordinates were also adjusted with respect to both positions of the pole and the balls to account for the effect of leaning of the pole. Following this, the original markers heights were reduced by the factors of leaning and increased by the factors of extra height (Table 3.3).

Table 3.4. Horizontal deviation of the markers

Ball number	The estimated horizontal deviation (m)	
	near side P1 to P3	far side P4 to P6
0	0.008	0.015
1	0.062	0.080
2	0.111	0.139
3	0.161	0.199
4	0.210	0.258
5	0.261	0.319
6	0.308	0.375

The x-y-z coordinates of seven balls of the pole were calculated from the measured distance of the ball and the amount of leaning (Figure 3.4). Following this, the amount of horizontal deviation of the markers (balls) were estimated (Table 3.4). The estimated horizontal deviation (a) and the length of the pole (l) were used to calculate the amount of leaning in each pole position (Table 3.5).

Table 3.5. The amount of leaning in each pole position

The estimated horizontal deviation (m)		
Pole position	Horizontal deviation	Leaning
1	0.308	2.9°
2	0.308	2.9°
3	0.308	2.9°
4	0.375	3.5°
5	0.375	3.5°
6	0.375	3.5°

The vertical height of the markers (z-coordinates) and horizontal distance (y-coordinates) were calculated and the lateral distance (x-coordinates) were adjusted as follows:

$$a_i = l_i \sin(\theta)$$

l_i = measured distance of the balls to the bottom of the calibration pole

$$x_i = [\text{the measured x coordinates}] - a_i$$

y_i = measured distance of the pole position after camera calibration

$$z_i = [l_i \cos(\theta)] + \text{extra height factor}$$

where $i = 0-6$ based on corresponding ball number

As the pole calibration with seven balls has been placed in six different places at near and far sides of the pool, it was expected to have 42 balls points available for calculation. However, there were 27 balls points within the field of view. At the near side of the pool only balls number 1 and 2 were visible, giving 6 balls within the field of view in positions 1, 2, and 3. At the far side of the pool, the ball number 4 at position 4 was not visible due to light reflection from a window on the back of the pool (Figure 3.3). Therefore, only 26 balls remained visible comprised 6 balls at the near side and 20 balls at the far side of the pool (Table 3.6).

Table 3.6. Space coordinates of the 26 control points for camera calibration (m)

Point*	x	y	z
P11	-6.15	-2.35	1.24
P12	-6.10	-2.35	2.23
P21	-6.15	-1.52	1.24
P22	-6.10	-1.52	2.23
P31	-6.15	-3.275	1.24
P32	-6.10	-3.275	2.23
P40	22.77	-1	0.25
P41	22.71	-1	1.34
P42	22.65	-1	2.32
P43	22.59	-1	3.32
P45	22.47	-1	5.32
P46	22.42	-1	6.25
P50	22.77	-5	0.25
P51	22.71	-5	1.34
P52	22.65	-5	2.32
P53	22.59	-5	3.32
P54	22.53	-5	4.30
P55	22.47	-5	5.32
P56	22.42	-5	6.25
P60	22.77	-10	0.25
P61	22.71	-10	1.34
P62	22.65	-10	2.32
P63	22.59	-10	3.32
P64	22.53	-10	4.30
P65	22.47	-10	5.32
P66	22.42	-10	6.25
Camera	-10.7	-2.23	2**

*point position P12:
1=pole position, 2=ball number
**height adjustment:
1.70 (original measure) + 0.288 (extra height) = 1.99 m

To measure the poles, the origin was set up on the board 0.1 m out from the centre where it is 2.55 m away from the board tip and 0.95 m above the ground (Figure 3.4). To have the position of the toes relative to the board tip, the origin was moved 2.55 m to the right towards the tip of the board and 0.95 above the pool to be at the end of the board. Since the diver's hip width was 0.319 m (Appendix 2), the left hip was 0.180 m away from the origin, the middle of the head is set off from the right foot, giving 0.1 m away from the origin (Figure 3.5).

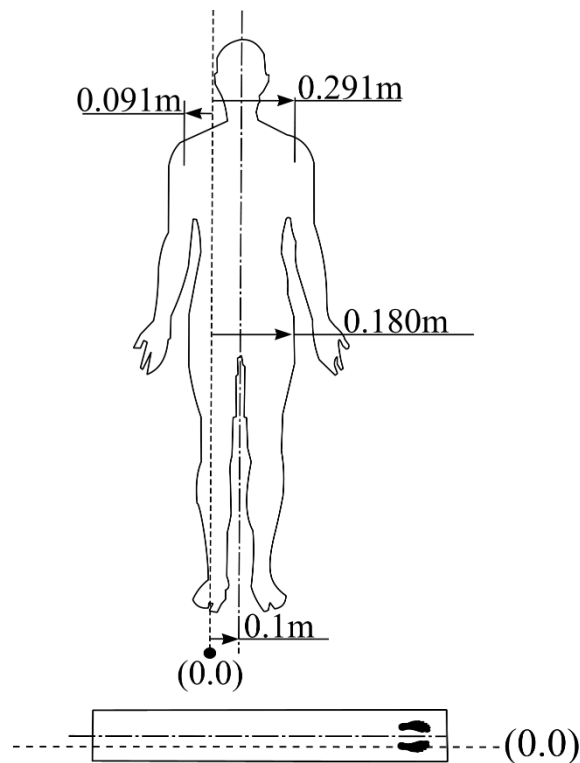


Figure 3.5. Generating pseudo-3D coordinates using anthropometric measurements with the assumption of symmetrical body movement.

3.1.5. Dynamic loading of the springboard

A dynamic loading of the springboard at the fulcrum number 8.5 that the diver normally used for all dives in forward group was applied to determine the springboard stiffness (Miller & Jones, 1999). This was performed by having the diver standing on the end of the board (Figure 3.1), setting the board into oscillation by raising his arms once, and keeping his body as rigid as possible with the arms adducted for 10 oscillations. The

same procedure was done to load the board with two different body weights (93.8 kg and 89.4 kg).

3.1.6. Anthropometric measurements

Since accurate segmental inertia values needed for each segment to use in this study, 45 anthropometric measurements of Yeadon (1990c) was used to determine the segmental inertia parameters (Appendix 2).

3.1.7. DLT image reconstruction

A three-dimensional (3D) direct linear transformation (DLT) method developed by Abdel-Aziz and Karara (1971) was used for camera calibration. The transformation of digitised coordinates into object space was performed using following equations:

$$u = \frac{L1x + L2y + L3z + L4}{L9x + L10y + L11z + 1} \quad (3.1)$$

$$v = \frac{L5x + L6y + L7z + L8}{L9x + L10y + L11z + 1} \quad (3.2)$$

Where

(u, v) = image coordinates

(y, z) = object space coordinates

L1 – L11 = DLT parameters.

The x-y-z-coordinates of the control points and the camera (Figure 3.4 and Table 3.6), giving 26 control points and 52 equations, were used to solve 11 unknown parameters. Since a single camera was used for data collection, giving a sagittal plane for calibration, three of the 11 DLT parameters were eliminated by setting x to 0, remaining only 8 parameters to reconstruct 2D coordinates. Thus, Equation (3.1) and (3.2) were re-arranged as follows:

$$\begin{bmatrix} L2 - L10u & L3 - L11u \\ L6 - L10v & L7 - L11v \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} u - L4 \\ v - L8 \end{bmatrix} \quad (3.3)$$

After reconstructing the 2D coordinates using Equation (3.3), a set of pseudo-3D space coordinates, based on the assumption of symmetrical movement of the left and right sides of the body was generated. All x-coordinates were projected onto the calibration plane according to the subject anthropometric measurements (Figure 3.5).

3.2. DIGITISATION

An AVI digitiser Version 1.2 was used to digitise the recorded videos. As body landmarks have been digitised to reconstruct 2D coordinates for each landmark, the software main advantage above other available programmes was not only to enable sub-pixel digitising to a 10th of a pixel but also the images can be magnified and interpolated to produce a smoother image when zoomed in and enhancing the user's ability to accurately locate the desired landmark and reduce blocky pixels.

Although the rear of the board was not in the field of view, the calibration data therefore was used to evaluate the consistency of the fulcrum position as the length of the board and fulcrum were known (Figure 3.6). This was done by digitising the joint of front rail and nearest board edge and checking the distance between these two points throughout all performances.

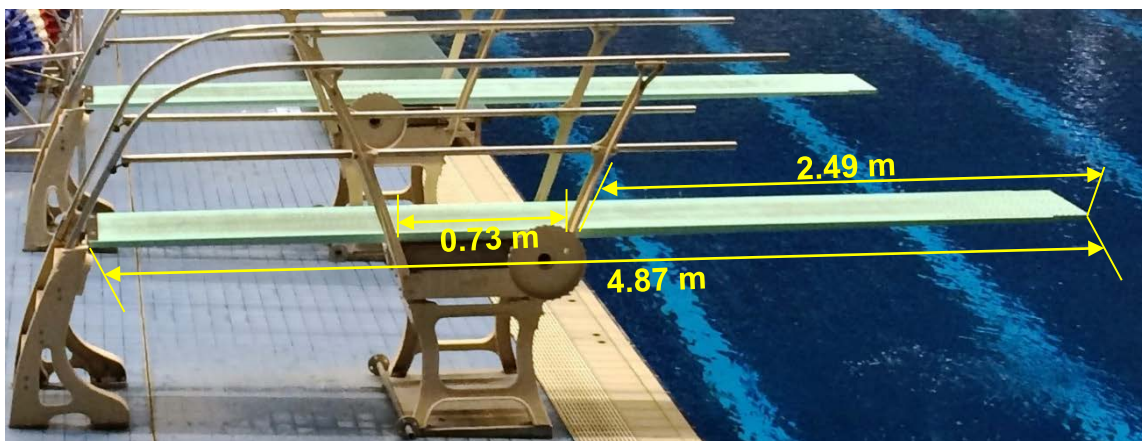


Figure 3.6. Measurements of 1-m springboard.

3.2.1. Board neutral position

The flight phase has been traditionally defined from the instant of losing contact with the board to entry into the water (Sanders & Gibson, 2000; Miller, 2001, 2002; Sanders & Wilson, 1988; O'Brien, 1992; Slobounov et al., 1997; Sinclair et al., 2012). However, the role of the board neutral position has not been taken into account. Once the springboard comes up to its own neutral position it is no longer exerting vertical force on the diver due to the elasticity of the board. Mechanically there would be no forces between the board and diver. Therefore, the hurdle, contact and flight phases were separated by the board neutral position as when the springboard reaches its neutral position at takeoff it moves at the same velocity as the feet. Similarly, when the diver lands on the board, depression starts at the neutral position.

Calibration data was used to identify the coordinates of the board neutral position (Figure 3.7). The tip of the board was digitised in 10 frames using the calibration video and an average of them was made to determine the digitised board neutral coordinates (Table 3.7). The average board neutral coordinates were used as the actual board coordinates in data analysis since the tip of the board was digitised in all phases. Once the video frame was identified, it was used to determine the start and the end of different phases.

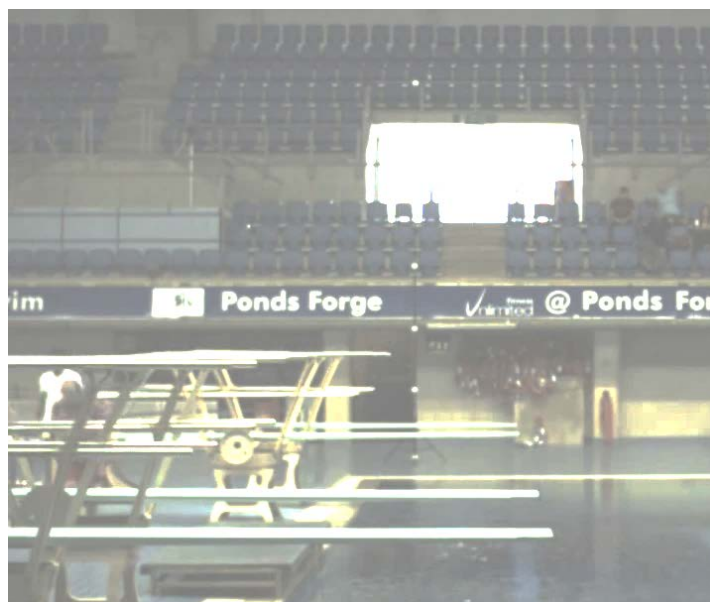


Figure 3.7. Calibration image.

Table 3.7. Board neutral coordinates calculated from 10 frames of the calibration data

Poles coordinates				
Clip	Frame number	Time(s)	Tip-X (pixel)	Tip-Y (pixel)
Clip1	1	0	591.0	266.4
Clip1	2	0.083	591.6	267.0
Clip1	3	0.166	591.0	267.0
Clip1	4	0.249	591.0	267.0
Clip1	5	0.332	591.0	267.0
Clip1	6	0.415	590.5	266.4
Clip1	7	0.498	590.5	266.4
Clip1	8	0.581	591.0	265.8
Clip1	9	0.664	591.0	266.4
Clip1	10	0.747	590.5	266.4
Average			590.9	266.6
SD			0.320	0.384

3.2.2. Foot placement

To investigate the variability in foot placement of the hurdle, when the drive foot (left foot) was flat on the board, the toe position distance from the end of the board at start, during and end of the hurdle takeoff and hurdle landing (for dive) were digitised (Figure 3.8).

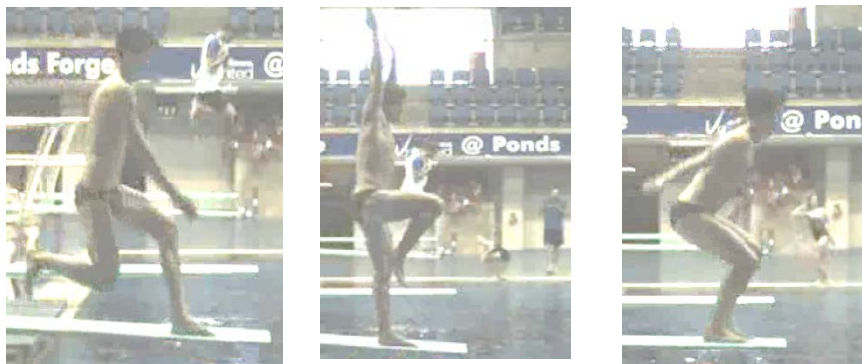


Figure 3.8. The diver position at hurdle touchdown (left), hurdle takeoff (middle) and hurdle landing (right) while the foot is flat on the board.

3.2.3. Flight phase

Digitisation of the hurdle flight phase started from the first frame where the board was in neutral position while the diver was moving upward. The hurdle landing was defined as the frame where the board was in neutral position while the diver was landing from the hurdle (Figure 3.9).

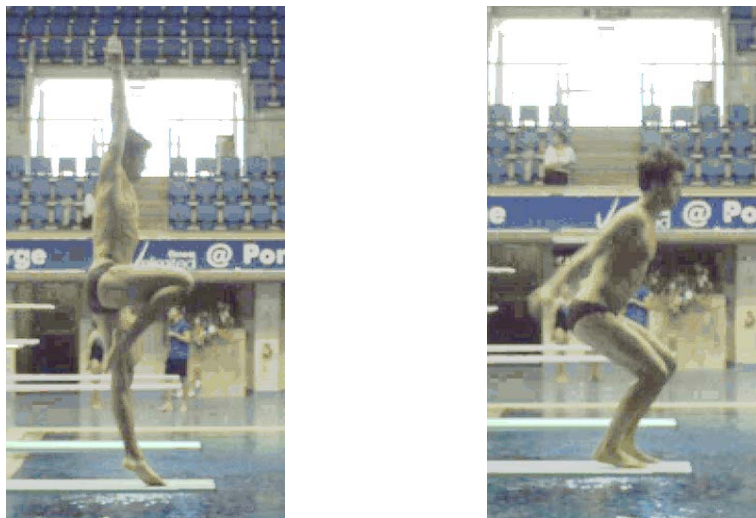


Figure 3.9. The diver position at hurdle takeoff (left) and hurdle landing (right) when the board is in neutral position.

As legs were moving differently in the hurdle flight phase, 20 body landmarks were used to digitise the whole-body movement plus the end of the board. The sequence of the body landmarks during the hurdle flight phase is shown in Table 3.8. Since the arms and legs were assumed to be symmetrical in Kong's (2005) simulation model, an average of both legs and arms were evaluated to minimise the asymmetrical movement.

Table 3.8. The sequence of the body landmarks

Point	Body landmark	Point	Body landmark
1	Right wrist	11	Left elbow
2	Right elbow	12	Left shoulder
3	Right shoulder	13	Left hip
4	Right hip	14	Left knee
5	Right knee	15	Left ankle
6	Right ankle	16	Left heel
7	Right heel	17	Left ball of foot
8	Right ball of foot	18	Left toes
9	Right toes	19	Middle of the head
10	Left wrist	20	Board tip at the right edge

The dive flight phase was digitised from takeoff position at board neutral and ended before entry into the water when the wrists were still visible (Figure 3.10). Since the arms were moved asymmetrically in dive flight phase, 14 body landmarks (points 1-12, 19 and 20 in Table 3.8) were chosen to digitise both arms and then an average of them was made.

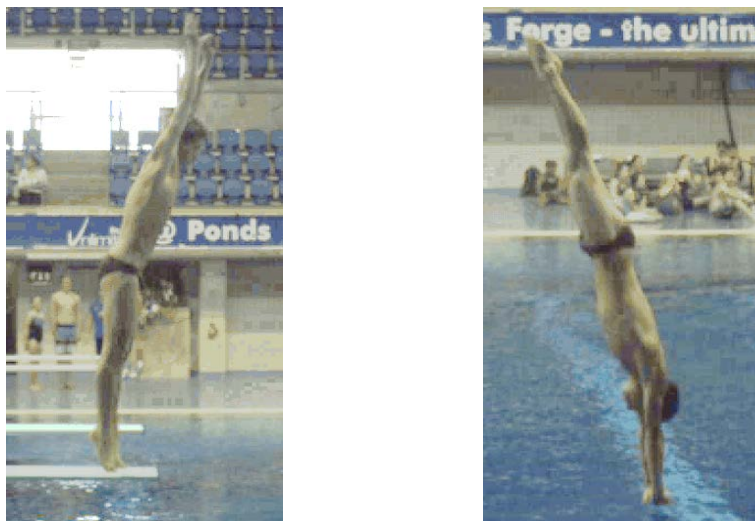


Figure 3.10. Dive takeoff position at board neutral (left) and entry position (right).

3.2.4. Takeoff phase

The takeoff phase started at touchdown at board neutral position where the diver was landing from the hurdle, and dive takeoff position as the last frame of takeoff at board neutral position (Figure 3.11). The diver's movement during contact phase was assumed to be symmetrical; so only 11 body landmarks (points 1-9, 19 and 20 in Table 3.8) were chosen to digitise the right-hand side (RHS) of the body. To avoid end-point errors in subsequent data processing, digitisation was started from 15 frames before the touchdown and ended 15 frames after the takeoff.

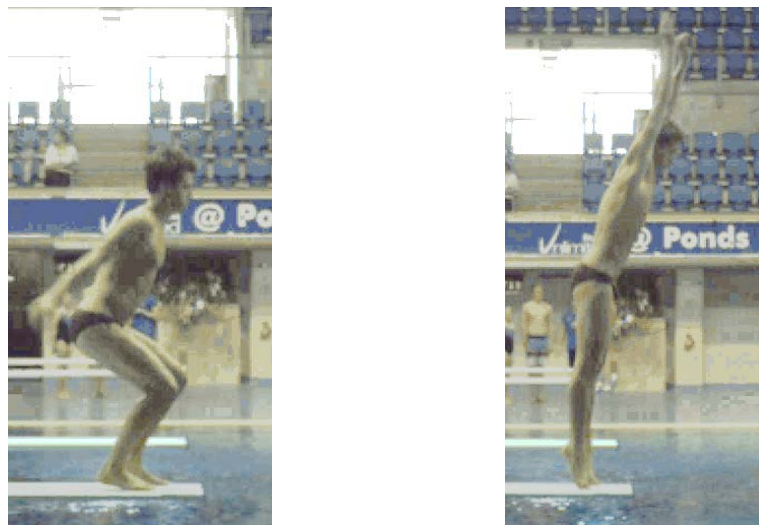


Figure 3.11. Diver position at board neutral position at landing from hurdle and dive takeoff

3.2.5. Order and frequency of digitising

The takeoff and flight phases were digitised after digitising and processing the foot placement of the hurdle. All phases were digitised separately with different sample rates. Since the interaction between the springboard and the diver produces unpredictable perturbation, the full sampling frequency of 250 Hz was applied to digitise the board contact phases. The sample rate of 50 Hz (every 5th frame) for digitising flight phases was found reasonable because it gives enough data points to fit a parabola to the centre of mass peak and calculate the takeoff velocity and angular momentum.

3.2.6. The accuracy of digitising

Measurement precision has been defined as the closeness of agreement between measured quantity values achieved by repeat measurements on the same or similar objects under specified conditions which are the repeatability conditions of measurement in the case of digitising in this study. It has been noted that the measurement precision is normally mentioned numerically by measures of imprecision such as standard deviation and variance (JCGM 200:2008).

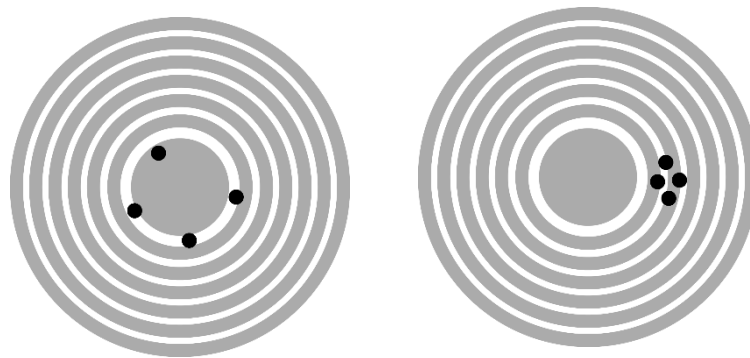


Figure 3.12. Low accuracy and poor precision (left), low accuracy and good precision (right), accuracy according to BIPM and ISO 5725.

Since “measurement accuracy” and “measurement precision” are not the same and according to the definition of accuracy (Figure 3.12), a human movement might be digitised precisely but not accurately. Thus, to assess closeness of agreement between the measured quantity values achieved by repeated measurements on digitisation of a series of similar movements, the term “measurement precision” is used in the present study.

The field of view was 5 metres wide and the maximum resolution of the TS3 camera was 1280 x 1024, thus 1 pixel represents 3.9 mm vertically and horizontally since the pixels are square. The mean and standard deviation of horizontal and vertical board neutral coordinates were 590.91 ± 0.32 and 266.58 ± 0.38 pixels respectively (Table 3.7), showing that the precision of digitisation of the board neutral coordinates is the standard deviation of 1.5 mm which is less than half a pixel.

After digitising all trials, the first and last trials of forward dives were re-digitised to determine the precision of digitising. The biggest difference in digitised coordinates appeared in the ball of the foot and the elbow because they had slightly blocky pixels and were not as identifiable as the other body landmarks in the digitising film (Table 3.9). The standard deviation of the difference between the re-digitised and the original of the first trial and the last trial were 1.598 pixels (7mm) and 0.436 pixels (2mm) respectively, showing improvement in the precision of digitising. The main factor responsible for this uncertainty could be the clearness of the images not only due to the lighting in the swimming pool but also a large field of view. The consistency of digitising each trial each time might also be a factor responsible for the difference. It was concluded that the precision of the digitisation, which is the standard deviation of the difference of 7 mm (Table 3.9), was reasonable as it was expected that the difference over the length of the body segments in the simulation model is unlikely to produce a large difference.

Table 3.9. RMS differences across all time points between trial 1 and re-digitised trial 1, trial 15 and re-digitised trial 15.

11 body joints*	pd01b (first trial)		pd15b (last trial)	
	average (pixel)	average (m)	Average (pixel)	Average (m)
Right Wrist-X	1.665	0.008	1.849	0.009
Right Wrist-Y	1.879	0.009	1.780	0.008
Right Elbow-X	3.382	0.016	2.339	0.011
Right Elbow-Y	3.556	0.016	2.145	0.010
Right Shoulder-X	2.316	0.011	1.427	0.007
Right Shoulder-Y	3.248	0.015	1.861	0.009
Right Hip-X	1.684	0.008	1.551	0.007
Right Hip-Y	1.942	0.009	0.946	0.004
Right Knee-X	3.702	0.017	1.325	0.006
Right Knee-Y	2.607	0.012	1.719	0.008
Rankle-X	2.508	0.012	1.096	0.005
Rankle-Y	3.539	0.016	1.262	0.006
Right heel-X	1.654	0.008	0.817	0.004
Right heel-Y	2.969	0.014	0.922	0.004
Right ball-X	2.203	0.010	1.195	0.005
Right ball-Y	7.864	0.036	0.976	0.004
Right toe-X	1.508	0.007	0.990	0.005
Right toe-Y	1.297	0.006	1.061	0.005
Mid-head-X	2.380	0.011	1.536	0.007
Mid-head-Y	6.505	0.030	1.450	0.007
Board tip-X	0.919	0.004	0.900	0.004
Board tip-Y	3.588	0.016	0.732	0.003
Max	7.864	0.036	2.339	0.011
Average	2.860	0.013	1.358	0.006
SD	1.598	0.007	0.436	0.002

*X: horizontal displacement Y: vertical displacement

3.3. KINEMATIC VARIABLES

3.3.1. Joint angles and mass centre locations

The digitised coordinates of body joint centres were used to calculate the diver's joint angles during each dive using Yeadon's (1990a) technique. The arm angle between each arm and the centre line of the trunk was calculated from the projected inclination and length of the arm on the vertical calibration plane. The body segmental inertia parameters (Chapter 4) were used to calculate the mass centre location for each digitised image using a segmental inertia method (Yeadon, 1990c). Since the arms were abducted to the side during the flight phase in forward pike dives, the arm

abduction angle was calculated by comparing the projected length of the arm on the sagittal plane with the measured length of the arm, allowing to use both elevation and abduction angles in data analysis.

3.3.2. Body orientation

The interaction between a surface of support and an organism is controlled by the act of orientation. Thus, the action of orienting the body is to maintain dynamic equilibrium with respect to the forces acting on the body. The orientation as an achievement of a particular kinematic state (Stoffregen, 1988) has been traditionally defined as a comparison between the body axes to some external reference frame (Howard & Templeton, 1966; Schone, 1984).

The trunk angle relative to horizontal was used as the body orientation in Kong (2005). However, Yeadon's (1984) definition which is the angle of a line between the knees and trunk relative to vertical was used as the orientation angle in this study. The whole body orientation was calculated as the angle between the vertical and a line from the midpoint of the knee centres to a point on the spine 60% of the distance from mid-hip centre to mid-shoulder centre (Figure 3.13). The use of such a body reference frame is less sensitive to changes in hip angle than a body frame embedded in a single segment (Yeadon, 1990a).

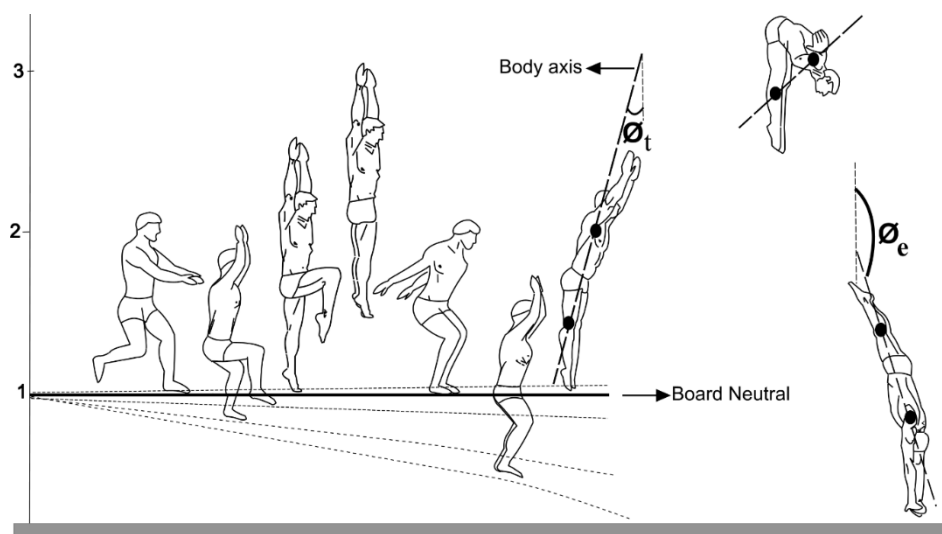


Figure 3.13. Body orientation angle as the angle between the vertical and a line from the midpoint of the knee centres to a fixed point on trunk and vertical.

3.3.3. Velocity, momentum and rotation potential

The centre of mass horizontal and vertical velocities were calculated using quintic spline fits to the displacement data. The touchdown velocity was calculated by obtaining mass centre position and fitting a parabola to hurdle flight phase using Yeadon (1990b) 3D reconstruction method. The same procedure was used to calculate the takeoff velocity by fitting a parabola to the dive flight phase. The last frame in the hurdle flight phase was used to determine the touchdown velocity and the first frame in flight phase was used to determine the velocity at takeoff. The angular momentum of each dive was calculated using the segmental inertia values and the time histories of the configuration and orientation angles in the aerial phase (Yeadon, 1990c). The rotation potential of each dive during the flight phase was calculated as the product of angular momentum ($\text{kg.m}^2.\text{s}^{-1}$) and flight time (s) divided by 2π times the moment of inertia of the body in the anatomical position, giving the equivalent number of straight somersaults when the arms are adducted in the flight phase (Hiley and Yeadon, 2008). Yeadon (1990c) used rotation potential to compare the rotational takeoff requirements for movements performed by subjects with different inertias and different flight times.

3.4. Summary

In this chapter, the procedure to collect data directly from experiment was explained. The protocol of digitising the recorded videos was described. Kinematic data were determined from video recording of 15 forward and 15 forward $2\frac{1}{2}$ somersaults pike dives.

CHAPTER 4

PARAMETER DETERMINATION

4.1. INTRODUCTION

Two simulation models that have already been developed were applied to investigate movement variability and control in springboard diving. In order to answer the research questions addressed in Chapter 1, an angle-driven simulation model of a diver and springboard (Kong, 2005) was used to investigate the diver's movement during the takeoff phase and an angle-driven simulation model of aerial movement (Yeadon et al., 1990b) was used to investigate the diver's movement during the flight phase.

4.1.1. Angle-driven model of a diver and springboard

4.1.1.1. The springboard model

During a springboard diving takeoff, there is a curvilinear path during board depression and recoil phases, propelling the diver into the air. The springboard was modelled as a rod with three degrees of freedom which represent the vertical, horizontal and rotational behaviour of the diving board (Figure 4.1).

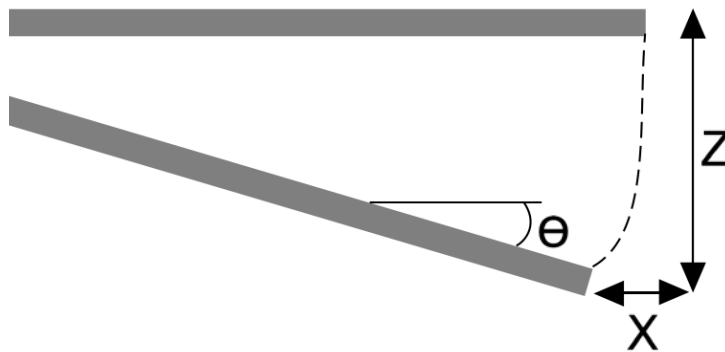


Figure 4.1. The springboard modelled as a rod with three degrees of freedom.

4.1.1.2. Foot-springboard interface

The elastic properties of a diver foot and the springboard are considered in the simulation model of diving takeoffs since the springboard is in contact with the diver at the foot during contact phase. Three pairs of perpendicular and parallel massless damped springs acting at the heel, ball and toes were used in the model of the foot-springboard interface (Figure 4.2).

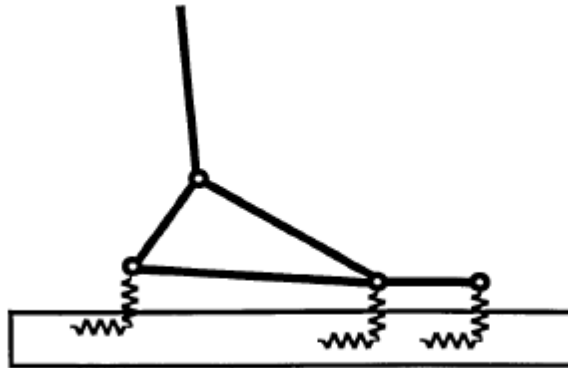


Figure 4.2. Model of the foot-springboard.

4.1.1.3. The diver model

The diver was represented as an eight-segment planar model comprising head including neck, upper arm, lower arm including hand, trunk, thigh, shank and a two-segment foot (Figure 4.3). A triangle and a rod connected at the metatarsal-phalangeal joint represented as the model of foot. The diver's orientation angle was defined by θ_t and the seven internal joint angles were θ_b (ball), θ_a (ankle), θ_k (knee), θ_h (hip), θ_s (shoulder), θ_e (elbow) and θ_d (head).

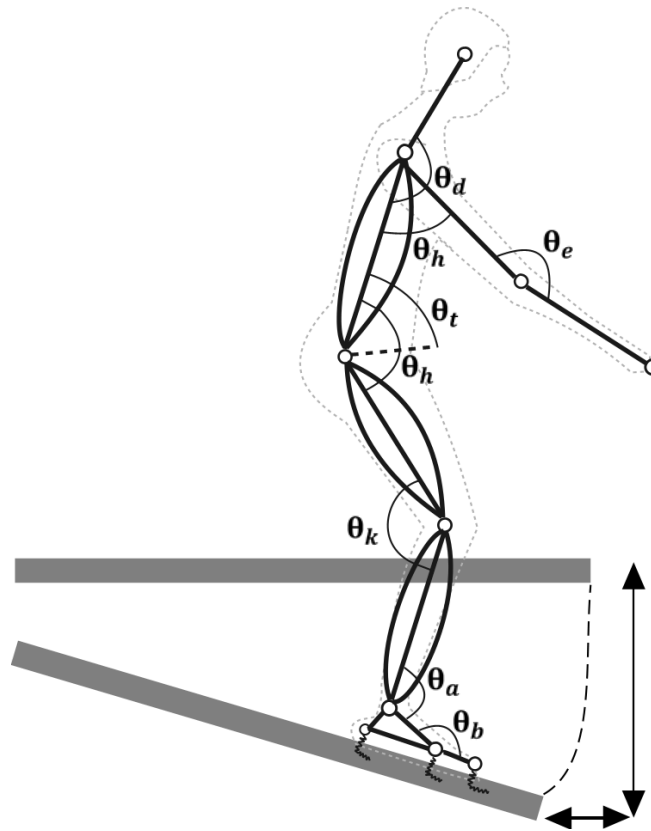


Figure 4.3. Model of a diver and springboard.

4.1.2. Angle-driven simulation model of aerial movement

The compute simulation model of human airborne movement comprised 11 rigid linked segments (Figure 4.4). The whole body orientation in space is specified by three external orientation angles corresponding to somersault, tilt and twist. The body configuration is defined by 14 internal orientation angles comprised three degrees of freedom for each shoulder joints, one degree of freedom for elbow flexion, two angles defined the movement of the chest-head segment C relative to the thorax T, one angle specified the movement of the shoulders in the frontal plane together with any lateral spinal flexion, two angles described the movement of thorax T relative to the pelvis P, flexion and abduction of the hip, somersault angle, abduction of the thighs and knee flexion. The angular momentum values for each movement as described in Yeadon (1990c) were calculated using the segmental inertia values and the time histories of the 3 orientations angles and 14 joint angles describing the relative movements of the

11 segments in the aerial phase. The angular momentum, initial vertical velocity and joint angle time histories in the flight phase were used as input into the model. The model output comprised the orientation angle time histories during the simulation of 15 forward pike (101B) and 15 forward 2½ somersault pike (105B) dives.

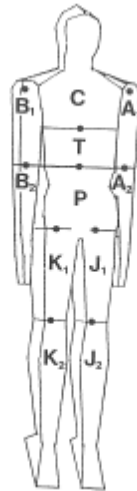


Figure 4.4. Model of human airborne.

In this chapter, the procedure to determine the subject-specific parameters, as input into the model, to customise the angle-driven simulation model of aerial movement (Yeadon et al.,1990b) and the angle-driven simulation model of a diver and springboard (Kong, 2005) is described. The process to calculate the mass, centre of mass location and the inertial values of the rigid and wobbling masses is explained. A description of the method applied to determine the strength parameters in the simulation of the board contact phase is given. The procedure to calculate the springboard parameters is provided and the determination of the associated parameters is described.

4.2. SEGMENTAL INERTIA PARAMETERS

The body segmental inertias (Appendix 3) of the diver were calculated from 45 anthropometric measures (Appendix 2) taken from the diver using Yeadon's (1990b) mathematical inertia model. Table 4.1 shows the value of mass, length, distance of CM

from the proximal joint and the moment of inertia about the transvers axis of each segment. For the limbs, the combined mass and moment of inertia and the average length and CM distance of the left (L) and right (R) side are displayed.

Table 4.1. Body segmental inertias calculated from anthropometric measurement

segment		Mass (kg)	Length (m)	CM* (m)	Inertia (kg.m ²)
Foot (L+R)	Whole foot	1.700	0.200	0.074	0.006
	Ankle/ball	1.459	0.138	0.059	0.002
	Ball/toes	0.242	0.062	0.026	0.000099
Shank (L+R)		8.038	0.424	0.174	0.110
Thigh (L+R)		18.525	0.411	0.178	0.281
Trunk		28.756	0.576	0.305	0.871
Upper arm (L+R)		3.926	0.291	0.130	0.030
Forearm (L+R)		2.344	0.259	0.110	0.013
Hand (L+R)		0.732	0.1945	0.076	0.002
Head		5.241	0.266	0.134	0.031

*from proximal joint

The foot was divided into a triangle and a rod segment connected at the ball (Kong, 2005), the dimensions (L_4 , L_7 , L_8 , L_9) of the diver's right foot from anthropometric measures were used to determine the segmental inertial parameters of the two-segment foot (Figure 4.5).

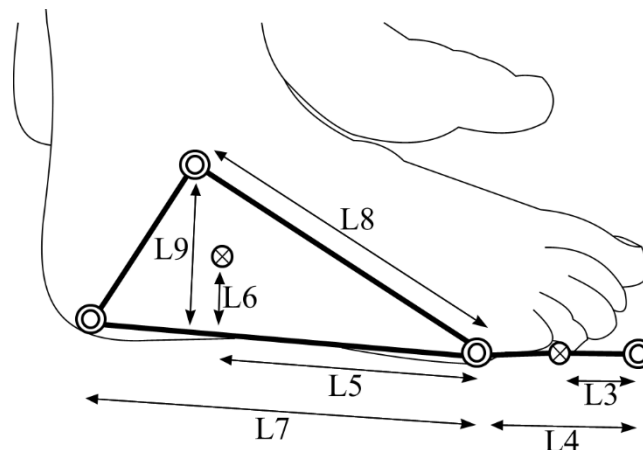


Figure 4.5. The two segments model of foot: a triangle and a rod segment.

The mass (m) and moment of inertia (I) of the whole foot, mass of rod (m_1) and triangle (m_2), moment of inertia of rod (I_1) and triangle (I_2) taken from the calculated values using Yeadon's (1990b) mathematical inertia model (Table 4.2).

Table 4.2. The mass and inertia values of the diver's right foot

Parameter	Value
m (foot) (kg)	0.8621
m ₁ (mass of rod)(right foot)(kg)	0.123
m ₂ (mass of triangle) (kg)	1.577
I (kg.m ²)	0.006
I ₁ (ball/toe) (kg.m ²)	0.000050
I ₂ (ankle/ball) (kg.m ²)	0.001

The remaining parameters, which were L₅ and L₆, estimated using Kong's (2005) method as shown below:

Taking horizontal moments about the tip of the toes, Equation (4.1) could be used to calculate L₅:

$$mgL_x = m_1gL_3 + m_2g(L_4 + L_5) \quad (4.1)$$

where g and L_x are gravitational acceleration and horizontal distance of whole foot mass centre to tip of the toes respectively, Equation (4.2) is used to calculate L_x:

$$L_x = (L_4 + L_7) \times 56\% \text{ (Chandler, 1975)} \quad (4.2)$$

Similarly, height of the triangle mass centre to the ground L₆ was calculated as:

$$mgL_z = m_1g(0) + m_2g(L_6) \quad (4.3)$$

where L_z is the vertical distance of whole foot mass centre to ground and L₆ is calculated as 32.9% of L₉ (Chandler, 1975). Table 4.3 summarises the measured and calculated dimensions for the two-segment foot. Tables 4.4, 4.5 and 4.6 show a comparison of the mass, moment of inertia and lengths of segments obtained in Kong (2005) and the present study.

Table 4.3. Measured and calculated dimensions for two-segment foot

Parameter	Description	Values (mm)
L3	$L_4/2$	31
L4	ball to toes	62
L5	equation 4.1	93.4
L6	$L_6/L_9 = 32.9\%$	21
L7	foot - L_4	197
L8	ankle to ball	138
L9	ankle height from floor	64

Table 4.4. A comparison of mass segments obtained in Kong (2005) and this study

Mass of segment	Description	Kong (2005)	Present study
MS	board	8.87	5.9
MA	toes	0.71	0.242
MB	foot	1.30	1.459
MC	knee (shank) (L+R)	7.73	8.038
MD	thigh (L+R)	17.96	18.525
ME	trunk	25.81	28.756
MF	upper arm (L+R)	3.66	3.926
MG	lower arm (L+R)	2.79	2.344
MH	head + neck	4.78	5.241

Table 4.5. A comparison of segments moment of inertias obtained in Kong (2005)

Moment of inertia	segment	Kong (2005)	Present study
Is	springboard*	0.066	0.044
IA	toes(L+R) ball to toe	0.000013	0.000099
IB	foot (L+R) ankle to ball	0.002	0.002
IC	knee (L+R)	0.101	0.110
ID	thigh (L+R)	0.261	0.281
IE	trunk	1.418	0.871
IF	upper arm (L+R)	0.023	0.03
IG	lower arm (L+R)	0.036	0.013
IH	head + neck	0.028	0.031

* $1/12 (8.87) (0.3)^2 = 0.0665$, $1/12 (5.9) (0.3)^2 = 0.04425$, where 8.87 and 5.9 are the effective board mass in Kong (2005) and the present study respectively

Table 4.6. Length of segments in Kong (2005) and this study

Segment	Description	Kong (2005)	Present study
L1	$L_1=L_2/2$ (middle board)	0.150	0.150
L2	springboard	0.300	0.300
L3	Two segment foot	0.039	0.031
L4		0.068	0.062
L5		0.102	0.093
L6		0.037	0.021
L7		0.152	0.197
L8		0.134	0.138
L9		0.073	0.064
L10	Shank (L+R)	0.403	0.424
L11	CM from proximal Thigh	0.169	0.178
L12	Thigh	0.411	0.411
L13	CM from Proximal Trunk	0.274	0.305
L14	Trunk	0.541	0.576
L15	CM from Proximal upper arm	0.115	0.130
L16	Upper arm	0.263	0.291
L17	CM from Proximal lower arm	0.164	0.110
L18	Lower arm	0.425	0.259
L19	CM from Proximal head + neck	0.136	0.134
L20	Head + neck	0.265	0.266

4.2.1. Wobbling mass parameters

Yeadon, Kong and King (2006) included visco-elastic elements in the model of the diver to represent soft tissue movement within body segments. However, they found that the torque-driven model was not sensitive to the wobbling mass stiffness parameter values. When the activation time histories were re-optimised for increased stiffness, a similar matching simulation was found with similar activation time history. It was concluded that there was no need to include wobbling masses in modelling takeoff phases from a compliant surface such as a springboard. Since the wobbling mass parameters were already included in Kong's (2005) model, they were excluded in this study by making the mass of wobbling component (m_w) very small and adding all of the masses into the mass of the fixed part of the segment (m_f).

Kong (2005) calculated the mass of the fixed (m_f) and wobbling (m_w) components using the fixed to wobbling ratio and the segmental mass obtained from Yeadon's (1990b) mathematical inertia model. To exclude the wobbling masses, the fixed mass ratio was

maximized and the wobbling mass ratio was minimised to determine the segmental inertia parameters (Table 4.7).

Table 4.7. Segmental inertia parameters for the fixed and wobbling component

Segment	Parameter	Definition	Kong (2005)	Present study
Shank*	mass	$m_f + m_w$ (kg)	4.063	4.019
	inertia	$I_f + I_w$ (kg.m ²)	0.0519	0.55
Thigh*	mass	$m_f + m_w$ (kg)	9.439	9.262
	inertia	$I_f + I_w$ (kg.m ²)	0.1197	0.1405
Trunk	mass	$m_f + m_w$ (kg)	25.819	28.756
	inertia	$I_f + I_w$ (kg.m ²)	0.715	0.871
*Values for shank and thigh are for one side of body				

4.3. SPRINGBOARD PARAMETERS

Since it has been shown that the effects of damping are negligible (Boda, 1992; Sprigings et al., 1989) in the model of a springboard and diver (Yeadon, Kong and King, 2006), a linear spring mass-system with no damping and the stiffness, defined by two regression equations obtained from experimental data, was used to model the vertical movement of the springboard. The stiffness was allowed to increase with the foot distance from the board tip. The two regression equations defined the vertical and horizontal deflection relationship and the vertical deflection and board rotation relationship. In order to determine the extent to which the board stiffness changes by landing at a certain distance from the end of the board, the equation of the board stiffness was calculated. The vertical stiffness, effective board mass, the relationship between the vertical board deflection and the horizontal deflection and board rotation angle alongside with the inertia of the springboard segment were calculated.

4.3.1. Vertical stiffness and effective board mass

Although mostly static loading has been used to determine the board stiffness for a given fulcrum setting and load location (Lanoue, 1936; Darda, 1972; Boda, 1992; Sprigings et al., 1989, 1990), a dynamic loading procedure was described by Stone (1987) to measure both the board vertical stiffness and effective board mass at the same time (Miller & Jones, 1999). According to the equations of harmonic motion:

$$\omega = \sqrt{\left[K_b / (m_e + m_l) \right]} \quad (4.4)$$

$$f = \frac{\sqrt{\left[K_b / (m_e + m_l) \right]}}{2\pi} \quad (4.5)$$

$$T_b = 2\pi \sqrt{\frac{(m_e + m_l)}{K_b}} \quad (4.6)$$

The oscillation period of the board (T_b) is equivalent to oscillation rate (ω in rad/s) and is determined by the effective board mass (m_e in kg), board stiffness (K_b in N/m), and load mass (m_l in kg). Thus, the board stiffness can be calculated if the oscillation period, effective board mass and added mass are known. The oscillations recorded on the data collection day was used to determine the oscillation period. Since progressive loading of the board, setting the loaded board in motion as well as measuring the oscillation period for each additive load are involved in this procedure, the equation for T_b therefore can be used to evaluate numerical values for both board stiffness (K_b in N/m) and effective board mass (m_e in kg). A linear relationship between the square of the oscillation period and load is gained by squaring and rearranging terms in equation of T_b as follows:

$$m_e + m_l = (T_b^2 \times K_b) / 4 \pi^2 \quad (4.7)$$

$$m_l = (K_b / 4 \times \pi^2) T_b^2 - m_e \quad (4.8)$$

where, m_l , which is the added load, can be plotted as a function of T_b^2 which is the square of the measured oscillation period, the constant ($K_b / 4 \times \pi^2$) therefore would be the slope of the line.

4.3.1.1. Dynamic loading of the springboard

By performing the dynamic loading of the springboard, the period of the oscillations of different loading masses were measured at two different positions of applied load (PAL). This was performed by having the diver ($m_{l2} = 69.7$ kg) stand at the end of the board and one step back from the board tip, setting the board into oscillation by raising

his arms once, and keeping his body as rigid as possible with the arms adducted for 10 oscillations (Figure 4.6).



Figure 4.6. Dynamic loading of the springboard on the board tip (left) and one step back from the board tip (right).

The same procedure was done to load the board with two different body weights ($m_{12} = 89.4 \text{ kg}$, $m_{13} = 93.8 \text{ kg}$). Five seconds of the oscillations were recorded to calculate the period of the oscillations.

The average value of T from all cycles was used to determine T^2 for each loaded mass. The period of oscillations during free load vibration was determined from the video data after the diver had taken off from the springboard. The oscillation period T was estimated by assuming the time during which the board was in contact with the fulcrum equalled $T/2$ because the board bounced off the fulcrum after takeoff (Tables 4.8 & 4.9).

Table 4.8. The oscillation period of different body weights and free load at board tip

	Feet at the board tip			Free load
	m_{l1}	m_{l2}	m_{l3}	
Number of frames*	2442	2846	2530	979
Number of period (T)**	12.5	13	11.5	9
Time of the whole oscillation	9.768	11.384	10.120	3.916
The average time of each oscillation (T)	0.781	0.876	0.880	0.435
$T/2$				0.218
T^2	0.611	0.767	0.774	0.047
* number of frames from the beginning until the end of oscillation				
** number of period during oscillation				

Table 4.9. The oscillation period of different body weights and free load one step back

	Feet one step back*			Free load
	m_{l1}	m_{l2}	m_{l3}	
Number of frames**	2894	2563	2357	979
Number of period (T)***	16	13	12	9
Time of the whole oscillation	11.576	10.252	9.428	3.916
The average time of each oscillation (T)	0.724	0.789	0.786	0.435
$T/2$				0.218
T^2	0.611	0.767	0.774	0.047
*1 step back: the position of the toes are 0.258 m (diver's foot length) far back from the board tip				
** number of frames from the beginning until the end of oscillations				
*** number of period during oscillation from the board tip				

The board stiffness K_b and board effective mass m_e were determined from Equation (4.4) by plotting T^2 against m_L (Figures 4.7 and 4.8).

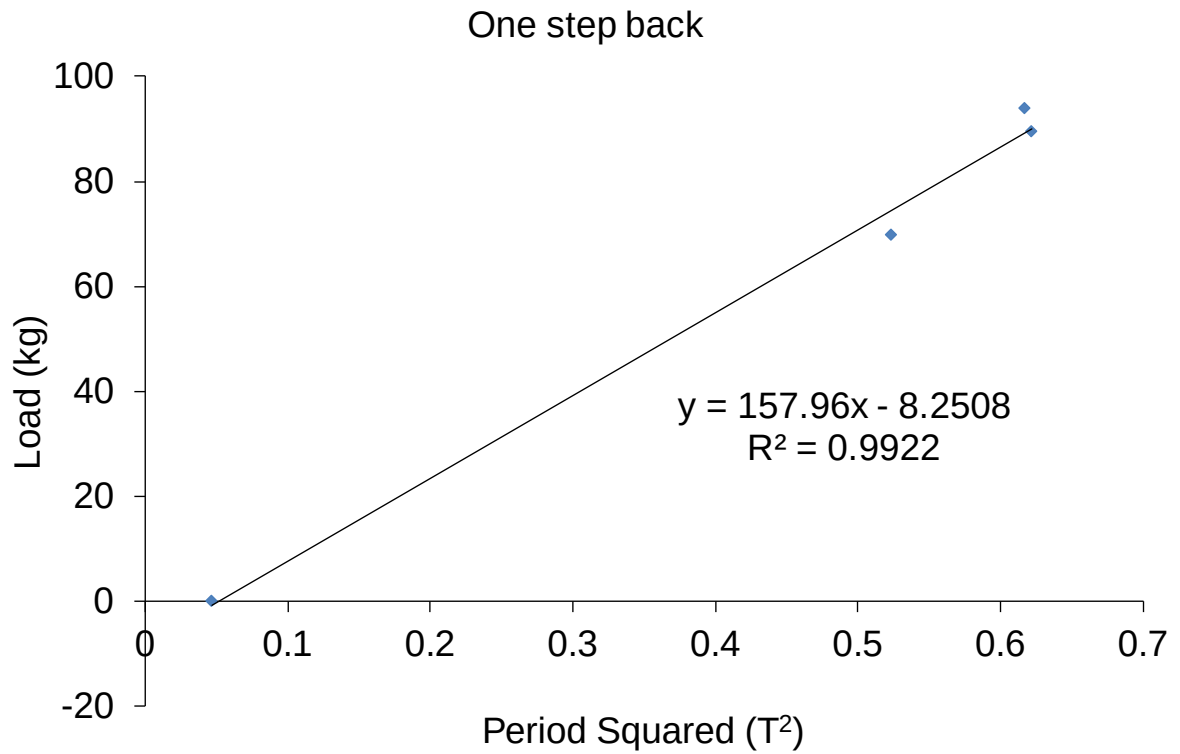


Figure 4.7. Determination of k and m_e at one step back from the board tip.

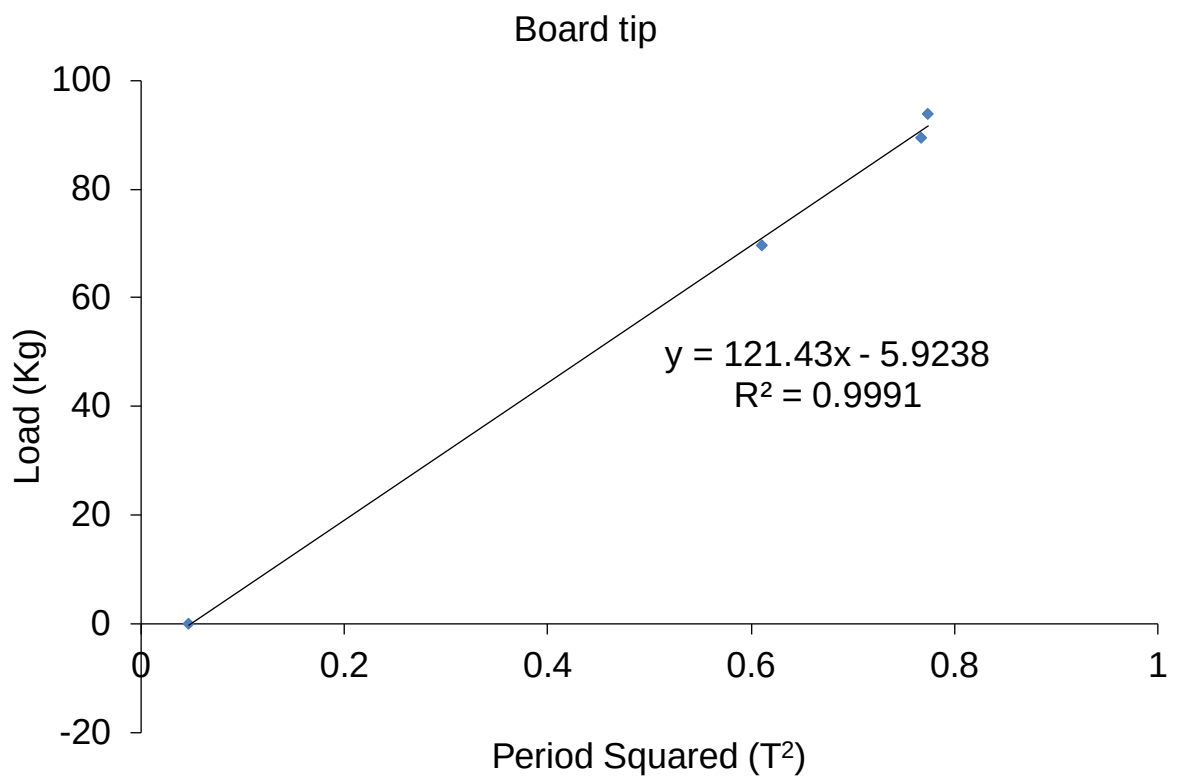


Figure 4.8. Determination of k and m_e at board tip.

4.3.2. Plotting T^2 against m_L

The regression equations obtained from plotting T^2 against m_L were used to estimate the board stiffness and effective board mass as follows:

$$Y_1 = 121.43 X - 5.92 \quad (4.7)$$

$$Y_2 = 157.96 X - 8.25 \quad (4.8)$$

Where Y_1 , Y_2 are the equations of PALs on the board tip and on one step back from the end of the board respectively. The effective board mass (m_c) is the absolute value of constant C , giving $m_{c1} = 5.92$ and $m_{c2} = 8.25$ as the effective board mass at one step back and board tip respectively.

Similarly, the slope of the lines $m = (K_b / 4 \times \pi^2)$ are used to calculate the board stiffness for the two PALs as follows:

$$m_1 = 121.43$$

$$m_2 = 157.96$$

$$K_1 = m_1 \times (4 \times \pi^2) \quad (4.9)$$

$$K_2 = m_2 \times (4 \times \pi^2) \quad (4.10)$$

Board stiffness at the board tip K_1 and at one step back from the end of the board K_2 are therefore 4794 N/m and 6236 N/m respectively. A comparison of the board stiffness and effective board mass obtained in this study and calculated in Kong (2005) is illustrated in Table 4.10.

Table 4.10. Comparison of the board stiffness and effective board mass

		Distance from the board tip (d)	Board stiffness (N/m)	Effective mass (kg)	Fulcrum number
Kong (2005)	At the tip	0	5446	8.87	7.5
Present study	At the tip	$d_1 = 0$	4794	5.9	8.5
	1 step back	$d_2 = 0.258$	6236	8.2	

4.3.3. Equation of the board stiffness

The two values of board stiffness K_1 and K_2 obtained from plotting T^2 against m_L (Equations 4.9 and 4.10) and their corresponding PALs at the board tip (d_1) and one step back (d_2) from the end of the board (Table 4.10), were used to determine the equation of a line created by these values. This was done by plotting board stiffness (K_1 and K_2) against the toes distance from the board tip (Figure 4.9).

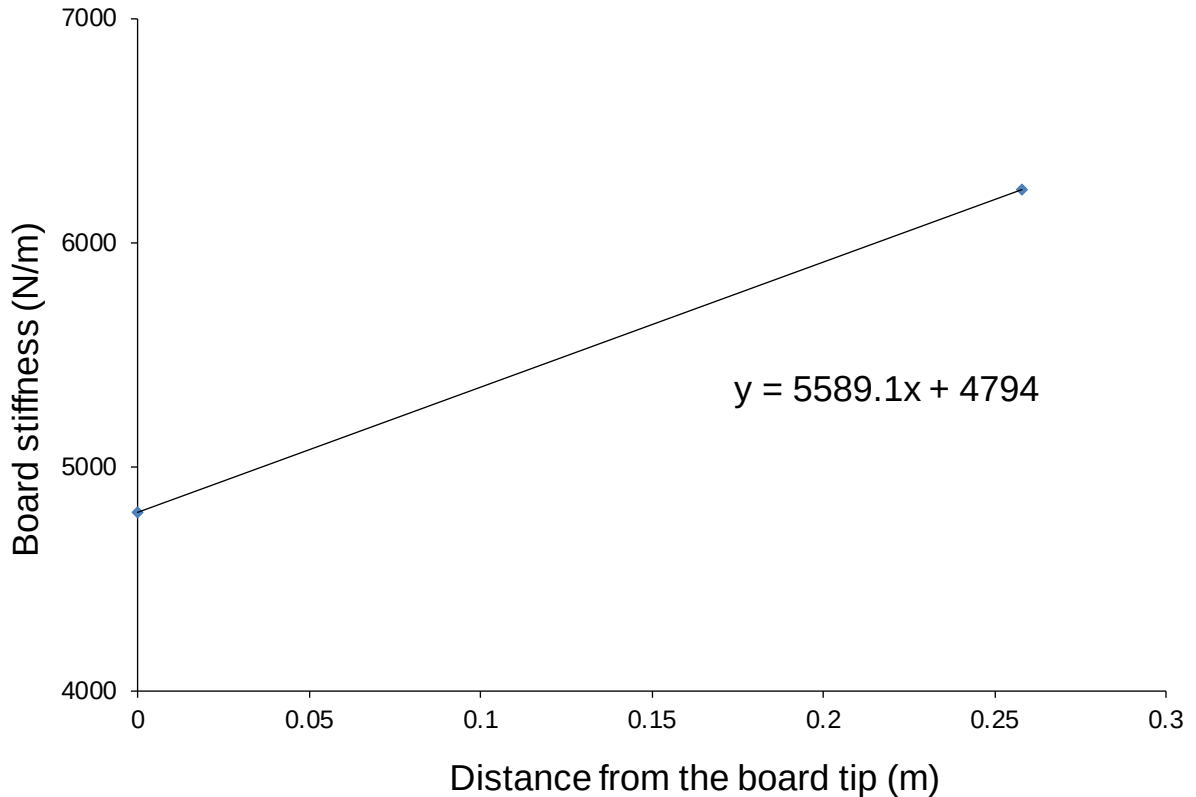


Figure 4.9. Plotting board stiffness against toe distance from the board tip.

Since the vertical stiffness varies depending on the foot position, the diver centre of mass when his toes are at the end of the board was assumed 0.15 m away from the board tip, the vertical stiffness is calculated as:

$$K = m (d + 0.15) + c \quad (4.11)$$

Where m = the slope

d = parallel distance between the toes and the board tip

$c = \text{constant}$

Taking the slope of the equation of line $m = 5589.1$ and substituting $K_1 = 4794$ and $d_1 = 0$ in equation (4.11), the constant c is calculated as 3955 N/m .

The equation of the vertical stiffness was therefore:

$$K = 5589.1 (d + 0.15) + 3955 \quad (4.12)$$

4.3.4. Board deflection

The average digitised coordinates of the board tip at its own neutral position at touchdown of all dives were used to calculate the board deflection. Table 4.11 shows the mean values of the horizontal and vertical digitised coordinates of 15 forward and 15 forward $2\frac{1}{2}$ somersaults pike dives. The mean digitised coordinates $(-2.52, 1.029)$ were used as the board starting point (time zero) at touchdown and takeoff for all dives.

Table 4.11. The board digitised coordinates at board neutral position at touchdown of 15 forward and 15 forward $2\frac{1}{2}$ somersaults pike dives

	Forward pike dives (m)		$2\frac{1}{2}$ somersaults pike dives (m)	
	Horizontal	Vertical	Horizontal	Vertical
1	-2.523	1.031	-2.524	1.033
2	-2.520	1.029	-2.528	1.026
3	-2.521	1.027	-2.521	1.031
4	-2.522	1.033	-2.521	1.024
5	-2.522	1.027	-2.521	1.023
6	-2.522	1.033	-2.523	1.033
7	-2.523	1.035	-2.524	1.026
8	-2.523	1.028	-2.525	1.036
9	-2.523	1.026	-2.527	1.026
10	-2.520	1.033	-2.520	1.030
11	-2.524	1.026	-2.518	1.028
12	-2.523	1.030	-2.521	1.025
13	-2.526	1.025	-2.523	1.033
14	-2.523	1.033	-2.520	1.032
15	-2.526	1.026	-2.522	1.035
Mean	-2.523	1.029	-2.522	1.029

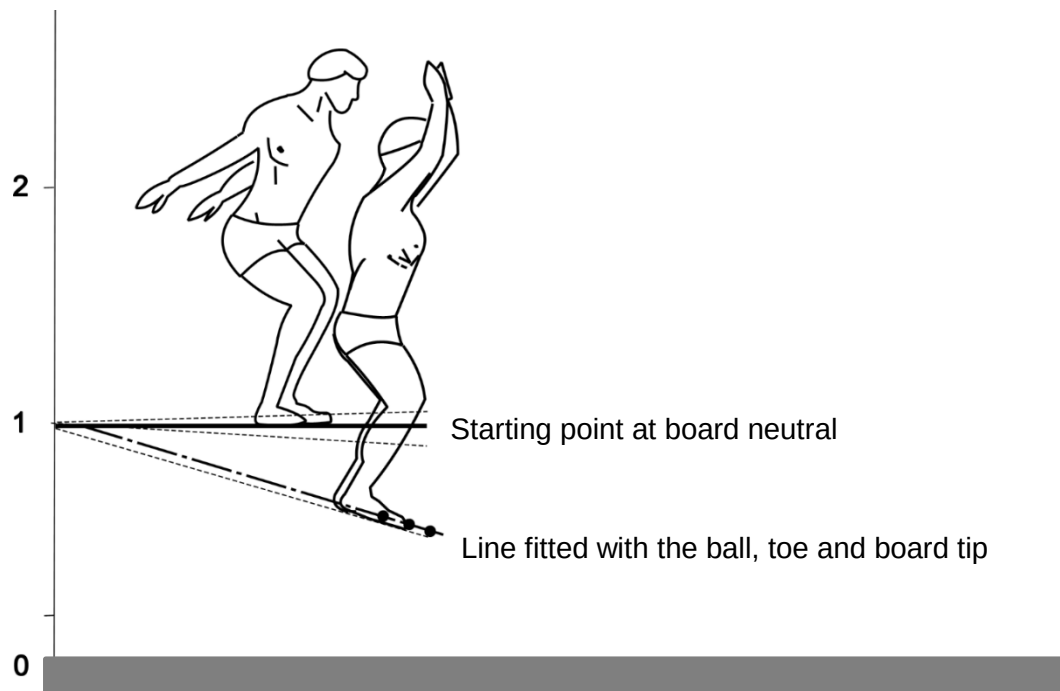


Figure 4.10. The board angle as the angle between a line fitted with the ball, toe and the board tip and the board at neutral position.

The board deflection was calculated as the board tip movement relative to the starting point. The board rotation angle was defined as a function of the vertical board tip deflection. From the digitised coordinates, the board angle (θ) was calculated as the angle between the average digitised coordinates of the board tip at neutral position at touchdown of 15 dives and a line fitted through the ball, toe and the board tip (Figure 4.10).

Since the heel was off the board at touchdown and takeoff and the ball was not clearly visible due to blocky pixels in some frames, the values of the board angle appeared noisy. This also might be because the diver is very close to the end of the board, increasing the error in the board angle. To minimise the error, the minimum values between the board-toe angle, the ball-board angle and the heel-board angle were used to regress the board angle against the vertical deflection.

4.3.5. Vertical deflection and board rotation relationship

The board angle was regressed against the vertical deflection using the film data of all dives. Figures 4.11 and 4.12 show the linear regression of the board angle and the vertical deflection of forward pike and forward 2½ somersaults pike dives respectively.

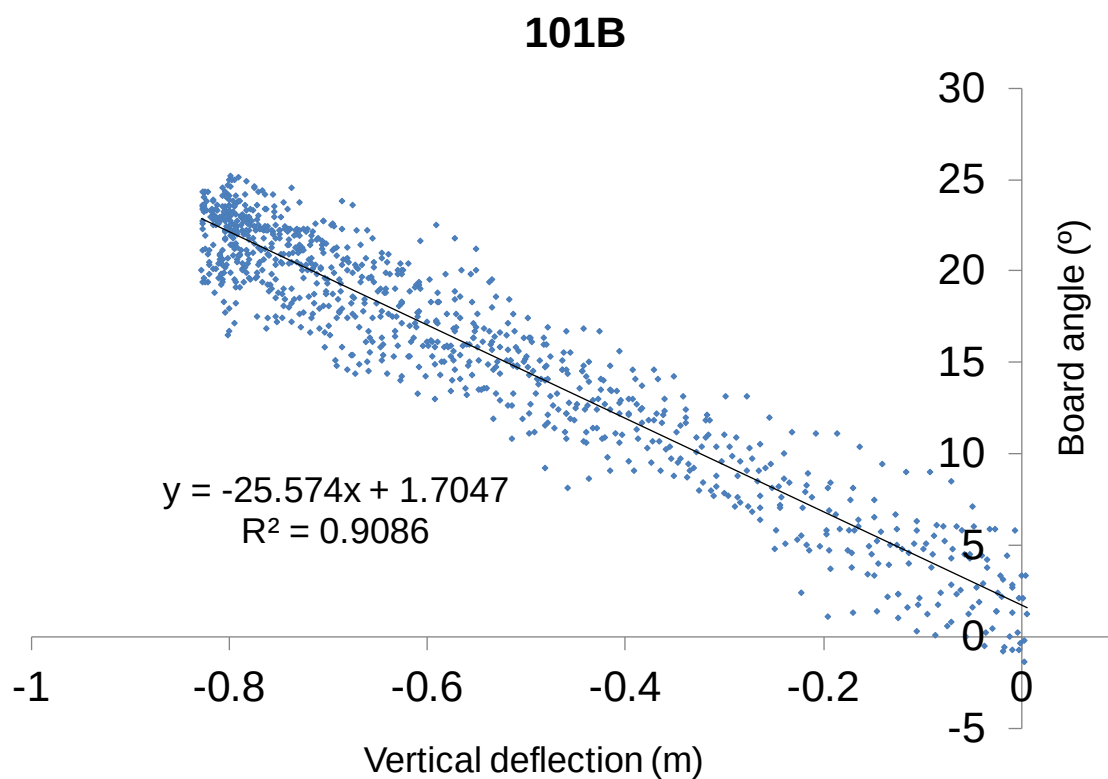


Figure 4.11. Linear regression of the board angle against the board vertical deflection using performances of 10 forward pike dives.

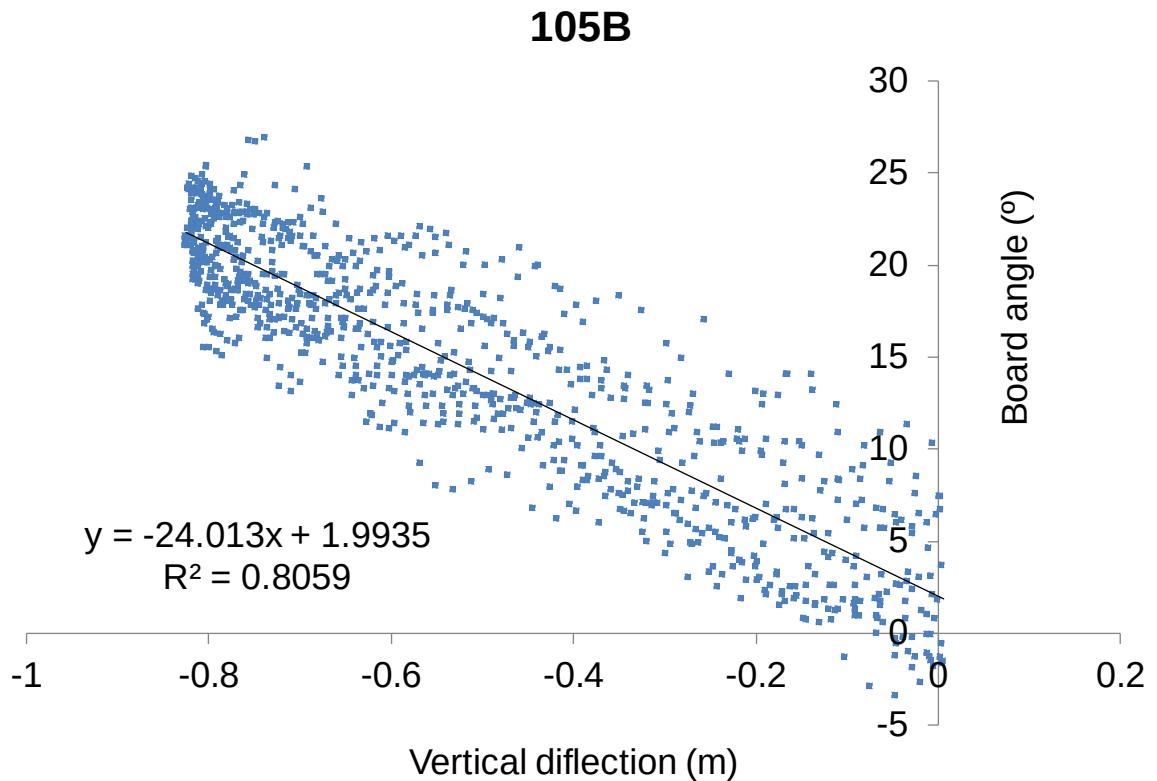


Figure 4.12. Linear regression of the board angle against the board vertical deflection using performances of 9 forward $2\frac{1}{2}$ somersaults pike dives.

It should be noted that the trials 11, 12, 13, 14 and 15 of forward pike dives and the trials 2, 6, 7, 8, 9 and 12 of forward $2\frac{1}{2}$ somersaults pike dives were excluded from the regressions in Figures 4.11 and 4.12 respectively due to unrealistic values. This might be partly because the toes were very close to the board tip and partly because of the side bending of the board. Table 4.12 shows the toe distance to the board tip during board contact phase for 101B and 105B.

Table 4.12. Mean toe distance to the board tip during contact phase of forward pike and forward 2½ somersaults pike dives

Toe distance from the board tip (m)		
	Forward pike	2 ½ forward pike
1	0.212	0.113
2	0.212	0.059
3	0.134	0.062
4	0.132	-0.042*
5	0.127	0.106
6	0.122	0.049
7	0.163	0.04
8	0.142	0.017
9	0.089	0.035
10	0.138	0.107
11	0.045	0.172
12	0.111	0.053
13	0.036	0.116
14	0.073	0.066
15	-0.005*	0.101
Mean	0.115	0.070
SD	0.060	0.051
*toes are over the board tip		

The average toe distance in forward pike 0.115 m, is greater than 0.070 m in 2½ somersaults pike dives. The smaller toe distance might be responsible for noisy board angle values in 2½ somersaults dives, making the linear regression of the board angle against the vertical deflection weaker in these dives, $R^2 = 0.80$, relative to the forward pike dives, $R^2 = 0.90$, (Figures 4.11 and 4.12). The average toe distance from the board tip in the excluded trials were less than 0.06 m, making the values of the board angle inaccurate. Although the toe distance in the trial 12 of the forward pike dives was sensible (0.111 m), it was excluded in the regression because the side bending of the board was observed clearly in the video data. Since the diver landed on the right side of the board, leaning the board to the side slightly, making the board angle very noisy in this trial. The results of the regressions demonstrated that a linear function was adequate to represent the board angle-vertical deflection relationship (Figures 4.11 and 4.12).

4.3.6. Vertical and horizontal deflection relationship

To calculate the regression equations of springboard, two quadratic functions were obtained using regression to relate the horizontal displacement to the vertical displacement of the springboard tip. Figures 4.13 and 4.14 show cubic fits to the springboard movement during the board contact phase of 15 forward and 15 forward 2½ somersaults pike dives.

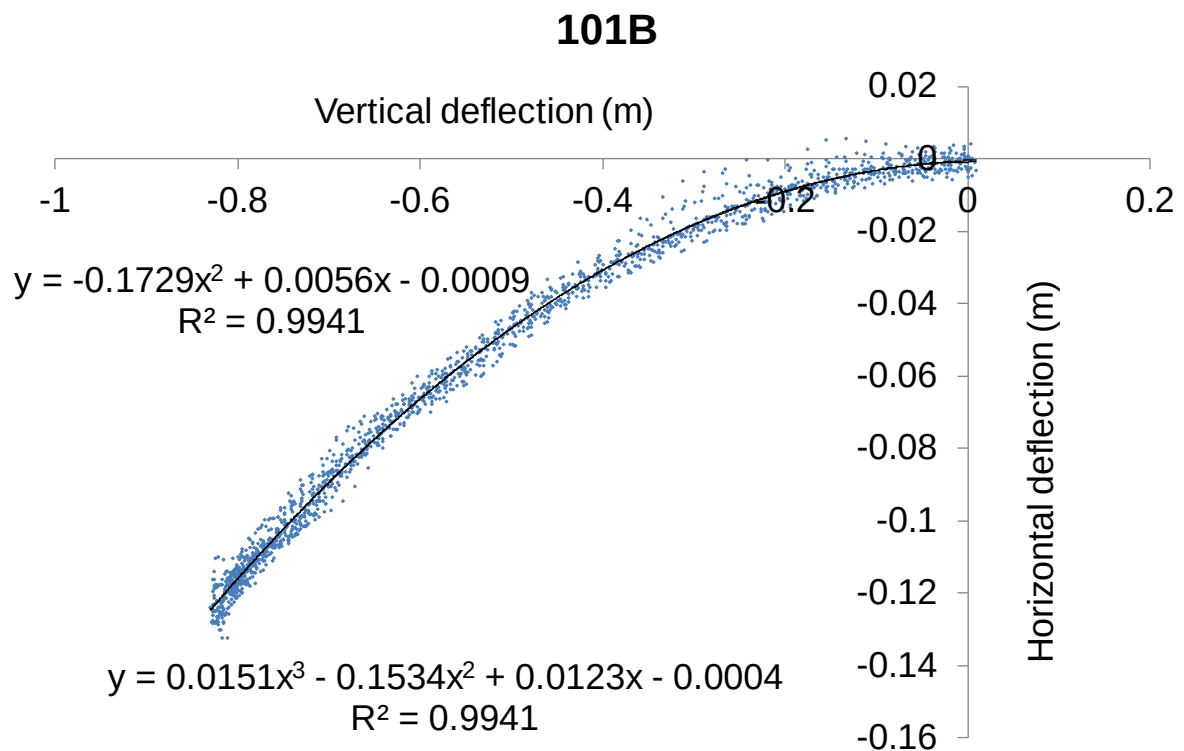


Figure 4.13. Regression of horizontal and vertical deflection of the board tip of 15 forward pike dives.

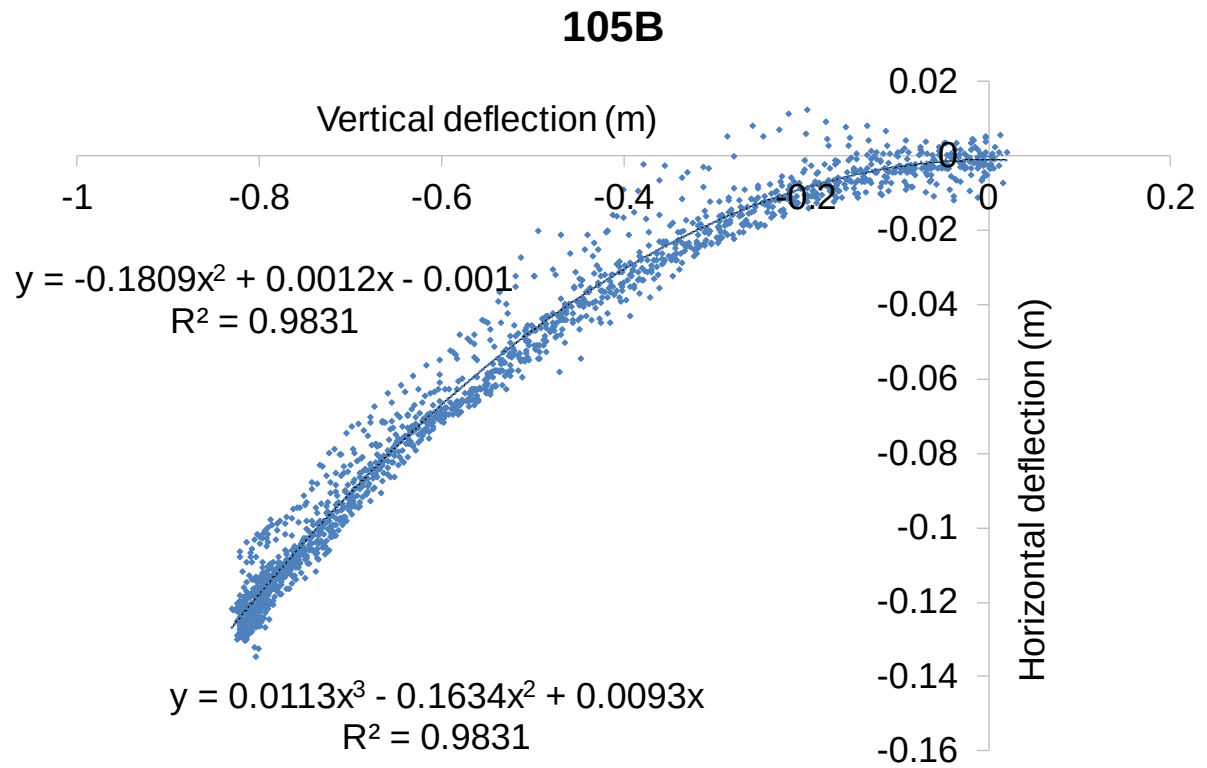


Figure 4.14. Regression of horizontal and vertical deflection of the board tip of 15 forward 2½ somersaults pike dives.

4.3.7. Moment of inertia

An inverse dynamics method described in Kong (2005) was used to calculate the moment of inertia of the springboard I_s which was required as an input to the model to calculate the torque provided by the springboard. Thus, the equation of a uniform rod was used to calculate I_s as follows:

$$\begin{aligned} I_s &= 1/12 m_e L^2 \\ &= 1/12 (5.9) (0.3)^2 \\ &= 0.04425 \text{ kg.m}^2 \end{aligned}$$

Where m_e and L are the effective board mass obtained in this study and the length of springboard modelled as a 0.3 m rod in Kong's (2005) simulation model of a diver and springboard. The moment of inertia of the diver I_{ss} in a standard straight position with adducted arms was calculated as 11.808 kg.m² using Yeadon's (1990b) mathematical inertia model.

4.4. Foot-springboard interface parameters

The angle-driven model was used to determine the stiffness and damping of the visco-elastic elements in the foot-springboard interface, along with the refined estimation of wobbling mass and springboard parameters. The input of the model was initial conditions at touchdown including the foot position, mass centre (CM) horizontal and vertical velocities, trunk angle and angular velocity. The model was derived with joint angle time histories obtained from video for optimising the values of the unknown parameters. The visco-elastic parameters were determined in an individual matching process. This was done by driving the model with known initial conditions and joint angle time histories, allowing to optimise the uncertain parameters to minimise the difference between simulation and performance.

4.5. Summary

In order to customise a simulation model to investigate the aims of the research, kinematic inputs and subject specific parameters are required. In this chapter, the segmental inertia parameters of the diver were determined. It was explained how the wobbling mass parameters were excluded. Springboard parameters were calculated from dynamic loading of the springboard experiment. The procedure how to determine the foot-springboard interface parameters was described and the springboard and the diver moment of inertias were determined. These parameters will then be used in subject-specific simulation models to answer the research questions.

CHAPTER 5 PERFORMANCE ANALYSIS

5.1. INTRODUCTION

In springboard diving, at the last step of the approach, the divers propel themselves up by jumping from the drive leg, moving the hurdle leg forward and upward simultaneously straightening the arms overhead to achieve hurdle height. They contact the board at the end of the hurdle flight using both feet to begin the contact phase and depress the board. The kinetic energy of the dive is stored during the board depression, and most of the stored energy is converted back into kinetic energy during the recoil phase to project the diver into the air to achieve the dive height, and then coming out of pike and finally enter into the water (Figure 5.1).

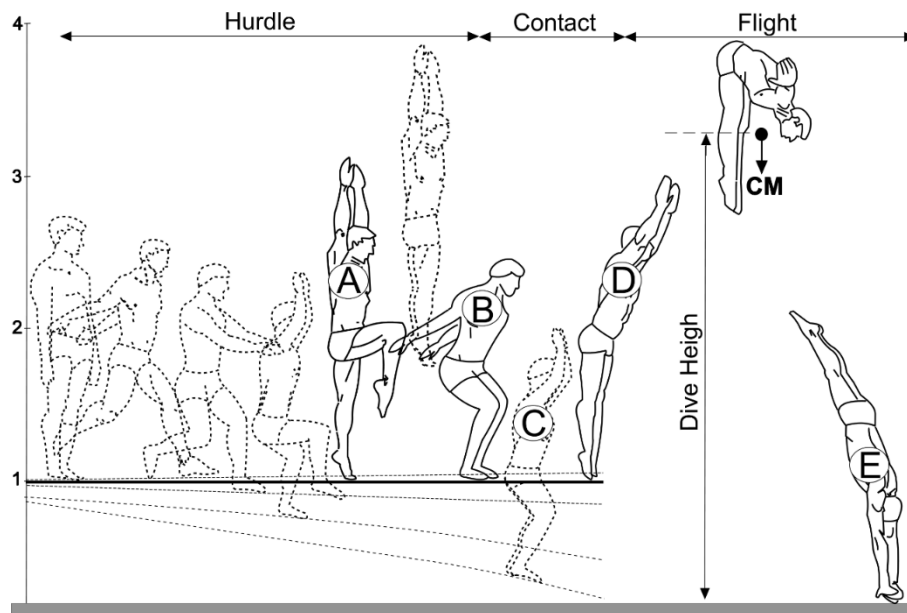


Figure 5.1. 1-m springboard forward pike dive. Positions: A= hurdle takeoff, B= touchdown, C= maximum board depression, D= takeoff, E= Entry, (B to C)= board depression, (C to D)= board recoil phase.

Generating sufficient angular momentum, obtaining maximum dive height and therefore time during flight phase, travelling safely with a reasonable board clearance distance and accuracy in entry orientation angle are the main objectives for competitive springboard diving (Miller and Munro 1985). The basic mechanics of factors involved

in the height of a dive have been identified as vertical velocity and height of centre of gravity at takeoff (Lanoue, 1940; Harper, 1966; Sanders & Wilson, 1988). Height is an important factor contributing directly to the dive score in springboard diving (McCormick, Subbaiah and Arnold, 1982). Superior divers have the ability of performing great height to provide more time to execute rotational movement in the air which is aesthetically pleasing part of the performance, leading to points awarded for a dive (Louganis, 1995; Miller and Munro, 1984, 1985).

Angular momentum is generated during the recoil phase using hip flexion in forward dives. Thus, the mass, shape and angular velocity of the diver play an important role for the initiation of rotation. However, losing energy by performing extra flexion may lead to a loss in dive height. The optimal flexion in hip and knees at touchdown greatly depends on the diver's skill and muscular strength in which extra flexion following the touchdown is minimised to avoid energy absorption (Sanders & Wilson, 1988; Sanders & Gibson, 2000). Although greater vertical landing velocity results in greater dive height, there is a little known to what extent the touchdown variables contribute to the prediction of dive height.

For competitive springboard diving, obtaining sufficient dive height and therefore time during flight phase is one of the main objectives (Miller and Munro 1985) which affects the orientation angle at entry into the water (performance outcome). In fact, the diver should aim for maximum dive height while generating sufficient angular momentum for rotation and keeping a safe distance away from the springboard. It is expected that vertical touchdown velocity and body configuration make a major contribution to the dive height. The relationships between dive height, foot distance from the end of the board, body configuration at touchdown and changes in configuration during the contact phase may determine the most significant factors responsible for variation in dive height.

The aim of this chapter is to investigate the extent to which the change in touchdown and takeoff variables can explain the variance in dive height and entry into the water. In order to understand the effect of a diver's kinematic variables at touchdown and during contact on dive height and entry orientation angle, a biomechanical analysis

was carried out to a) quantify the kinematic variables to investigate the movement variability b) determine the factors contributing to variation in dive height and entry orientation angle.

5.2. METHOD

The recorded performances of 15 forward pike and 15 forward 2½ somersault pike dives (Chapter 3) and inertia parameters (Chapter 4) were used to determine the movement pattern of the mass centre location and velocity, the time histories of the joint configuration and body orientation angles and the duration of each phase. Regression analyses using SPSS were performed to investigate the extent to which the change in touchdown and takeoff variables can explain the variance in the dive height and entry orientation angle. The maximum vertical displacement travelled by the centre of mass relative to the water was calculated as the peak dive height (Figure 5.1). The significant correlations between kinematic variables were determined. Two analyses were carried out to examine the contributions to dive height and entry orientation angle. In the first analysis, the dive height was chosen to be dependent variable. Two stepwise multiple regressions were performed to investigate the touchdown variables only against dive height and all variables against dive height. In the second analysis, the process of the first analysis repeated but the entry orientation angle was chosen as the dependent variable.

5.3. RESULTS

5.3.1. Peak dive height

The mean movement pattern depicted in Figure 5.2 represents the body centre of mass location during hurdle flight, dive takeoff and flight phases among 15 forward (101B) and 15 forward 2½ somersault pike dives (105B), demonstrating the ability of the diver to reproduce the same pattern of the movement repeatedly. Although the centre of mass location at hurdle takeoff and the peak hurdle height are comparable in both 101B and 105B, the diver became closer to the board tip during the board contact phase, had lower dive height and entered into the water further away from the end of the board in 105B.

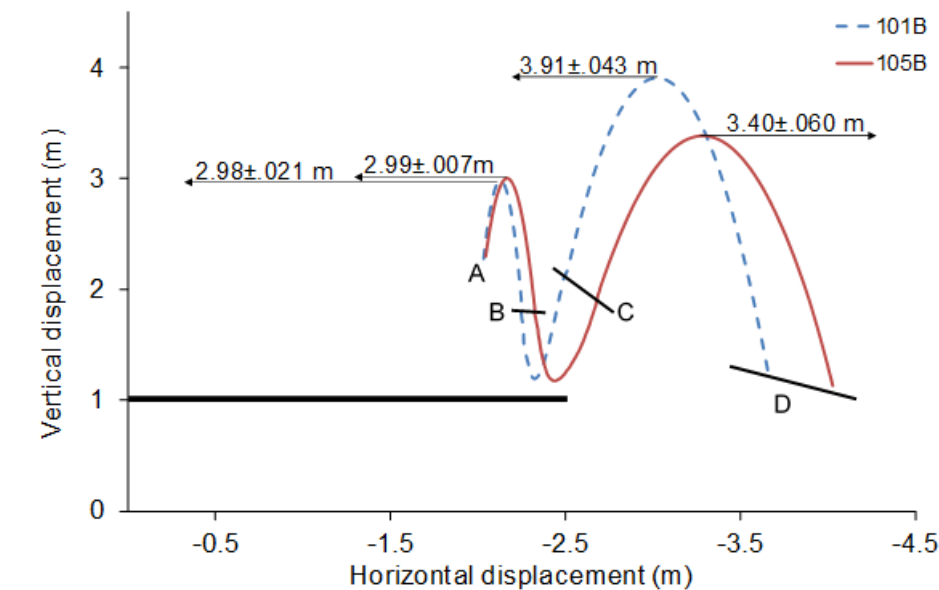


Figure 5.2. Centre of mass movement pattern in hurdle flight (A to B), board contact phase (B to C) and aerial (C to D) phases of 15 forward pike (101B) and 15 forward 2 ½ somersault pike dives (105B).

5.3.2. Angle and time variability

The angles variability displayed in Table 5.1 shows the mean and standard deviation of the joint and orientation angles at hurdle touchdown, hurdle takeoff, landing from hurdle, dive takeoff and entry into the water in 101B and 105B. The standard deviations demonstrate the consistency in body configuration at the start and the end of each phase. The mean and standard deviation of the joint angles in the hurdle are comparable in 101B and 105B but, there is more variability in all joint angles at takeoff and entry in forward 2½ somersault pike dives (105B). Although the mean knee angles in 101B and 105B are similar, the corresponding values of the hip, trunk and arms at dive takeoff and entry are significantly greater in 101B.

Table 5.1. Mean and standard deviation of the Kinematic variables of 15 forward pike and 15 forward 2½ somersault pike dives.

Variable (°)	Dive	Hurdle		Dive		
		Touchdown	Takeoff	Touchdown	Takeoff	Entry
Hip	101B	108.5 ±1.6	166.7 ±2.0	92.2 ±4.9	168.2 ±3.3	180.3 ±3.6
	105B	105.8 ±2.3	166.0 ±2.3	94.1 ±3.1	109.2 ±5.0	141.7 ±13.1
Knee	101B	130.9 ±1.6	177.5 ±1.8	98.6 ±2.6	178.9 ±0.8	179.4 ±0.6
	105B	131.0 ±1.3	176.6 ±1.6	99.0 ±2.2	178.8 ±1.8	176.8 ±1.2
Trunk	101B	82.4 ±1.1	86.5 ±1.2	59.2 ±3.1	72.4 ±2.0	-75.5 ±1.7
	105B	79.9 ±1.6	86.1 ±1.3	57.9 ±1.6	18.0 ±4.2	-59.0 ±4.7
Arm elevation	101B	-16.2 ±3.8	171.6 ±5.5	-32.5 ±5.1	178.7 ±4.5	165.2 ±2.5
	105B	-17.1 ±4.5	172.1 ±4.6	-16.0 ±3.1	121.4 ±7.5	142.1 ±4.3
Orientation	101B	-63.7 ±2.2	-5.9 ±1.6	-12.2 ±1.3	13.1 ±1.3	165.0 ±2.3
	105B	-62.1 ±1.6	-5.6 ±1.4	-10.8 ±1.0	40.2 ±1.8	130.5 ±4.7

*angle between arm and midline of trunk

The orientation angles indicate that the body is near the vertical and consistent at takeoff and entry in 101B but not in 105B as the diver performed 31° more rotation at takeoff and 34° less rotation at entry (Table 5.1). The mean and standard deviations of the centre of mass velocity at touchdown, takeoff and entry in 101B and 105B are shown in Table 5.2. Despite the similarity in vertical and horizontal velocities in hurdle between the 101B and 105B, the diver achieved slightly greater vertical velocity and smaller horizontal velocity at takeoff in 101B.

Table 5.2. Mean and standard deviation of vertical and horizontal velocity of 15 forward pike dives (101B) and 15 forward 2½ somersault pike dives (105B)

Variable	Dive	Hurdle		Dive		
		Touchdown	Takeoff	Touchdown	Takeoff	Entry
V _z * (m/s)	101B	-2.33 ± .065	3.73 ± .06	-4.80 ± .050	5.84 ± .08	-7.38 ± .05
	105B	-2.31 ± .078	3.76 ± .02	-4.88 ± .036	5.19 ± .08	-6.80 ± .08
V _y * (m/s)	101B	2.28 ± .039	0.253 ± .047		0.859 ± .064	
	105B	2.25 ± .064	0.31 ± .030		1.06 ± .070	

*V_z=centre of mass vertical velocity, V_y=centre of mass horizontal velocity

Table 5.3. Mean and standard deviations of time in different phases.

Variable	Dive	Hurdle Contact	Hurdle flight	Dive takeoff	Dive flight
Time (ms)	101B	426 $\pm$ 6.55	870 $\pm$ 9.06	446 $\pm$ 9.42	1349 $\pm$ 13.81
	105B	422 $\pm$ 4.24	881 $\pm$ 5.16	421 $\pm$ 10.12	1221 $\pm$ 15.40

Small variations in the times of different phases show that the diver was very consistent in timing the movement. Despite the consistency within both 101B and 105B, the time of flight phase was shorter in 105B (Table 5.3).

5.3.3. Angle time history

5.3.3.1. Takeoff phase

The mean and standard deviation of the joint angle time histories during the dive takeoff phase in 101B and 105B depicted in Figure 5.3. In 101B, there is arm flexion with a large range of motion (mean SD=6.2°), hip (mean SD=3.7°) and knee (mean SD=4.0°) extensions with similar ranges of motion whereas the trunk angle (mean SD=1.6°) appeared to plateau with a small range of motion. A similar pattern was observed in 105B but, the hip (mean SD=4.4°), arm (mean SD=4.5°) and trunk (mean SD=2.8°) angles decreased in the second half of the contact phase. Performing the arm extension (closing the shoulder angle) at about 0.25 s, the hip flexion and bending the trunk at about 0.3 s indicate that the body flexion was executed at the end of the takeoff phase in 105B to generate angular momentum. The knee angle showed similar pattern and range of motion in both 101B and 105B.

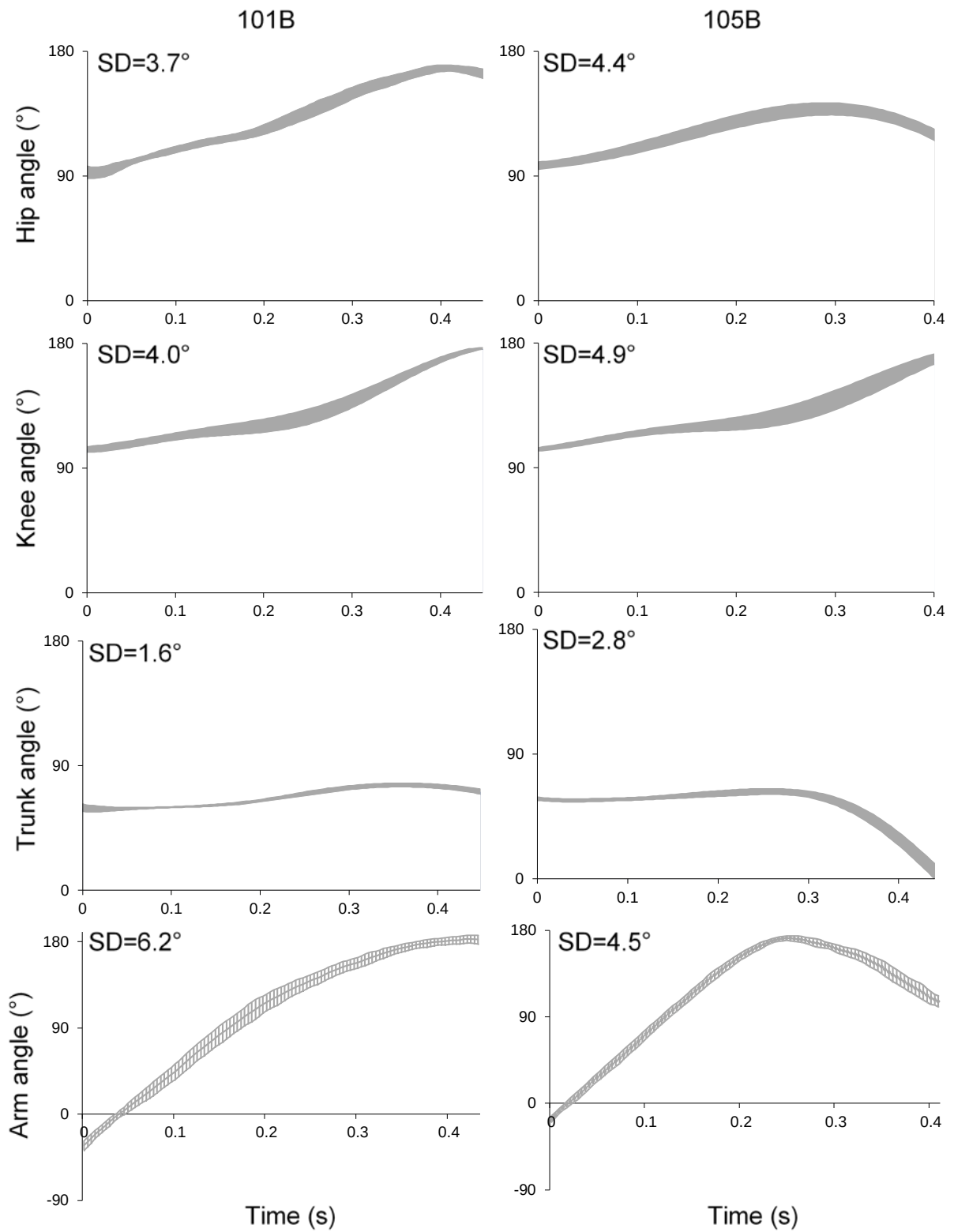


Figure 5.3. Average standard deviation and time histories of joint angles during dive takeoff phase of 15 forward and 15 forward 2½ somersault pike dives.

5.3.3.2. Flight phase

The mean and standard deviation of the joint angle time histories during the dive flight phase in 101B and 105B are represented in Figure 5.4. In 101B, the knee angle (mean SD=1.4°) was constant because the legs are straight in pike position throughout the flight phase. The hip angle (mean SD=10.7°) decreased to minimum at about 0.5 s and extended to maximum at entry. During the first half of flight, the arm angle (mean SD=6.5) decreased to a minimum at about 0.8 s, in this period the diver went into the pike position by moving the arms down while flexing the hip so that at the peak of the dive arms are on the side. In the second half of flight, the arm and the hip angle increased to maximum as the body was opened in preparation for entry. In the last 0.2 s of the flight, the hip angle levelled off indicating that the body achieved straight alignment prior to entry into the water whereas the arm angle decreased slightly in this period. Similarly, in 105B, the knee (mean SD=1.9°) was almost constant due to performing in the pike position. The hip angle (mean SD=5.6) decreased to a minimum at 0.25 s and maintained plateaued for about 0.5 s and started increasing at about 0.75 s and finally reached 141° at entry. The arm angle (mean SD=4.6°) decreased in the first 0.2 s and relatively constant for 0.8 s and increased in the last 0.2 s to reach the maximum angle of 142° at entry into the water.

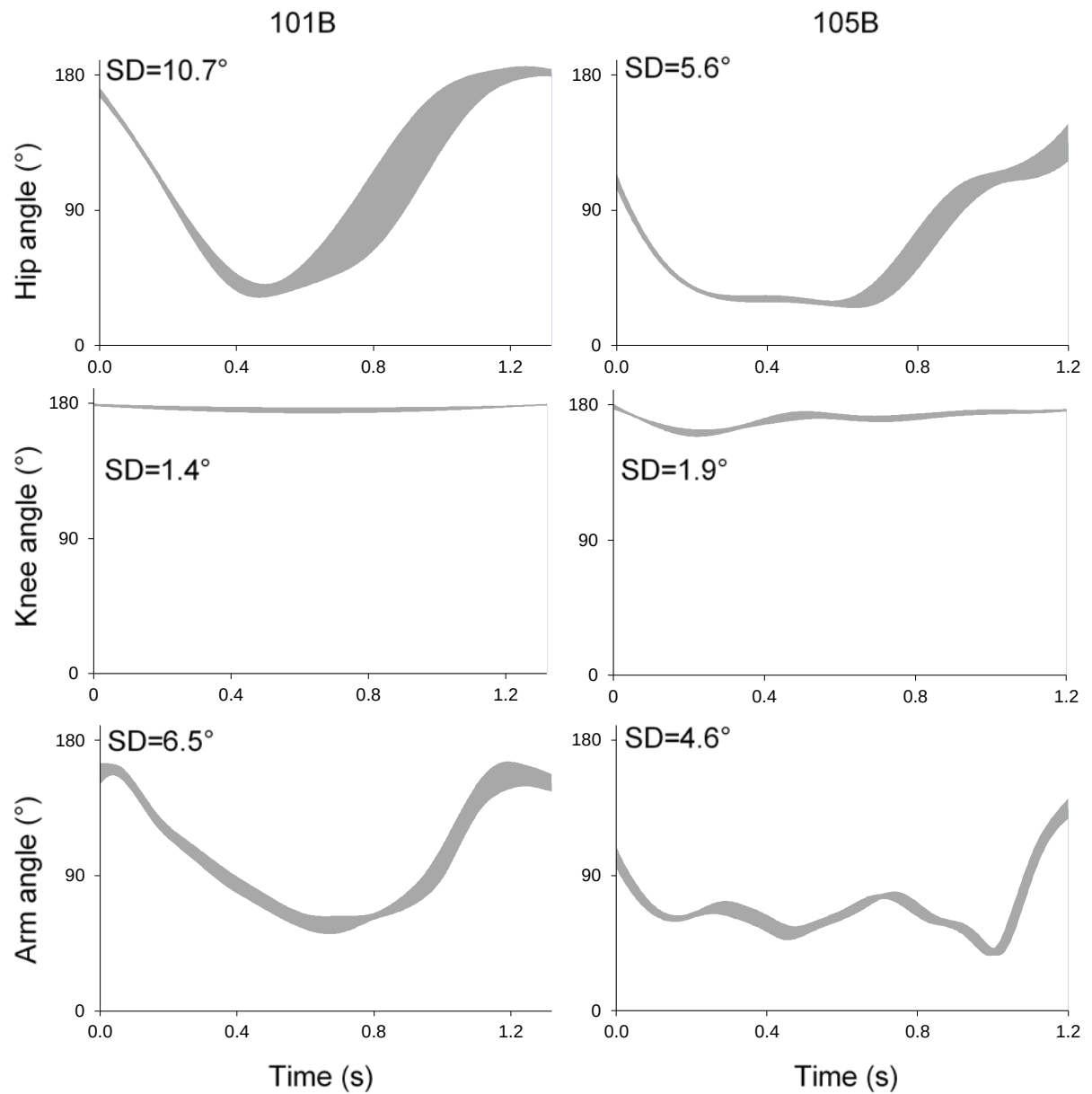


Figure 5.4. Average standard deviation and time histories of joint angles during dive flight phase of 15 forward and 15 forward 2½ somersault pike dives.

5.3.4. Comparison between 101B AND 105B

5.3.4.1. Takeoff phase

A comparison in joint angles between 101B and 105B showed that the average knee angle time history throughout the dive takeoff phase was similar in both 101B and 105B (Figure 5.3). Although the average hip and arm angles time histories followed a similar pattern in the first half of flight in both 101B and 105B, the arm angle reached a peak value of 171° at about 0.25 s and decreased to 101° at takeoff in 105B whereas it increased throughout the dive contact phase and reached a maximum of 182° at takeoff in 101B. Similarly, the hip angle started decreasing at about 0.3 s in 105B whereas it increased during the whole dive contact phase in 101B. The difference in the joint angle time histories between 101B and 105B is apparent in the last half of the takeoff phase due to the difference in the degree of difficulty since there is more angular momentum required for $2\frac{1}{2}$ somersault dives.

5.3.4.2. Flight phase

Figure 5.4 shows a comparison of hip and arms angle time histories between forward pike and forward $2\frac{1}{2}$ somersault pike dives. The knee angles showed similar pattern in both 101B and 105B due to the execution of pike dives. The hip and arm angles in both dives followed similar patterns but, there was greater hip and arm angles in 101B.

5.3.5. Orientation angle

The average orientation angles in takeoff phase at the instant of contact landing from hurdle were -12° and -11° in 101B and 105B respectively, leaning forward faster in 105B such that the body became vertical (perpendicular to the horizontal) at about 0.12 s and reached the maximum of 35° at takeoff whereas in 101B the body became upright at about 0.22 s, which appeared at about maximum board depression, and reached to the maximum of 11° at takeoff (Figure 5.5). Similarly, in 105B the orientation angle during the flight phase shows that the requirement to perform $2\frac{1}{2}$ somersault

increased the amount of orientation angle to 800° (Figure 5.5) whereas there was less than 160° rotation in 101B.

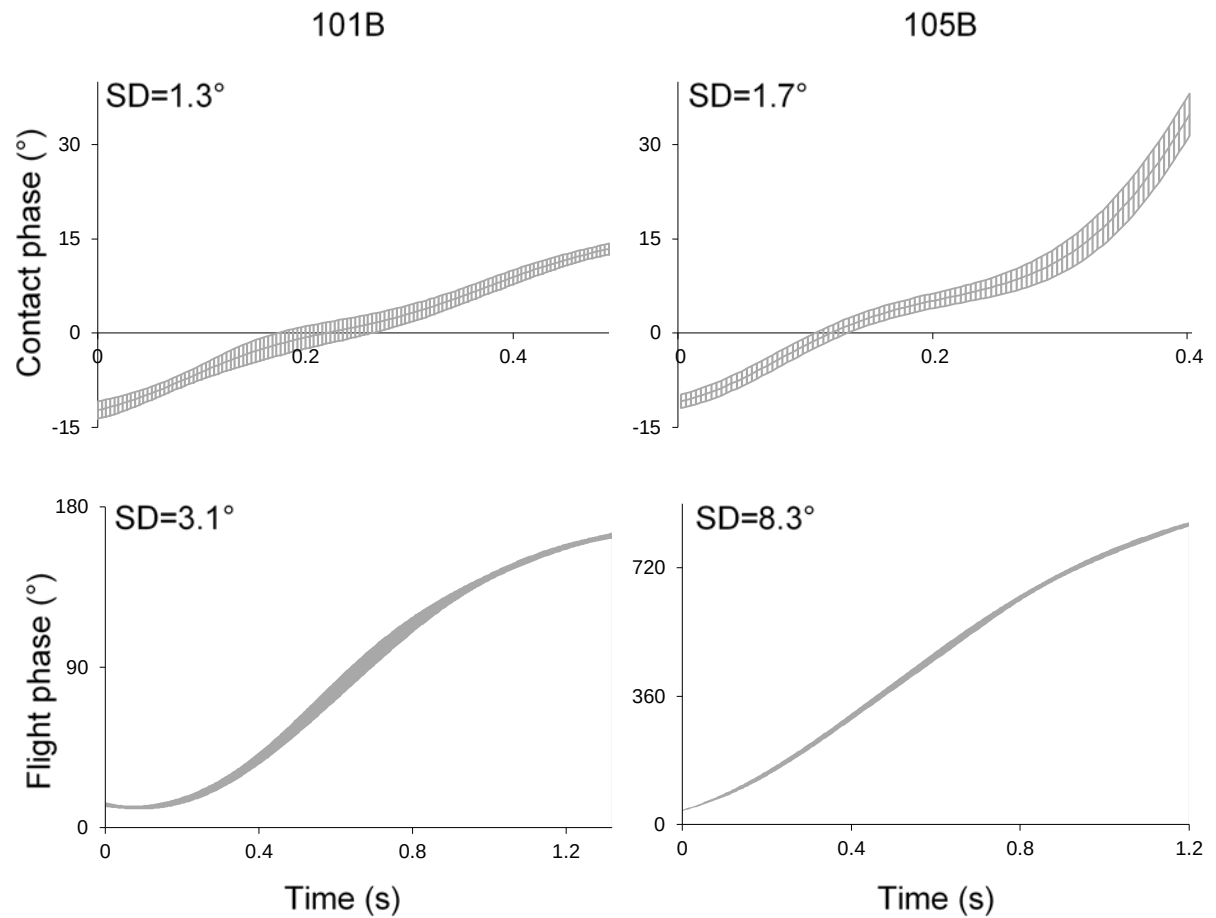


Figure 5.5. Average standard deviation and time histories of body orientation angle during dive contact and flight phases of 15 forward pike (left) and 15 forward $2\frac{1}{2}$ somersault pike (right) dives.

5.3.6. Factors contributing to dive height

The kinematic variables showed in Table 5.4 represent the variables used in correlation and regression analyses.

Table 5.4. A list of the kinematic variables used in SPSS to perform regression analysis

Variables	Location
Hip angle	Touchdown, Takeoff, Entry
Knee angle	
Trunk angle	
Orientation angle	
Peak dive height	The diver's mass centre peak height relative to the water
Peak hurdle height	
Board depression	The lowest vertical displacement of the board tip relative to the board neutral position
Centre of mass vertical and horizontal velocity	at touchdown, takeoff and entry
Hip extension from maximum hip flexion to maximum hip extension	During board contact phase
Knee extension from maximum knee flexion to maximum knee extension	
Toe distance to board tip	

*A line between the knee to trunk relative to vertical

5.3.6.1. Correlation

A Pearson's linear correlation for all variables was performed using SPSS. Tables 5.5 and 5.6 illustrate the correlation of all significant variables against peak dive height of 15 forward pike (101B) and 15 forward 2½ somersault pike dives (105B) respectively.

Table 5.5. Pearson Correlation for all significant variables against peak dive height of 15 forward pike dives (101B)

Variable	Mean (SD)	Peak Dive height	
		R	Sig. (2-tailed)
CM vertical dive takeoff velocity	5.84 (0.08)	0.954**	0.000
Hip extension***	76.22 (5.31)	0.900**	0.000
CM vertical touchdown velocity from hurdle	4.80 (0.05)	0.885**	0.000
Hip angle at hurdle takeoff	166.70 (2.05)	-0.773*	0.001
Peak Hurdle height	2.98 (0.021)	0.559*	0.030
*Correlation is significant at the 0.05 level (2-tailed)			
**Correlation is significant at the 0.01 level (2-tailed)			
*** from max flexion to max Extension during dive takeoff phase			

Table 5.6. Pearson Correlation for all significant variables against peak dive height of 15 forward 2½ somersault pike dives (105B)

	Mean (SD)	Peak Dive height	
		R	Sig. (2-tailed)
CM vertical velocity at takeoff	5.18 (0.08)	0.871**	0.000023
Trunk angle at hurdle touchdown	79.97 (1.61)	0.581*	0.02322
Peak hurdle height	2.99 (0.007)	0.521*	0.04634
*Correlation is significant at the 0.05 level (2-tailed).			
**Correlation is significant at the 0.01 level (2-tailed).			

Although the centre of mass vertical velocity at takeoff was significantly correlated with the peak dive height in both 101B and 105B, the takeoff variables are dependent on the contact phase which are the effect of mechanical work done in the previous stage. In fact, the hip extension during the board contact phase from maximum hip flexion to maximum hip extension was the most significant variable correlated with peak dive height in 101B ($R = 0.900$, $p < 0.01$) and the trunk angle at hurdle touchdown was significantly correlated with peak dive height in 105B ($R = 0.581$, $p < 0.05$). The other variables correlated with dive height were the peak hurdle height in both 101B and 105B and the centre of mass vertical touchdown velocity and hip angle at hurdle takeoff in 101B.

5.3.6.2. Stepwise multiple regression

The body configuration angles at touchdown and during the board contact phase were chosen to be the independent variables comprised the centre of mass vertical velocity and joint angles at touchdown and the diver's toe's distance from the end of the board at takeoff along with the net hip and knee extension (from maximum flexion to maximum extension) during board contact phase and the peak dive height as the dependent variable. In the 1st analysis, the touchdown variables were regressed against peak dive height since they are the causes of the diver's mechanical work during the board contact phase whereas the other variables in later stages such as takeoff variables are the effect of touchdown conditions. In the second analysis all variables were included.

A summary of the stepwise multiple regression analysis is depicted in Table 5.7. In all regression models the entry significance level was selected to be 5%. In the regression models 1, 2 and 5 the touchdown variables only were chosen against peak dive height. In the regression models 3 and 4 all variables were included in the regression.

Table 5.7. Model summary of a stepwise multiple regression analysis of kinematic variables against peak dive height in forward pike and forward 2½ somersault pike dives

Dive	Variable	Model	R ²	Model Summary ^a			
				Unstandardised		standardised	Sig.
				B		Beta	
101B	Touchdown variables only	1	0.78	Constant	-0.029		0.961
				V _z TD	0.821	0.885	0.000
		2	0.84	Constant	-0.278		0.602
				V _z TD	0.807	0.870	0.000
	All variables			Orientation TD	-0.005	-0.249	0.049
		3	0.81	Constant	3.354		0.000
				Hip extension	0.007	0.900	0.000
		4	0.90	Constant	1.483		0.024
105B	Touchdown only			Hip extension	0.004	0.537	0.003
				V _z TD	0.437	0.471	0.007
		5	0.33	Constant	1.676		0.027
				Trunk TD	0.022	0.581	0.023

a. Dependent Variable: Peak dive height relative to the water (m)

Note: V_z is centre of mass vertical velocity, TD refers to touchdown and hip extension is from maximum hip flexion to maximum hip extension.

The model 1 shows that 78% of the variation in the peak dive height in forward pike dives was explained by vertical touchdown velocity whereas in model 2 the body orientation at touchdown comes into the regression equation and together with vertical touchdown velocity accounted for 84% of the variance in the dive height. When all variables were included in model 3, the hip extension from maximum flexion to the maximum extension during board contact phase explained 81% of the variance in dive height. In model 4, the centre of mass vertical touchdown velocity came into the regression and together with the hip extension predicted 90% of variance in dive height. This demonstrates that the centre of mass vertical touchdown velocity and the hip extension during the board contact phase are the most significant predictors of the peak dive height in forward pike dives (101B). Model 5 shows that when touchdown variables only were included in the regression equation, the trunk angle at touchdown explained 33% of variance in dive height in forward 2½ somersault pike dives. There was no change in the regression outcome when all variables were included in the model in 105B.

5.3.7. Factors contributing to entry angle

5.3.7.1. Correlation

In order to investigate the relationship between the takeoff variables and entry orientation angle, Pearson's linear correlation was calculated for each variable using SPSS. The kinematic parameters at takeoff and in the flight were used to investigate the factors influencing the entry angle. Table 5.8 illustrates the correlation between all significant variables and entry orientation angle in 101B and 105B. Although the hip extension in 101B and vertical takeoff velocity and orientation angle at touchdown in 105B are correlated with the entry orientation angle, the peak dive height in both 101B and 105B was significantly correlated with the entry orientation angle.

Table 5.8. Stepwise multiple regression of takeoff variables and dive height against entry orientation angle in forward pike and forward 2½ somersault pike dives

Dive	Model	R ²	Model Summary ^a			
			Unstandardised		standardised	Sig.
			B		Beta	
101B	1	0.29	Constant	52.157		0.304
			Peak dive height	28.862	0.541	0.037
105B	2	0.33	Constant	-24.206		0.695
			Peak dive height	45.479	0.579	0.024

a. Dependent Variable: Body orientation angle at entry into the water (°)

Table 5.9. Pearson correlation of all significant variables with entry orientation angle in forward pike and forward 2½ somersault pike dives

Dive	Variable	Entry orientation angle	
		Pearson Correlation (R)	Sig. (2-tailed)
101B	Hip extension	0.615	0.015
	Peak dive height	0.541	0.037
105B	Peak dive height	0.579	0.024
	Vertical takeoff velocity	0.562	0.029
	Orientation touchdown	-0.535	0.040

5.3.7.2. Stepwise multiple regression

The model summary of performing stepwise multiple regressions of takeoff variables and dive height against entry orientation angle in 101B and 105B are depicted in Table 5.9. The peak dive height accounted for 29% and 33% of variance in entry orientation angle in 101B and 105B respectively. The positive coefficient shows that the orientation angle at entry increased with greater dive height.

5.4. DISCUSSION AND CONCLUSION

The standard deviation of joint and orientation angles at entry compared with the mean standard deviation during the flight phase showed that the diver was highly consistent in performing forward pike and forward 2½ somersault pike dives (Table 5.1, Figures 5.3, 5.4 and 5.5). Although there was kinematic variability in different phases

of the dives, low variability at the entry orientation angle, (SD=2.3° in 101B, SD=4.7° in 105B) (Table 5.1) compared to the mean variability during the flight phase, (mean SD=3.1° in 101B, mean SD=8.3°) (Figure 5.5) demonstrated that the diver was consistent in performance outcome in both 101B and 105B. The greater mean and relatively lower standard deviation at the hip, arm and trunk angles at takeoff and entry in 101B indicated that the diver was more upright and less variable in the dives with lower degree of difficulty (Table 5.1). This was expected because the diver had to execute hip flexion to generate angular momentum in the recoil phase of 105B, leading to lower dive height, shorter flight time, a short of vertical entry and longer board clearance distance in more difficult dives (Figure 5.2 and Tables 5.1, 5.2 and 5.3). Greater variability in 105B might be because contact is more difficult for performing faster rotational movement.

At entry in 105B, the standard deviation of the hip angle was 13° whereas the variability of the orientation angle was 4.7°, indicating that although the dive was short of vertical, the diver entered into the water with the right body alignment as was expected for such a difficult dive. Bernstein (1967) in the study of hammer strokes noted that consistency in performance outcome is achieved by varying movement characteristics, referred to “repetition without repetition”.

Huber (2016) reported that for forward pike dives the diver should come out before seeing the water on the last somersault unless the dive is slow. If the diver waits until seeing the water to come out, the body extension will be too late and the dive will rotate past vertical. Since the generation of angular momentum in 105B causes lower dive height relative to 101B, inevitably there is less time available to perform 2½ somersault, indicating that coming out of the pike before seeing the water may be responsible for greater variability in more difficult dives. It could be interpreted that the divers rely on their vestibular system in 105B and use more visual information in 101B.

To investigate factors influencing the dive height, stepwise multiple regression analyses were performed. In the 1st regression model in 101B only vertical touchdown velocity was significant and this accounted for 78% of the variation in dive height. When the orientation angle at touchdown came into the regression in the 2nd regression

model, together with vertical touchdown velocity explained 84% of the variance in dive height (Table 5.7). It can be hypothesised that the remaining variation comes from body configuration changes during board contact. The 3rd regression model could use any of the variables calculated, but choose only to include hip extension with 81% of the variation in dive height explained (Table 5.7). Although the amount of hip extension and vertical touchdown velocity together explains 90% of the variance in dive height, the hip action is highly dependent on the touchdown variables in 101B not only because it has been measured from body configuration at touchdown but also the body configuration variability during contact comes from the variability at touchdown. Since only hip extension during contact explained more of the variance in dive height than the touchdown variables, it can be identified as the most important variable for explaining dive height variability. This suggests that the amount of hip extension during board contact could explain the amount of energy that the diver has put into the diving board. Therefore, instead of using the amount of board depression as a function of mechanical work done by the diver, the amount of hip extension during board contact can explain the extent to which the energy has been used to achieve a great dive height in forward pike dives.

In 105B, the trunk angle at touchdown was the only significant variable and accounted for 33% of variance in peak dive height (Table 5.7). The generation of more angular momentum might have affected the contribution of all variables to achieving optimal dive height. It is likely that performing the required rotational movement is the key performance factor in forward 2½ somersault dives rather than peak dive height. In fact, larger hip extension during board contact increases the dive height but, the execution of hip flexion at the end of the board contact phase in order to generate greater angular momentum degrades the dive height in more difficult dives. It was expected that this is less observable in forward pike dives as there is less need for a large angular momentum.

Dive height in forward pike dives relative to more complicated dives is greater because less somersault angular momentum is required. This suggests that lower dive height is to be expected in more complicated dives such as 1½, 2½ or 3½ somersault dives due to the requirement for more angular momentum. When the dive height for a

number of different dives was optimised using a simulation model, optimal solutions were only able to increase the dive height by an average of 0.16 m above that of the recorded performances (Kong, 2005). However, no increase in dive height was found for forward pike dives. Although Kong (2005) reported that the reason for the lack of increase in dive height may have been associated with a limitation of the simulation, the diver might have already been using close to optimum technique in forward pike dives. This may be related to the fact that this dive is relatively simple and not so technically challenging as multiple somersault dives.

The knee extension during board contact, the knee angle at touchdown and takeoff were not significantly correlated with dive height. This may be a consequence of the small variation in these knee angle parameters (Table 5.1).

Traditionally the mechanical work done on the springboard is defined as the amount of board depression. The amount of energy that the diver has put into the diving board depends on both vertical touchdown velocity and body extension performed during the contact phase. The amount of hip extension from maximum hip flexion to the maximum hip extension during board contact determines the amount of additional energy that the diver put into the board. Greater hip extension increases the dive height. Variation in hip extension primarily accounts for the variation in dive height which is small (< 3%) indicating the consistency of performance of this diver.

In order to investigate what variability at entry might be explained, a multiple regression analysis of takeoff variables and peak dive height against entry orientation angle was performed. The peak dive height accounted for 29% and 33% of variance in entry angle in 101B and 105B (Table 5.8). This indicates that greater dive height provides more time to complete the rotational movement in the air and lead to more vertical entries. Although this is in agreement with a number of studies who acknowledged that dive height is an important factor contributing to the performance outcome (Lanoue, 1940; Harper, 1966; McCormick, Subbaiah and Arnold, 1982; Miller and Munro 1985; Sanders & Wilson, 1988; Miller and Munro, 1984, 1985; Sanders & Wilson, 1988; Louganis, 1995; Sanders & Gibson, 2000), not only peak dive height but also the

execution of correction during the flight phase may affect the entry angle. The diver's adjustment in the air is investigated in Chapter 8.

5.5. Summary

In this chapter, the variability in the joint and body angles at the start and the end of each phase were quantified. Determination of variation in the peak dive height and time of flight phases was provided. In terms of body configuration, a comparison between forward pike and forward 2½ somersault pike dives were given. The relationships between variables at hurdle landing, during board contact, dive height and entry orientation angle were determined. Factors influencing dive height and entry orientation angle were described. The variability of orientation angle at water entry (performance outcome) compare to the mean standard deviation during the flight phase indicated that the diver was consistent in performing both dives. There was relatively more variability in more difficult dives. The amount of hip extension from maximum hip flexion to the maximum hip extension during board contact phase determines the amount of additional energy that the diver put into the board, indicating that greater hip extension increases the dive height.

CHAPTER 6

VARIABILITY IN FOOT PLACEMENT

6.1. INTRODUCTION

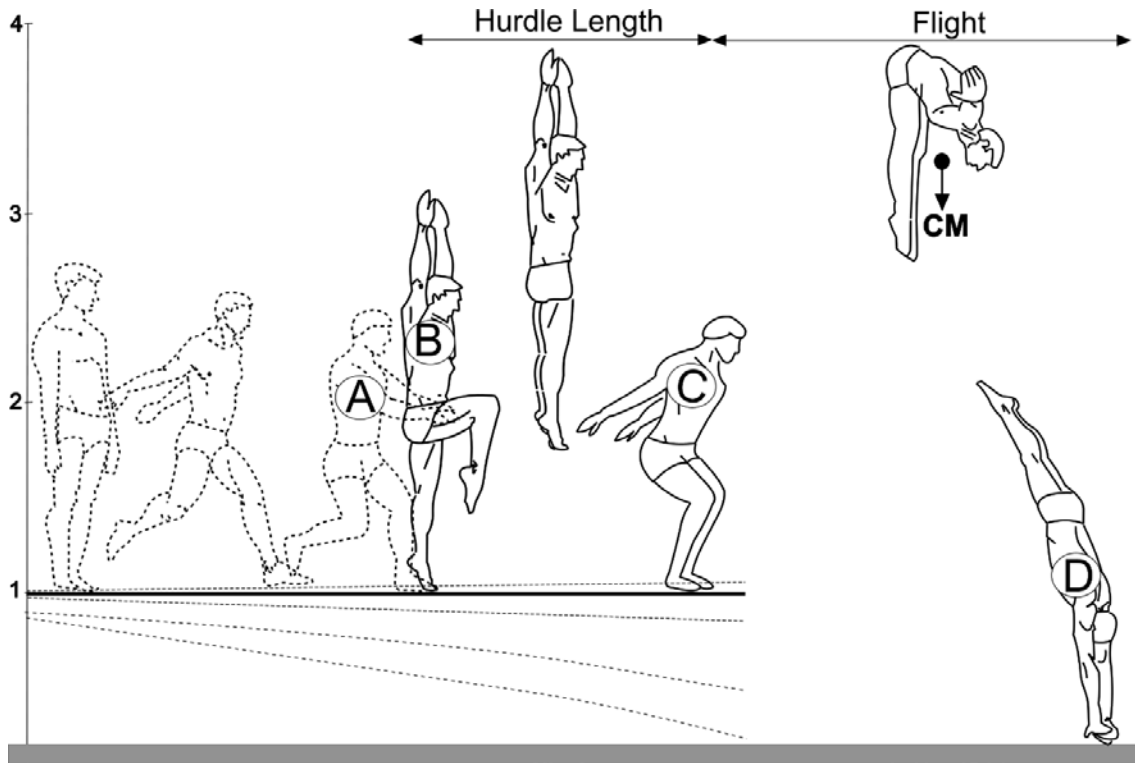


Figure 6.1. Forward dive, A= hurdle touchdown, B= hurdle takeoff, C= landing from hurdle, (B to C) = hurdle length.

Springboard divers can set the board with the desired vertical stiffness, based on practice, by manipulating the position of the adjustable fulcrum forward to increase the stiffness or back further away from the board tip to make the board more compliant (Miller & Jones, 1999). For a given fulcrum setting, the board deflection and oscillation characteristics are also affected by the location and magnitude of the diver's vertical force on the board. If the diver's foot placement on the board is closer to the fulcrum, more of the board mass comes into play increases the effective board mass as greater impact loses more energy, the board becomes effectively stiffer (Sprigings et al., 1987; Boda, 1992; Sprigings et al., 1989, 1990; Stone, 1987). Conversely, a springboard will deflect more if loads are applied closer to the board tip.

Most springboard divers start the hurdle with four or five steps to have more time for creating a good rhythm for the last stride prior to the hurdle contact phase even though the rule is a minimum three steps (O'Brien, 1992). The first step during the approach stage is always shorter and the last one is longer and faster than the earlier steps. During the hurdle touchdown, the board is depressed by the full weight of the diver's body and the execution of the knee flexion of the supporting leg further increases the amount of the board depression as it lowers the body mass. During the hurdle recoil phase, the diver jumps by extending the knee of the supporting leg along with moving the arms and the knee of the non-supporting leg forward and upward simultaneously to achieve a hurdle height. There is inevitably variation in the pattern of the preparatory approach steps, hurdle and takeoff phases (Sanders & Wilson, 1988; Miller, 1983 and 1984; Batterman, 1968; Rewt, 1993; O'Brien, 1992).

Landing from a hurdle further away from the board tip makes it much more difficult for divers to depress the board as far as when they land close to the board tip. Mechanically most of the potential energy stored during the board depression is converted to the kinetic energy during the board recoil phase to project the diver into the air (Yeadon, 1997; Ward-Smith AJ, 1983, 1986). Furthermore, more hip extension during the board contact phase further amplifies the amount of energy into the system to increase the dive height (Chapter 5). Therefore, there is a difference on the board vertical stiffness depending on not only the fulcrum setting but also whether the diver lands on the end of the board or a certain distance from the board tip at hurdle landing.

It has been assumed traditionally that the approach and hurdle phases, as preparatory movements in springboard diving, are the forerunners that facilitate the actual performance of dives. Miller (1984) reported that although women had comparable hurdle length, they achieved lower hurdle height than men, indicating that the hurdle length was adjusted by changing centre of mass velocity to compensate for inadequate hurdle height. Sanders & Wilson (1987) reported that most diving coaches agree that divers need to perform a good vertical jump in the hurdle to execute a great dive even though there is no guarantee that a good vertical jump ends with a great dive. Although the effect of horizontal and vertical velocities at takeoff are independent, the resultant velocity at hurdle takeoff determines the diver's movement in the hurdle flight. The

vertical velocity determines the dive height and the horizontal velocity influences the horizontal displacement, indicating that not only a good vertical jump but also landing from the hurdle as close to the board tip as possible is essential to execute a great dive.

Since divers perform the hurdle by jumping from one leg and it is beneficial for them to have a hurdle landing as close to the board tip as possible to make the board more compliant, there is inevitably an amount of variability in foot placement in the hurdle. It was expected that there might be an adjustment for the variability of foot placement in the hurdle and the centre of mass velocity may be used as a controlling variable for correction. It is hypothesised that if the foot placement at hurdle touchdown is further away from the board tip, the divers make the hurdle length (horizontal displacement) longer to compensate for variability in hurdle touchdown. The aim of this chapter is to investigate whether the diver makes an adjustment to compensate for the variability of foot placement in hurdle.

6.2. METHOD

Movement variability of 15 forward pike (101B) and 15 forward 2½ somersault pike (105B) dives were determined using video analysis (Chapter 3). IBM SPSS Statistics (version 23) software was used to investigate the variability of foot placement. A linear regression analysis of toe distance at hurdle landing from the board tip of all dives against dive number was performed. Since 101B was performed in the morning and 105B in the afternoon, the similarity between the sessions was investigated. The last 8 dives in the morning and the first 8 dives in the afternoon were used to investigate the variability in foot placement in this Chapter. A correlation analysis was run to identify the relationship between the variables which comprised the hurdle landing distance (distance between the foot placement in hurdle landing and board tip), hurdle touchdown distance (distance between foot placement in hurdle touchdown and the board tip), centre of mass horizontal velocity at hurdle takeoff, hurdle length (horizontal displacement in the hurdle flight calculated as the distance between the foot placement in hurdle touchdown and hurdle landing) and dive number (Figure 6.1). Stepwise multiple regression analyses of a) hurdle landing distance against all significant

variables b) hurdle length against all significant variables were carried out to investigate the variability of the foot placement in hurdle.

6.3. Results

Figures 6.2 and 6.3 show that the dive number explained 58% ($p < .01$) of variance in the hurdle landing distance from the board tip in the last 8 dives in the morning (101B) and the first 8 dives in the afternoon (105B). However, no significant correlation found for the first 7 dives in the morning and the last 7 dives in the afternoon.

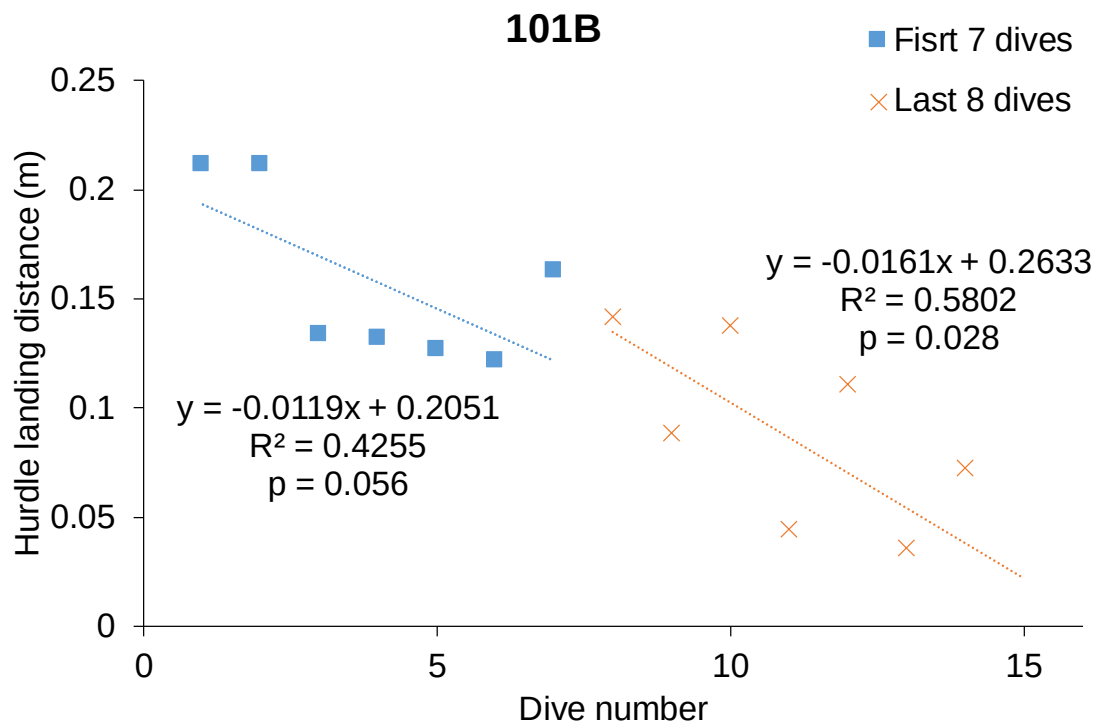


Figure 6.2. Linear regression of foot placement at takeoff of forward pike dives (morning session), the first 7 dives (squared points), the last 8 dives (crossed points).

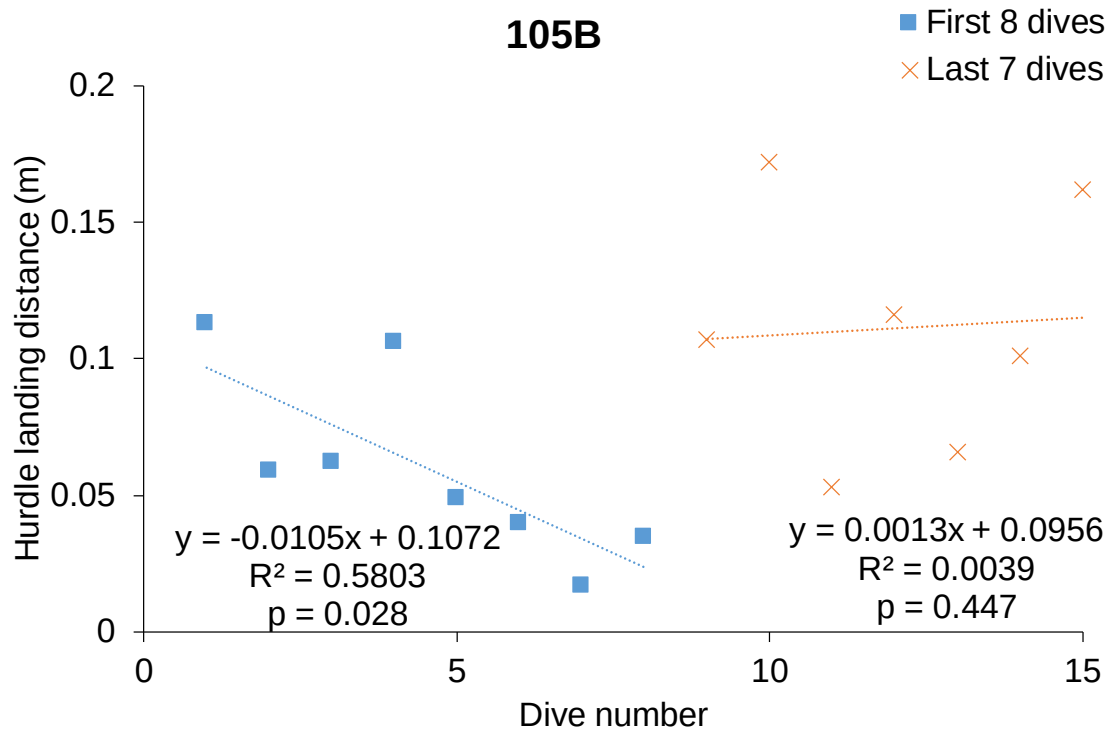


Figure 6.3. Linear regression of foot placement at takeoff of forward 2½ somersault pike dives (afternoon session), the first 8 dives (squared points), the last 7 dives (crossed points).

Since the forward pike dives (101B) were performed in the morning and the 2½ somersault pike dives (105B) in the afternoon, this might be viewed that the difference between 101B and 105B might be due to performing them in two different sessions. However, the significant correlation between the dive number and the hurdle landing distance in the last 8 dives in 101B and the first 8 dives in 105B (Figure 6.4) showed that there was no significant difference between the morning (101B) and the afternoon 105B sessions. Therefore, the last 8 dives of 101B and the first 8 dives of 105B were used in this chapter to investigate the variability in foot placement. The coefficient of determination 58% ($p < .05$) of dive number against hurdle landing distance showed that the diver generally became closer to the board tip with later dives (Figure 6.4).

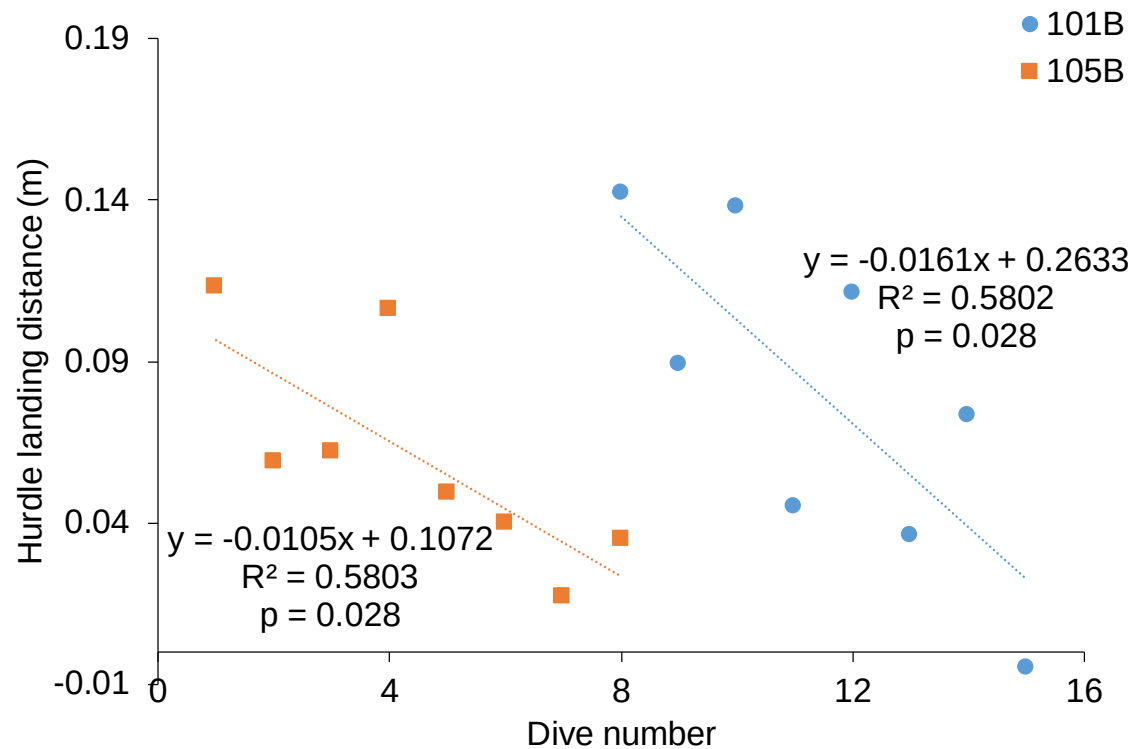


Figure 6.4 Linear regression of foot placement at takeoff of the last 8 dives in the morning (squared points) and the first 8 dives in the afternoon (circle points).

Figure 6.5 shows the variability of foot placement in hurdle contact and dive takeoff of 8 forward pike and 8 forward 2½ somersault pike dives. There is more variability in 101B than 105B. The variability at takeoff is greater at takeoff than the hurdle in both 101B and 105B.

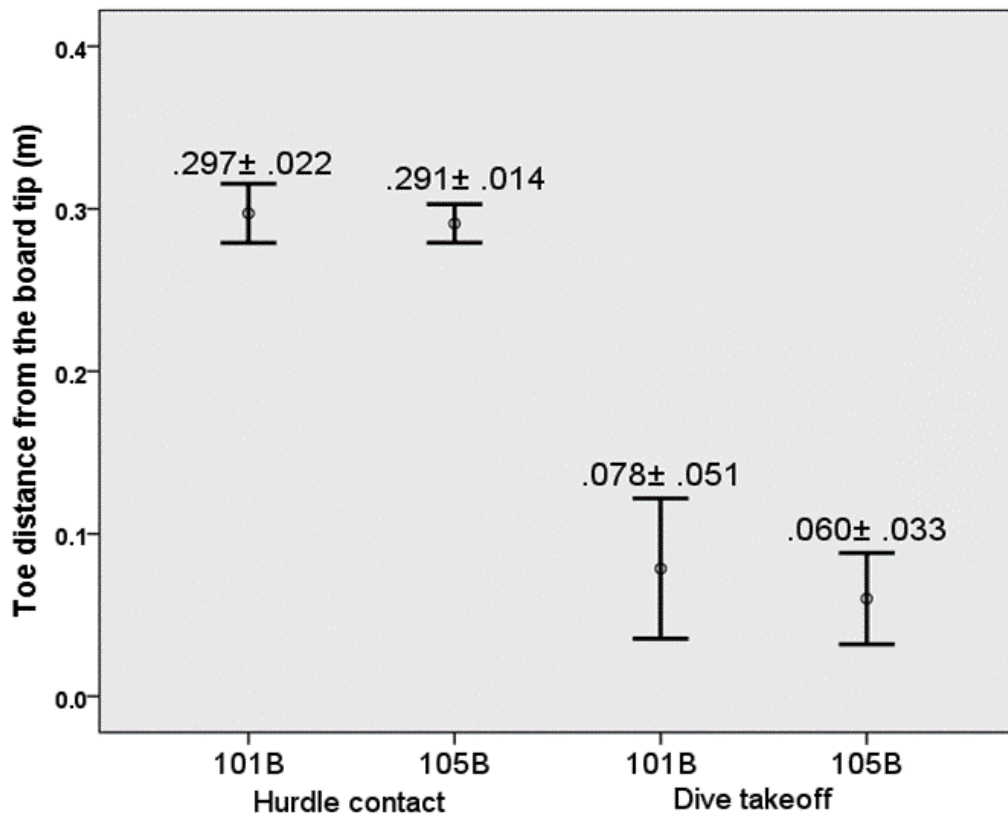


Figure 6.5. Mean and standard deviations of toe distance from the board tip at hurdle contact and dive takeoff of the last 8 dives in the morning (101B) and the first 8 dives in the afternoon (105B).

6.3.1. Correlation

Table 6.1 shows the correlations between the dive number, hurdle landing distance from the board tip, hurdle flight CM horizontal velocity and hurdle touchdown distance from the board tip of the last 8 dives in the morning (101B) and the first 8 dives in the afternoon (105B). The dive number explained 58% of variance in the landing distance in both 101B and 105B with the same level of significance, $p = 0.014$. Similarly, the dive number accounts for 38% ($p = .052$) and 57% ($p = .014$) of horizontal velocity in 101B and 105B respectively. There was a significant relationship 49% ($p = .025$) between the dive number and hurdle touchdown distance in 105B but, no correlation found in 101B.

Table 6.1. Correlations between variables of the last 8 dives in 101B and the first 8 dives in 105B

		Dive number		Hurdle landing		Horizontal velocity		Hurdle TD ^a		Hurdle length	
		101	105	101	105	101	105	101	105	101	105
Dive number	R ²	1	1	.58*	.58*	.38	.57*	.08	.49	.39*	.65**
	Sig			.014	.014	.052	.014	.246	.025	.048	.007
Hurdle landing	R ²	.58*	.58*	1	1	.79	.92	.025	.40*	.83**	.93**
	Sig	.014	.014			.002	.000	.351	.046	.001	.000
Hurdle TD ^a	R ²	.08	.49*	.025	.40*	.06	.53*	1	1	.06	.65**
	Sig	.246	.025	.351	.046	.268	.020			.271	.008
Horizontal Velocity	R ²	.38	.57*	.79	.92	1	1	.06	.53*	.95**	.94**
	Sig	.052	.014	.002	.000			.268	.020	.000	.000

a: Touchdown

*. Correlation is significant at the 0.05 level (1-tailed).

**. Correlation is significant at the 0.01 level (1-tailed).

The coefficients of determination (R-squared) indicate that 65% ($p < 0.01$) of the hurdle length is predicted by hurdle touchdown distance in 105B, but no correlation found in 101B. The centre of mass horizontal velocity accounts for 95% and 94% of variance in the hurdle length in 101B and 105B respectively ($p < .01$) (Table 6.1).

6.3.2. Stepwise multiple regression analysis

6.3.2.1. Hurdle landing

A summary of the multiple regression analyses of hurdle landing distance against dive number, hurdle flight CM horizontal velocity and hurdle length in 101B and 105B using stepwise method at the entry significance 5% level is shown in Table 6.2. The hurdle length explained 83% ($p < .01$) and 93% ($p < .01$) of variance in the hurdle landing distance from board tip in 101B and 105B respectively. The negative coefficient of the hurdle length, -0.894 and -0.741 in 101B and 105B respectively, indicates that the foot placement at hurdle landing becomes closer to the board tip by increasing the hurdle length in both forward pike and 2½ somersault pike dives. The adjusted R-square 0.80 and 0.92 in 101B and 105B respectively, indicates that if the models were derived from the population rather than a sample, they would account for approximately 3% and 1% less variance in the outcome.

Table 6.2. Stepwise multiple regression analysis of hurdle landing distance against dive number, hurdle touchdown distance and hurdle flight CM horizontal velocity in 101B and 105B

Model Summary ^b						
Dive	Model	R Square	Adjusted R Square	Unstandardised B	standardised Beta	Sig.
101B	1	.834 ^a	.806	Constant	.274	.000
				Hurdle length	-.894	.002
105B	1	.938 ^a	.928	Constant	.231	.000
				Hurdle length	-.741	.000
a. Predictors: (Constant), Hurdle length						
b. Dependent Variable: Hurdle landing distance to the board tip						

6.3.2.2. Hurdle length

The model summary and coefficients of the stepwise multiple regression analysis of hurdle length against dive number, hurdle touchdown distance from the board tip and hurdle flight CM horizontal velocity are shown in Table 6.3. The hurdle flight centre of mass horizontal velocity accounted for 96% and 94% of variance in the hurdle length in 101B and 105B respectively (both $p < .01$). The positive coefficients of horizontal velocity 1.244 and 1.282 in 101B and 105B respectively, indicate that the diver made the hurdle flight longer by increasing his horizontal velocity at hurdle takeoff. The adjusted R-square indicates that if the models were derived from the population rather than a sample, they would account for approximately 0.7% and .9% less variance in the outcome.

Table 6.3. Model summary of the multiple regression of hurdle length against dive number, hurdle touchdown distance and hurdle flight CM horizontal velocity of the last 8 trials in the morning (101B) and the first 8 trials in the afternoon (105B)

Model Summary ^b						
Dive	Model	R Square	Adjusted R Square	Unstandardised		standardised
				B	Beta	Sig.
101B	1	.959 ^a	.952	Constant	-.134	.004
				Horizontal velocity	1.244	.979
105B	1	.945 ^a	.936	Constant	-.187	.004
				Horizontal velocity	1.282	.972
a. Predictors: (Constant), Hurdle flight CM horizontal velocity						
b. Dependent Variable: Hurdle length						

The b-value 1.282 specifies that as the CM horizontal velocity increases by one unit, hurdle length increases by 1.282 units. The velocity and flight distance were measured in metres per second (m/s) and metres (m) respectively; therefore, for every 1 m/s increase in horizontal velocity, an extra 1.282 metres was achieved in hurdle length. The b-values and their significance are important statistics; however, the standardized versions of the b-values are in many ways easier to interpret because they are not dependent on the units of measurement of the variables (Field, 2009).

The standardised coefficients beta in 101B, $\beta = 0.979$, shows that as the horizontal velocity at hurdle takeoff increases by 1 standard deviation (0.041 m/s), flight length increases by 0.979 standard deviations. The standard deviation for the flight length is 0.052 m and so this constitutes a change of 0.051 m in the hurdle length (0.979 times 0.052). Therefore, if the horizontal velocity increases an extra 0.041 m/s, 0.051 m extra distance in hurdle length is expected. Similarly, in 105B, an extra distance of 0.0426 m requires an extra 0.033 m/s. This interpretation is true only if the effect of dive number and hurdle touchdown distance from the board tip are held constant.

6.4. DISCUSSION

There was larger variability in foot placement at dive takeoff than hurdle contact in both 101B, $p < 0.05$ and 105B, $p < 0.05$ (Figure 6.5). Although it has been recognised that the divers should perform a high hurdle to generate high vertical velocity at landing to generate greater momentum at touchdown (Harper, 1966; Sanders & Wilson, 1988), faster movement requires greater control and may have increased the variability on landing. The regression analysis showed that the diver came closer to the board tip with later in both forward pike and $2\frac{1}{2}$ somersault pike dives (Figure 6.4), indicating that there is feedforward fine tuning of the hurdle takeoff technique. The variability at hurdle landing may also be due to the execution of adjustment in the hurdle contact.

There was significant correlation between the dive number and the hurdle flight CM horizontal velocity (Table 6.1), indicating that the diver increased his horizontal velocity in both 101B and 105B. This means that the diver attempted to become closer to the end of the board by moving faster along the springboard with later dives. The knowledge of result (learning) may have been used in each dive to improve his distance to the board tip at landing from hurdle in the following dive, making the diver more confident to move faster and become closer to the end of the board. Schmidt (1975) provided evidence that learning is very difficult without presenting the correct movement on each trial or without post-response error information. To strengthen the motor program, the sensory information of the correct response is required and if knowledge of results is not provided, the subject has no way of learning the correct location. Although this suggests that divers learn the drive foot position in long term practice, it was found that the knowledge of results is used based on feedback in process during the performances of $2\frac{1}{2}$ somersault pike dives that corrects the hurdle technique slightly in later dives, providing post-response error information in each dive to learn the correct foot placement at takeoff in the following dive.

The multiple regression analysis showed that the hurdle length predicted 83% ($p < .01$) and 93% ($p < .01$) of variance in toe's distance from the board tip in 101B and 105B respectively (Table 6.2), indicating that the diver came closer to the board tip by making hurdle length longer. The centre of mass horizontal velocity at hurdle takeoff was

significantly correlated with foot position at hurdle touchdown in 105B ($p < 0.05$). This was expected that if diver is further away from the board tip at hurdle contact he makes adjustment in the hurdle flight phase to compensate for variation in foot position in hurdle to land as close to the board tip as possible in dive takeoff phase. The horizontal velocity at hurdle takeoff was not correlated with foot placement at hurdle touchdown. This might be because the forward pike dives are relatively easier and not so technically challenging as multiple somersault dives. Although the hurdle flight centre of mass horizontal velocity explained 96% and 94% of variance in hurdle length in both 101B and 105B respectively (Table 6.3), there was significant relationship between the horizontal velocity and hurdle touchdown in 105B (Table 6.1), indicating that the diver adjusted his hurdle length by changing his centre of mass horizontal velocity at hurdle takeoff to correct his hurdle length in $2\frac{1}{2}$ somersault dives. This was also expected since the generation of the horizontal velocity at hurdle takeoff determines the diver horizontal displacement. This suggests that the diver controlled his foot position in dive takeoff phase depending on how far his foot position was from the board tip at hurdle contact such that when the diver was further away from the board tip, he made greater horizontal velocity during hurdle recoil phase to make longer horizontal displacement. This is in agreement with Miller (1984) who found that centre of mass velocity is changed to compensate for inadequate hurdle height.

The average time of hurdle contact phase was 426 ms and 422 ms in 101B and 105B respectively (Chapter 5), indicating that there is enough time to use feedback. Thus, it is likely that the visual system provides post-response error information about the distance from the board tip to make an adjustment by changing the horizontal velocity to reduce the hurdle landing distance to the board tip.

There was more variability in the first 7 dives in the morning and the last 7 dives in the afternoon (Figures 6.2). The distance from the end of the board at takeoff was greater at the beginning of the session than the end of the session in the morning. It is likely that the diver used feedback in the early dives in the morning to reach his accuracy in foot position in the later dives in the morning. In the first 8 dives in the afternoon the diver improved his distance from the board tip but he didn't maintain his accuracy in

the last 7 dives and became more variable (Figure 6.3). The effect of fatigue may be responsible for the greater variability in the last 7 dives in the afternoon session.

The mean and standard deviations of the foot placement showed that although there is more variability in hurdle landing than hurdle touchdown in both 101B and 105B (Figure 6.5), the variation in 101B was greater than 105B. This is in agreement with Slobounov et al. (1997) who found that the variability of the foot placement at hurdle decreased in more difficult dives.

6.5. Summary

The variability of foot position during board contact phase at hurdle contact and dive takeoff was investigated in this chapter. It was found that there was an adjustment in more difficult dives (105B) made based on feedback control that corrected the hurdle technique slightly in later dives, providing post-response error information in each dive to learn the correct foot placement in the following dive. The diver controlled his distance to the board tip at dive takeoff phase in 105B depending on how far his foot position was from the end of the board at hurdle contact. The diver varied his centre of mass horizontal velocity at hurdle contact to adjust the hurdle length and compensate for the variability of foot position at hurdle touchdown to control his distance on the board at hurdle landing, leading to land safely and as close to the board tip as possible in forward 2½ somersault pike dives.

CHAPTER 7

VARIABILITY AND CONTROL IN TAKEOFF

7.1. INTRODUCTION

Board contact phase begins when the diver contacts the diving board from the hurdle flight using both feet to depress the board. Board recoil phase starts at maximum board depression until the board reaches its neutral position at takeoff to propel the diver into the air (Figure 7.1).

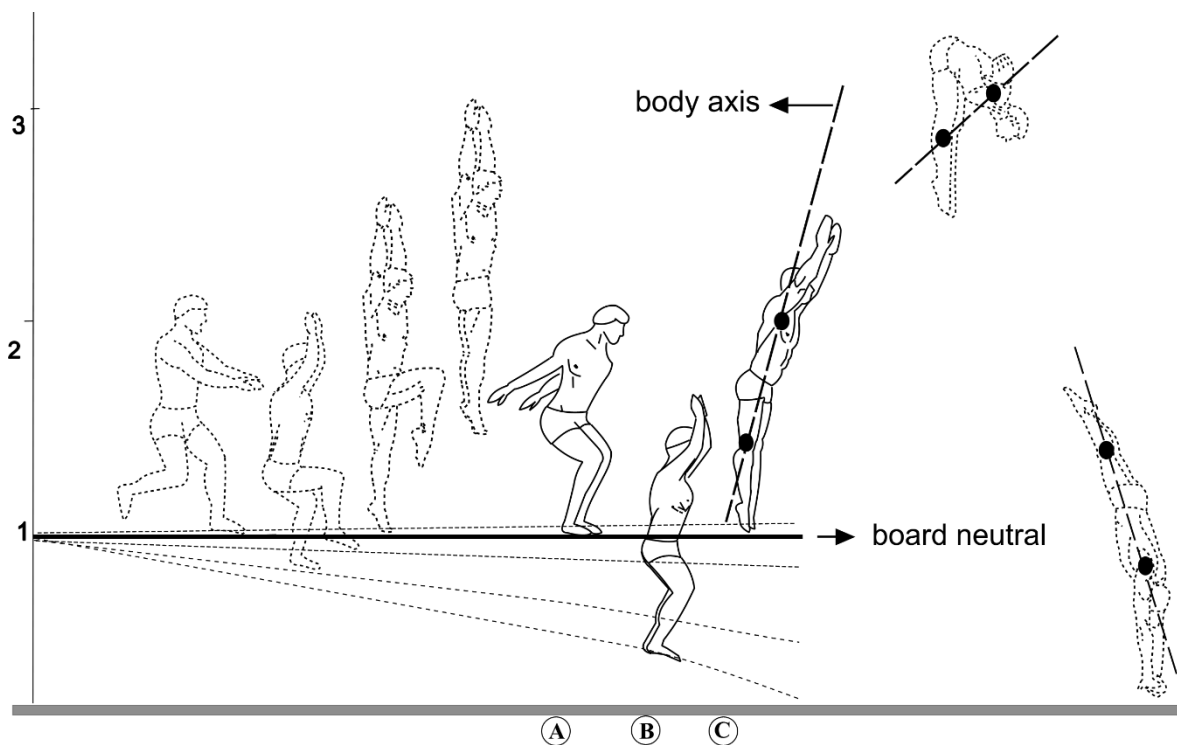


Figure 7.1. 1-m springboard forward pike dives. Board contact phase (A to C), Hurdle landing (A), Maximum board depression (B), Takeoff (C), Board depression phase (A to B), Board recoil phase (B to C).

In forward pike dives, the angular momentum is generated during the takeoff phase by hip flexion. The execution of extra flexion absorbs energy and might result in a loss in dive height. Thus, the optimal flexion in hip and knees at touchdown greatly depends on the diver's skill and muscular strength in which extra flexion following the touchdown would be minimised to avoid energy absorption (Sanders & Wilson, 1988; Sanders & Gibson, 2000). In addition, since the surface of support is the springboard, the act of

orienting controls the interaction between the diver's body and the board. Hence, the perception of information about the board characteristics and responses is essential for the divers to control the action of orientation (Stoffregen, 1988).

There are many movements that can be executed in far less time than is required for using feedback control (Keele & Posner, 1968). If there is any error in the initiation of the movement, the processes involving the generation of sensory error information, perceiving it, and initiating corrections in response to those errors requires about 0.12 s to 0.2 s to start the correction (Schmidt, 1975). When the motor program is initiated, it runs itself as planned but, if there is any error in the initial conditions demanding some new movement to be planned, the performer cannot complete any such changes until the program has run its course approximately 0.2 s. The control of the movement in these circumstances is called open loop because stimuli from the periphery cannot initiate a new program until the present one has run its course for one reaction time. The amount of possible feedback involvement in a movement longer than 0.2 s is dependent mainly on the movement time (Schmidt, 1975).

The average takeoff time 0.446 s and 0.421 s in 101B and 105B respectively (Chapter 5) suggests that there is enough time to use feedback control during the board contact phase. Although previous studies have investigated the relationship between different variables in different phases in springboard diving and optimal takeoff techniques in terms of generating both linear and angular momentum have been developed such as Rewt (1993) who investigated the control of body movements relative to head position and the environment and found that springboard divers regulated their braking by keeping the rate of change of tau function constant when approaching takeoff, little is known about movement variability and control in the takeoff phase.

The role of the board contact phase is fundamental to the flight phase and the whole dive because the dive height, board clearance distance and angular momentum are all determined during this mechanically important phase. Although it has been reported that a diver can perform the same dive with slightly different techniques and make corrections to compensate for any mistakes during the takeoff, flight, and/or entry, it is not clear how divers control their takeoff movement and to what extent changes in the

initial touchdown conditions can affect the takeoff conditions. It is hypothesised that divers make adjustment during the contact phase to cope with variation of the initial conditions leading to consistency at takeoff. A subject-specific angle driven simulation model was used to investigate the effect of variation at touchdown and maximum board depression on the takeoff conditions.

7.2. METHOD

In order to answer the research question, an angle-driven simulation model of a diver and springboard (Chapter 4, section 4.1) was applied. Since initial conditions and model parameters were required to be used as input to the model, video recordings of actual performances were used to specify body and joint kinematics as the initial conditions. Kinematic data for the diver's movement during the board contact phase were obtained from digitisation of video recordings followed by appropriate smoothing (Chapter 3). Model parameters were determined directly from experiments and indirectly through an optimisation process when direct measurement were not possible. Subject-specific segmental inertia parameters calculated from anthropometric measurement (Chapter 4, section 4.2) and springboard parameters calculated from experiment (Chapter 4, section 4.3). The model was customised based on the data obtained in this study as described in Kong (2005) by using all initial conditions, the joint angle time histories obtained from the diver's performances and quantic spline fit to displacement data to match the performances with the model and achieve a simulation that does the same as the dives. This was done to simulate 15 forward pike and 15 forward 2½ somersault pike dives.

A Simulated Annealing Optimisation method (Corana et al., 1987) was applied to determine unknown model parameters. The angle-driven model was used to determine the stiffness and damping of the visco-elastic elements in the foot-springboard interface. This was achieved by driving the model with joint angle time histories obtained from video data and optimising the values of the unknown visco-elastic parameters to minimise the difference between simulation and performance. The visco-elastic parameters were calibrated in a matching process such that nine

parameters (Table 7.3) were varied until the best match between simulation and performance was found.

An objective score was used to quantify the difference between the simulation and performance. The matching process was achieved by minimising the score using the optimisation method. Since the vertical velocity, horizontal velocity and angular momentum are the outcome variables in takeoff phase, after matching a simulation closely to the performance, the three outcome variables were used to compare rotation potential variability at takeoff in the actual performances (original matching simulations) and simulations arising from perturbing the initial conditions. This was carried out by performing two different simulation analyses. In the 1st analysis, the initial conditions of each of the three variables were perturbed individually. In the 2nd analysis, a combination of perturbing all three variables at a time was used. In both analyses plus and minus one standard deviation of the actual performance were used for perturbation. The two analyses were performed for two different initial conditions (a) at touchdown and (b) maximum board depression.

The input to the angle-driven model included initial conditions at touchdown and joint angle time histories throughout the simulation. The initial conditions at touchdown including mass centre (CM) horizontal and vertical velocities, angular velocity, trunk angle and foot positions. The diver's movement was driven by joint angle time histories calculated from video analysis. The model output included the diver's CM velocities, springboard displacement time histories, trunk angle and whole body angular momentum.

In the study of simulation, there are two types of optimisations, one is to minimise the RMS difference and the other is to produce the best solution for the outcome variable. Since the aim of this chapter required a simulation that does the same as the dives, the outcome takeoff conditions were matched such that the model does the same dives as in the actual performances. It needs to be mentioned that the wobbling masses were excluded from the model as there was no need to include them in modelling takeoff phases from the compliant springboard (Chapter 4, section 4.2.1).

7.2.1. Model input

The initial conditions at touchdown and the actual joint angle time histories were used to drive the model. The initial touchdown conditions for forward pike and forward 2½ somersault pike dives showed in Tables 7.1 and 7.2 respectively.

Table 7.1. Initial touchdown variables used in the simulation model in forward dives

Dive	V _y (m/s)	V _z (m/s)	ω (rad/s)	θ (°)	d (m)
1	0.26	-4.7	-3.0	66.9	0.21
2	0.24	-4.7	-1.4	59.5	0.21
3	0.19	-4.7	-0.8	55.5	0.13
4	0.21	-4.7	-1.1	58.1	0.13
5	0.27	-4.8	-1.0	58.3	0.12
7	0.32	-4.7	-0.6	57.5	0.16
9	0.28	-4.7	-0.7	57.5	0.08
10	0.31	-4.7	-0.8	58.8	0.13
11	0.25	-4.8	-1.2	56.1	0.04
12	0.31	-4.7	-1.7	63.9	0.11
13	0.25	-4.8	-0.2	55.6	0.03
14	0.25	-4.7	-1.2	59.9	0.07
15	0.30	-4.9	-1.0	57.6	-0.005

where V_y = centre of mass horizontal velocity, V_z = centre of mass vertical velocity, ω = angular velocity, θ = trunk angle as orientation angle and d = toe distance from the board tip.

Table 7.2. Initial touchdown variables used in the simulation model in forward 2½ somersault pike dives

Dive	V _y (m/s)	V _z (m/s)	ω (rad/s)	θ (°)	d (m)
1	0.40	-4.9	-2.0	57.7	0.11
2	0.30	-4.9	-1.0	59.0	0.06
3	0.36	-4.9	-0.7	55.9	0.06
4	0.41	-4.9	-1.6	57.4	0.11
5	0.47	-5.0	-1.1	57.8	0.05
7	0.38	-5.0	-1.2	56.6	0.04
9	0.52	-4.9	-1.6	57.4	0.02
10	0.37	-5.0	-1.3	56.2	0.04
11	0.45	-5.0	-0.6	57.1	0.11
12	0.27	-4.8	-1.2	59.6	0.17
13	0.39	-5.0	-0.3	57.5	0.05
14	0.39	-4.9	-1.9	62.1	0.12
15	0.38	-5.0	-1.3	57.0	0.07

7.2.2. Matching simulation

Since Kong's (2005) estimation of the peak vertical springboard reaction force at maximum board depression, from a preliminary matching simulation, was 3600 N, by assuming the foot-spring displacement was 5 mm, the vertical (perpendicular) stiffness was estimated as follows:

$$F_z = -K_z Z$$

$$3600 = -K_z (-0.005)$$

$$K_z = 0.72 \times 10^6 \text{ N/m}$$

Since the heel, ball and toes were in contact with the board, it was assumed that F_z is shared equally between the three points of contact, $1/3$ of K_z equals 0.24×10^6 was initially estimated for each foot-spring. Equally, it was assumed that the stiffness of the parallel foot-springs K_x is similar to the perpendicular stiffness and calculated as follows:

$$K_z = |Z|K_x$$

$$0.24 \times 10^6 \text{ N/m} = (0.005) K_x$$

$$K_x = 48 \times 10^6 \text{ N/m}^2$$

The lower bound (LB) and upper bound (UB) in the matching process were set to be $1/3$ and 3 times of their initial estimates based on the two values K_z and K_x . In addition to vary the damping parameters using the same bounds as the stiffness values, the springboard contact C, ball correction angle, triangular foot angle, initial trunk angle and initial angular velocity were allowed to be varied. To compensate for digitisation error, small flexibility of $\pm 1^\circ$ for initial trunk angle and ± 1 rad/s for initial trunk angular velocity was allowed to minimise the difference between the simulation and performance. A summary of the lower and upper bounds of the parameters that were varied are shown in Table 7.3. During the matching process, the nine parameters were varied until the best match between the simulation and performance was achieved.

Table 7.3. Lower and upper bounds of the parameters in the matching process

Parameters	LB	UB
K_x ($\times 10^6$ N/m ²)	16	144
C_x ($\times 10^6$ Ns/m ³)	16	144
K_z ($\times 10^6$ Ns/m)	0.08	0.72
C_z ($\times 10^6$ Ns/m ²)	0.08	0.72
Springboard contact C (N/m)*	2800	3700
Ball correction angle	+12°	+13%
Triangular foot angle	22°	23°
Initial trunk angle θ_t (x 15 dives)	-1°	+1°
Initial trunk angular velocity ω_t (x 15 dives)	-1 rad/s	+1 rad/s

*according to the equation of board stiffness, $m = 5589$

7.2.3. Objective score

An objective score S_{ANG} was calculated as a measure of the average percentage (%) difference between the simulation and the performance. The average of six equally weighted measures comprised S1: orientation angle, S2: horizontal velocity, S3: vertical velocity, S4: board depression S5: time of contact, and S6: angular momentum.

The orientation angle score S1 was determined as the RMS difference in the trunk angle (θ_t) time history. The horizontal and vertical velocity scores S2 and S3 calculated as the average % difference in CM horizontal V_y and vertical V_z at takeoff. The springboard score S4 was the average of the takeoff time difference and the % difference in maximum vertical board depression. The time of contact score S5 was the RMS difference in the time of takeoff. The angular momentum score S6 determined as the % difference of the angular momentum H at takeoff. A 1° difference in orientation angle and 0.001 s difference in the time of contact was allocated as equivalent to 1% difference in other variables to compare each of the six variables equally. To constrain the foot spring displacements to be within 15 mm in the parallel and 10 mm in the perpendicular directions, penalty scores were allocated such that if the spring deformation exceeded the bounds, 1 mm difference generated 1% penalty point. This was done by applying a function (d) using the Simulated Annealing optimisation algorithm (Corana et al., 1987), minimising the score for each individual dive. The

maximum number of simulations to match the performances was 40000 and it took one day to complete each individual dive separately.

$$d = \frac{S_1 + S_2 + \dots + S_6}{6} + P_1 + P_2$$

7.2.4. Accuracy of the simulation

The root mean square (RMS) was determined to compare the simulation output and video data. Table 7.4 shows the scores of matching the takeoff phase of forward pike (101B) and forward 2½ somersault pike (105B) dives. The simulation of dive 6 and 8 in 101B and dive 7 in 105B were removed from analysis because they didn't match the performance, leaving 13 matching simulations for 101B and 14 for 105B. When the output variables were matched individually, the 13 simulations of 101B and 14 simulations of 105B matched the performance very well with mean scores of 2.7% and 3.1% respectively (Table 7.4). This demonstrates that the simulations closely matched the performances.

Table 7.4. Root mean square differences of the six scores between simulation and video data (S_{ANG} score) in forward pike and forward 2½ somersault pike dives

Dive	101B	Dive	105B
1	1.03	1	5.83
2	3.25	2	4.58
3	3.51	3	0.57
4	1.28	4	2.08
5	1.89	5	1.91
7	2.26	6	2.75
9	1.71	8	2.77
10	1.38	9	3.52
11	1.78	10	1.56
12	2.6	11	2.56
13	5.51	12	4.13
14	4.43	13	1.71
15	4.47	14	4.56
		15	4.95
Mean	2.7		3.1
SD	1.42		1.52

7.3. Perturbation

The variability in the initial conditions of the performances were used to perturb each individual simulation. Since during the contact phase the CM horizontal and vertical and angular velocities are variables which determine dive height, board clearance distance and rotational requirements during the flight phase, the initial touchdown conditions of these three variables were used to vary the simulation. The mean and standard deviation of initial touchdown conditions of the three variables in 101B and 105B are shown in Table 7.5. The standard deviations of the initial conditions 0.040 m/s, 0.069 m/s and 0.418 rad/s were used to perturb the simulations in 101B and the corresponding values of 0.054 m/s, 0.063 m/s and 0.714 rad/s were used to perturb the simulations in 105B. The amount of variability in the simulation outcome was compared with the variability in the original matching simulation to determine the difference between the variability of the rotation potential in the actual performance and that arising from perturbing initial conditions.

Table 7.5. Initial touchdown conditions of CM horizontal and vertical velocities and angular velocity of 13 forward and 14 forward 2½ somersault pike dives

101B				105B			
Dive	V _y	V _z	ω	Dive	V _y	V _z	ω
1	0.26	-4.76	-2.72	1	0.4	-4.93	-1.05
2	0.24	-4.78	-2.26	2	0.3	-4.85	-1.44
3	0.19	-4.77	-1.91	3	0.36	-4.88	0.09
4	0.21	-4.77	-2.16	5	0.41	-4.88	-0.59
5	0.27	-4.84	-2.02	6	0.47	-4.99	-0.10
7	0.32	-4.72	-1.7	7	0.38	-5.01	-1.06
9	0.28	-4.73	-1.72	9	0.37	-4.96	-1.50
10	0.31	-4.75	-1.3	10	0.45	-4.96	0.36
11	0.25	-4.85	-1.77	11	0.27	-4.81	-1.98
12	0.25	-4.83	-2.78	12	0.39	-4.98	0.19
13	0.32	-4.94	-1.67	13	0.39	-4.85	-0.95
14	0.25	-4.79	-2.22	14	0.38	-4.95	-1.52
15	0.30	-4.92	-1.96	15	0.32	-4.94	-0.72
				16	0.34	-4.85	-0.84
Mean	0.265	-4.80	-2.01		0.374	-4.92	-0.794
SD	0.040	0.069	0.418		0.054	0.063	0.714

7.3.1. Individual perturbation

For each individual simulation, the effect of varying each variable on the rotation potential at takeoff were determined. Plus, and minus one standard deviation of initial touchdown horizontal, vertical and angular velocities were used to vary the simulation, giving one perturbation at a time. There are 13 dives for 101B and 14 dives for 105B, giving a total number of 162 simulations ($6 \times 13 + 6 \times 14 = 162$). The rotation potential for each individual simulation shown in Tables 7.6 and 7.7 represent the effect of perturbing the initial conditions by one standard deviation in 101B and 105B respectively. The mean and standard deviation of the original matching simulation (without perturbation) for all 13 dives in 101B were 0.347 ± 0.018 straight somersault (ss) whereas decreasing the horizontal velocity separately by one standard deviation decreased the mean rotation potential to 0.302 ± 0.058 ss and by increasing it by one standard deviation the rotation potential was raised to 0.392 ± 0.040 ss. The mean rotation potential arising from perturbation in vertical velocity obtained 0.364 ± 0.033 ss and 0.335 ± 0.034 ss by varying plus and minus standard deviation respectively. The corresponding values for perturbed angular velocities were 0.370 ± 0.124 ss and 0.296 ± 0.067 ss respectively (Table 7.6). A similar effect occurred for 105B (Table 7.7). These indicate the extent to which each individual perturbation affected the mean rotation potential at takeoff. The standard deviation of rotation potential for each individual perturbation in both 101B and 105B are greater than the variation of the original matching simulation.

Table 7.6. Rotation potential at takeoff obtained in simulation outcome by varying the initial condition using $\pm$ SD of the initial CM horizontal, vertical and angular velocities in forward pike dives

Dive	Matching	V _y		V _z		ω	
		(-SD)	(+SD)	(-SD)	(+SD)	(-SD)	(+SD)
1	-0.332	-0.306	-0.406	-0.405	-0.319	-0.511	-0.395
2	-0.358	-0.316	-0.399	-0.381	-0.339	-0.403	-0.318
3	-0.337	-0.305	-0.371	-0.348	-0.325	-0.371	-0.305
4	-0.358	-0.318	-0.398	-0.372	-0.345	-0.412	-0.307
5	-0.386	-0.359	-0.395	-0.375	-0.417	0.001	-0.318
7	-0.334	-0.356	-0.495	-0.431	-0.308	-0.321	-0.189
9	-0.340	-0.301	-0.310	-0.300	-0.333	-0.341	-0.244
10	-0.331	-0.294	-0.371	-0.347	-0.315	-0.477	-0.268
11	-0.343	-0.304	-0.378	-0.357	-0.331	-0.386	-0.386
12	-0.360	-0.326	-0.382	-0.358	-0.353	-0.355	-0.312
13	-0.321	-0.122	-0.390	-0.331	-0.271	-0.393	-0.171
14	-0.340	-0.299	-0.378	-0.352	-0.330	-0.389	-0.288
15	-0.364	-0.324	-0.418	-0.382	-0.363	-0.455	-0.352
Mean	-0.347	-0.302	-0.392	-0.364	-0.335	-0.370	-0.296
SD	0.018	0.058	0.040	0.033	0.034	0.124	0.067

Table 7.7. Rotation potential at takeoff obtained in simulation outcome by varying the initial condition using $\pm$ SD of the initial CM horizontal, vertical and angular velocities in forward 2½ somersault pike dives

Dive	Matching	V _y		V _z		ω	
		(-SD)	(+SD)	(-SD)	(+SD)	(-SD)	(+SD)
1	-1.275	-1.230	-1.334	-1.299	-1.263	-1.370	-1.200
2	-1.277	-1.231	-0.825	-0.845	-1.259	-0.846	-0.716
3	-1.316	-1.253	-1.392	-1.338	-1.295	-1.561	-1.224
5	-1.342	-1.311	-1.147	-1.143	-1.333	-1.194	-1.022
6	-1.319	-1.208	-1.494	-1.440	-1.310	-1.366	-1.195
7	-1.337	-1.297	-1.275	-1.356	-1.319	-1.409	-1.298
9	-1.372	-0.566	-1.641	-0.591	-1.354	-1.596	-0.534
10	-1.359	-1.304	-1.450	-1.394	-1.336	-1.427	-1.302
11	-1.327	-1.286	-1.377	-1.361	-1.317	-1.171	-1.277
12	-1.313	-1.219	-1.388	-1.358	-1.273	-1.432	-1.130
13	-1.377	-1.319	-1.166	-1.409	-1.354	-1.487	-1.487
14	-1.279	-1.198	-1.463	-1.321	-1.242	-1.200	-1.166
15	-1.349	-1.197	-1.464	-1.404	-1.281	-1.531	-1.128
16	-1.289	-1.224	-1.359	-1.307	-1.271	-1.434	-1.146
Mean	-1.324	-1.203	-1.341	-1.255	-1.301	-1.359	-1.130
SD	0.034	0.188	0.196	0.243	0.037	0.199	0.242

Tables 7.8 and 7.9 show the root mean square difference in rotation potential between each individual perturbation and the original matching simulation for horizontal, vertical and angular velocities. The rotation potentials greater than two standard deviations were removed, as outliers, to minimise the RMS differences. As it can be seen in Tables 7.8 and 7.9, the RMS difference of all variables are greater than the standard deviation of the original matching simulation in both 101B and 105B. The vertical velocity and angular velocity had the smallest and greatest effect on the rotation potential respectively. The root mean square difference of the SDs associated with all individual perturbation together obtained 0.082 ss and 0.211 ss in 101B and 105B respectively, indicating that the effect of individual perturbation on the rotation potential is (4½ times) greater than the original matching performance (0.018 ss) in 101B. The corresponding values in 105B was 0.211 which more than 6 times greater than the original matching performance (0.034 ss).

For instance, in 101B increasing horizontal velocity by one standard deviation 0.04 m/s arose the rotation potential by 0.036 ss whereas the variability in the rotation potential across all 15 dives was 0.018 ss, demonstrating that if no correction is made during the contact phase changing one variable only will significantly increase the variability at takeoff. In other words, varying the initial conditions of contact phase makes a great change in the rotation potential at the end of the takeoff.

Table 7.8. Root mean square difference between rotation potential obtained in each individual perturbation and the original matching simulation of 13 forward pike dives

Dive	V _y		V _z		ϑ		RMS**
	-SD	+SD	-SD	+SD	-SD	+SD	
1	-0.026	0.074	>SD*	-0.013	0.179	0.063	
2	-0.042	0.041	0.023	-0.019	0.045	-0.040	
3	-0.032	0.034	0.011	-0.012	0.035	-0.032	
4	-0.040	0.040	0.013	-0.013	0.053	-0.051	
5	-0.028	0.009	-0.012	0.031	>SD*	-0.068	
7	0.022	>SD*	>SD*	-0.026	-0.013	>SD*	
9	-0.038	-0.030	-0.040	-0.007	0.001	-0.096	
10	-0.036	0.040	0.016	-0.015	0.146	-0.062	
11	-0.040	0.035	0.014	-0.012	0.042	0.042	
12	-0.034	0.022	-0.003	-0.008	-0.005	-0.048	
13	>SD *	0.069	0.010	-0.050	0.073	>SD*	
14	-0.041	0.038	0.011	-0.010	0.049	-0.052	
15	-0.040	0.053	0.018	-0.002	0.091	-0.012	
rmsd	0.036	0.044	0.018	0.021	0.080	0.056	0.082
% diff	-10	-13	-5	-6	-23	-16	
*removed because they were greater than 2 standard deviations							
**perturbation effect as one single RMS for the effect of all three variables							

Table 7.9. Root mean square difference between rotation potential obtained in each individual perturbation and the original matching simulation of 14 forward 2½ pike dives

Dive	V _y		V _z		ϑ		RMS**
	-SD	+SD	-SD	+SD	-SD	+SD	
1	-0.046	0.059	0.024	-0.013	0.095	-0.075	
2	-0.046	>SD*	-0.432	-0.018	>SD*	>SD*	
3	-0.063	0.076	0.021	-0.021	0.245	-0.092	
5	-0.031	-0.195	-0.198	-0.009	-0.147	-0.319	
6	-0.111	0.175	0.121	-0.009	0.046	-0.125	
7	-0.039	-0.062	0.020	-0.017	0.072	-0.038	
9	>SD*	0.269	>SD*	-0.018	0.224	>SD*	
10	-0.054	0.091	0.035	-0.023	0.068	-0.057	
11	-0.042	0.049	0.034	-0.010	-0.156	-0.050	
12	-0.094	0.075	0.044	-0.040	0.119	-0.184	
13	-0.058	-0.210	0.032	-0.022	0.110	0.110	
14	-0.081	0.185	0.042	-0.037	-0.079	-0.113	
15	-0.152	0.115	0.055	-0.069	0.182	-0.222	
16	-0.064	0.071	0.018	-0.018	0.145	-0.142	
rmsd	0.075	0.143	0.139	0.028	0.142	0.149	0.211
% diff	-6	-11	-11	-2	-11	-11	
*removed because they were greater than two standard deviations							
**perturbation effect as one single RMS for the effect of all three variables							

7.3.2. Combined perturbation

For each individual simulation, eight combinations of plus and minus one standard deviation was used to perturb the initial conditions, giving three perturbations at a time. The three perturbations are in order of horizontal velocity, vertical velocity and angular momentum in each combination. The eight combinations depicted in Tables 7.10 and 7.11 represent the root mean square (RMS) difference between the rotation potential arising from perturbing the initial conditions and the rotation potential in the original matching simulation. To minimise the RMS difference, values greater than two standard deviations were removed. For instance, increasing all three variables at once by one standard deviation gave an RMS difference of 0.043 ss and 0.115 ss in 101B and 105B respectively which are the combined effect of increasing horizontal, vertical and angular velocities at once on the rotation potential.

Table 7.10. Root mean square difference between rotation potential obtained in each combined perturbation of initial touchdown conditions and the original matching simulation of 13 forward pike dives

Dive	+++	+--	+-+	++-	-++	-+-	--+	---	RMS
1	0.054	0.185	>SD*	0.209	-0.008	0.191	0.055	0.153	
2	-0.020	0.083	0.001	0.066	-0.079	0.015	-0.064	0.029	
3	-0.003	0.094	0.025	0.066	-0.089	-0.019	-0.070	0.007	
4	-0.021	0.102	-0.001	0.076	-0.100	0.004	-0.081	0.035	
5	-0.028	>SD*	-0.070	>SD*	-0.047	>SD*	-0.120	>SD*	
7	>SD*	0.188	-0.069	0.128	>SD*	-0.066	>SD*	-0.035	
9	-0.063	0.034	-0.057	0.007	-0.145	-0.038	-0.142	-0.032	
10	-0.042	0.199	-0.009	0.186	-0.109	0.094	-0.085	0.090	
11	-0.030	0.142	-0.012	0.102	-0.095	-0.024	-0.079	0.004	
12	-0.022	0.035	-0.018	0.025	-0.108	-0.034	-0.081	-0.034	
13	-0.098	0.118	-0.045	0.086	>SD*	-0.004	>SD*	0.016	
14	-0.026	0.094	-0.029	0.076	-0.098	0.005	-0.078	0.021	
15	-0.001	0.129	0.075	0.139	-0.125	0.055	-0.033	0.044	
rmsd	0.043	0.128	0.043	0.113	0.098	0.069	0.085	0.057	0.085
%diff	-12	-37	-13	-33	-28	-20	-25	-17	

*removed because they are greater than two standard deviations

Table 7.11. Root mean square difference between rotation potential obtained in each combined perturbation of initial touchdown conditions and the original matching simulation of 13 forward pike dives

Dive	+++	+--	+-+	++-	-++	-+-	--+	---	RMS
1	-0.045	0.196	-0.010	0.175	-0.139	0.196	-0.106	0.050	
2	>SD*	-0.390	>SD*	>SD*	-0.170	-0.390	-0.124	-0.450	
3	-0.067	0.307	-0.015	0.274	-0.163	0.307	-0.128	0.184	
5	-0.294	-0.090	-0.262	-0.121	-0.372	-0.090	-0.351	-0.177	
6	-0.076	0.136	-0.048	0.125	-0.231	0.136	-0.180	-0.014	
7	-0.011	0.069	0.073	0.086	-0.145	0.069	-0.074	0.063	
9	>SD*	0.288	>SD*	0.249	>SD*	0.288	>SD*	0.192	
10	-0.014	0.181	0.019	0.135	-0.126	0.181	-0.095	0.020	
11	-0.024	-0.144	0.024	-0.127	-0.135	-0.144	-0.113	0.173	
12	-0.092	0.190	-0.147	0.149	-0.341	0.190	-0.286	0.084	
13	-0.055	-0.118	-0.015	-0.154	-0.301	-0.055	-0.255	0.084	
14	-0.084	-0.015	-0.043	-0.061	-0.178	-0.015	-0.146	-0.103	
15	-0.172	0.261	-0.105	0.199	-0.310	0.261	-0.268	0.150	
16	-0.098	0.240	-0.050	0.200	-0.212	0.240	-0.185	0.093	
rmsd	0.115	0.212	0.098	0.168	0.217	0.210	0.196	0.169	0.178
%diff	-9	-16	-7	-13	-16	-16	-15	-13	
*removed because they are greater than two standard deviations									

The estimates of combined perturbations gave 2½ and 3½ times greater variability than in the whole rotation potential 0.018 ss and 0.034 ss in the original matching simulations (actual performances) in 101B and 105B respectively. The root mean square difference of the all eight combined perturbations for 13 simulations in 101B obtained 0.085 ss and for 14 simulations in 105B was 0.178 ss (Tables 7.10 and 7.11). This demonstrates the validity of the perturbation process in (Analysis 2) since it is comparable with the root mean square 0.082 ss and 0.211 ss obtained in the individual perturbation (Analysis 1) in 101B and 105B respectively (Tables 7.8 and 7.9), indicating that there were no errors in the perturbation processes.

7.3.3. Lowest point perturbation

The board contact phase can be subdivided into two phases, the board depression and board recoil phase. At maximum board depression where the board is in its lowest point, the vertical velocity is zero. The lowest point conditions were perturbed to investigate the effect of varying the initial recoil conditions on the rotation potential at takeoff. For each individual simulation, four combinations of plus and minus one standard deviation of horizontal and angular velocities at maximum board depression were used to perturb the simulation. The values greater than two standard deviations were removed to avoid the effect of outliers. Tables 7.12 and 7.13 show the effect of varying the lowest point conditions on the rotation potential at takeoff. The root mean square difference of the all four combined perturbations for 13 simulations in 101B and 14 simulations in 105B were 0.076 ss and 0.127 ss respectively. This demonstrates that the effect of varying the initial recoil conditions on the rotation potential is also greater than the variability in the actual rotation potential in both 101B and 105B, which are 0.018 ss and 0.034 ss respectively.

Table 7.12. Root mean square difference between rotation potential obtained in each combined perturbation of initial lowest point and the original matching simulation of 13 forward pike dives

Dive	++	+-	-+	--	RMS
1	0.007	0.101	0.137	0.101	
2	0.011	0.054	0.096	0.054	
3	-0.074	0.008	0.068	0.008	
4	-0.029	0.028	-0.059	0.028	
5	-0.437	-0.368	0.005	-0.368	
7	0.165	0.047	0.105	0.047	
9	-0.065	-0.070	-0.025	-0.070	
10	-0.067	-0.003	0.066	-0.003	
11	-0.034	0.076	0.127	0.076	
12	-0.131	0.000	0.048	0.000	
13	-0.178	-0.171	0.088	-0.171	
14	-0.076	-0.018	0.051	-0.018	
15	-0.038	0.055	0.119	0.055	
rmsd	0.091	0.071	0.068	0.071	0.076
% diff	-26	-20	-20	-20	
Vx at lowest point					
*removed as greater than 2 standard deviations					

Table 7.13. Root mean square difference between rotation potential obtained in each combined perturbation of initial lowest point and the original matching simulation of 14 forward 2½ pike dives

	++	+-	-+	--	RMS
1	0.0144	0.0967	-0.072	0.0181	
2	-0.552	-0.436	-0.51	-0.486	
3	0.1003	0.1117	-0.047	-0.014	
5	-0.183	-0.218	-0.247	-0.28	
6	0.0676	-0.064	-0.064	-0.163	
7	0.0323	-0.045	-0.003	0.0772	
9	-0.733	0.2873	-0.828	-0.801	
10	0.0761	0.099	-0.076	-0.04	
11	0.033	0.0691	-0.074	-0.045	
12	0.0415	0.0962	-0.16	-0.035	
13	-0.195	-0.193	-0.24	-0.247	
14	0.0297	0.2001	-0.111	-0.03	
15	0.0542	0.1551	-0.171	-0.041	
16	0.0063	0.1294	-0.122	-0.011	
rmsd	0.091	0.152	0.136	0.123	0.127
% diff	-7	-11	-10	-9	

7.4. Actual performance

The mean and standard deviation of the angles, velocities and takeoff time represented in Table 7.14 show that the diver was consistent at touchdown and takeoff. The mean orientation angles indicate that the body was near the vertical and consistent at touchdown in both 101B and 105B but, the diver leaned 40.2° forward at takeoff in 105B. Although the mean and standard deviation of the angles and CM velocities at touchdown are comparable in both 101B and 105B, there is greater range of movement in 101B.

Table 7.14. Mean and standard deviation of the angles at touchdown and takeoff, flight time and CM velocity in contact phase of 15 forward and 15 forward 2½ somersault pike dives

Variable	Dive	Touchdown	Takeoff
Hip (°)	101B	92.2 ± 4.9	168.2 ± 3.3
	105B	94.1 ± 3.1	109.2 ± 5.0
Knee (°)	101B	98.6 ± 2.6	178.9 ± 0.8
	105B	99.0 ± 2.2	178.8 ± 1.8
Trunk (°)	101B	59.2 ± 3.1	72.4 ± 2.0
	105B	57.9 ± 1.6	18.0 ± 4.2
Arm elevation (°)	101B	-32.5 ± 5.1	178.7 ± 4.5
	105B	-16.0 ± 3.1	121.4 ± 7.5
Orientation (°)	101B	-12.2 ± 1.3	13.1 ± 1.3
	105B	-10.8 ± 1.0	40.2 ± 1.8
Horizontal velocity	101B	0.25 ± 0.047	0.86 ± 0.06
	105B	0.31 ± 0.03	1.06 ± 0.07
Vertical velocity	101B	-4.8 ± 0.05	5.84 ± 0.08
	105B	-4.9 ± 0.036	5.19 ± 0.08
Takeoff time	101B	446 ± 9.42	
	105B	421 ± 10.12	

Figure 7.2 shows the angles variability standard deviation time history during the takeoff phase of 15 forward and 15 forward 2½ somersault pike dives. The average time of the takeoff phase was used as a common end point for the joint angle time histories. The average time of maximum board depression was 0.217 s. The variation in the orientation angle reached a peak at 0.176 s and decreased by the end of the takeoff in 101B whereas it reached a peak at the end of the takeoff phase in 105B. The peak variation of knee and hip angles appeared at about 0.256 and 0.264 s in 101B and 0.252 s and 0.356 s in 105B, indicating that the peak variations occurred in the recoil phase and decreased by the end of takeoff in both 101B and 105B.

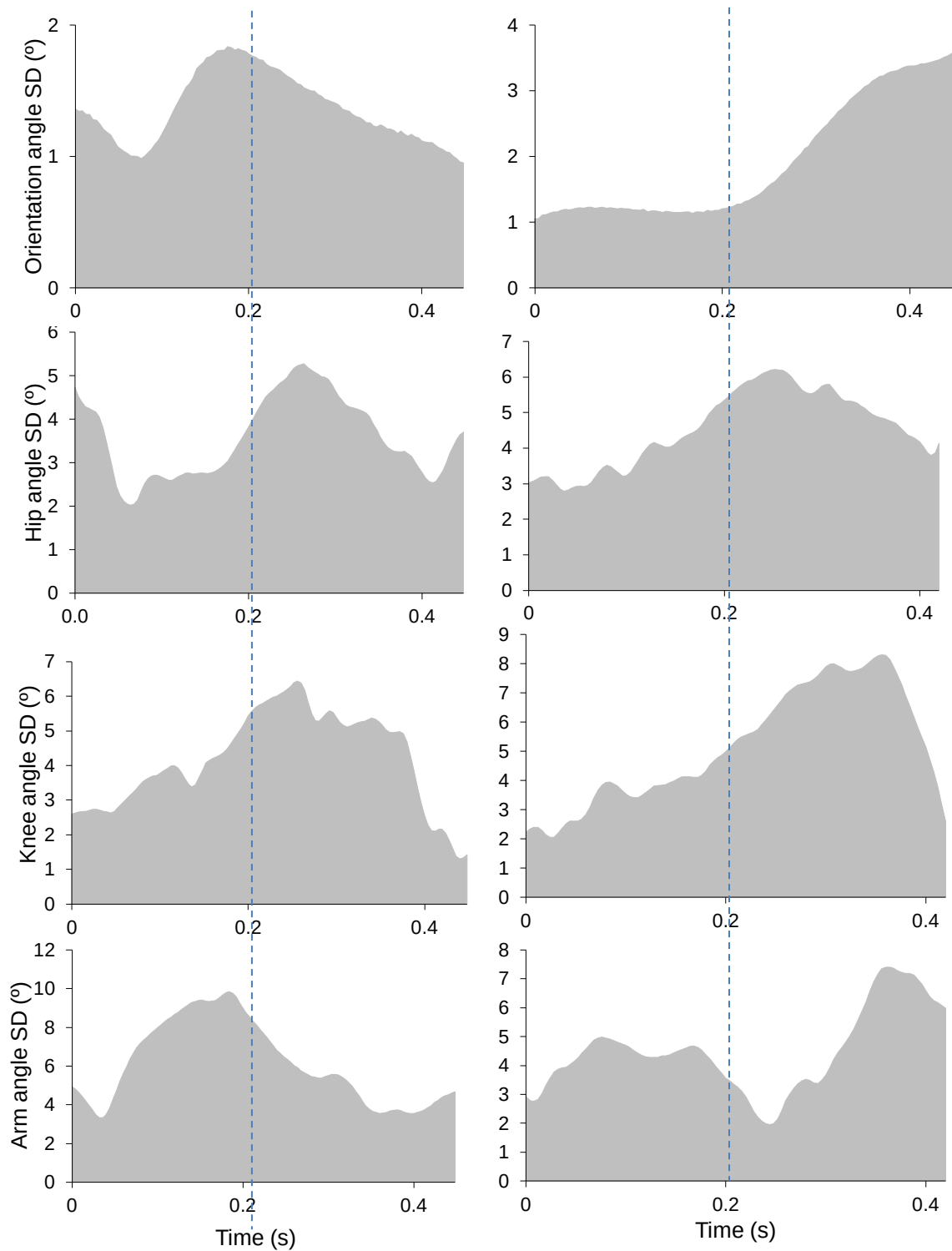


Figure 7.2. Standard deviation time history of orientation, hip, knee and arm angles during the takeoff phase of 15 forward and 15 forward 2½ somersault pike dives. The dash-line is the average time at maximum board depression.

7.5. DISCUSSION

The aim of this chapter was to investigate whether the diver made any correction during the contact phase in 1-m springboard forward pike dives. The mean score between the simulation and performance in forward pike and forward 2½ somersault pike dives obtained 2.7° and 3.1° respectively, demonstrating that the simulation model closely matched the performance. Two simulation analyses were carried out to examine whether movement variability during the contact phase control the rotation potential at takeoff. In the first analysis, the effect of varying the initial touchdown conditions on the rotation potential at takeoff was investigated. In the second analysis, the effect of varying the initial recoil conditions on the rotation potential at takeoff was examined. The perturbation of mass centre horizontal and vertical velocities and angular velocity at touchdown in 101B in analysis 1 gave an RMS difference of rotation potential 0.085 ss at takeoff (Table 7.10) and in analysis 2 the variation arising from maximum board depression (lowest point) gave an RMS difference of 0.076 ss in rotation potential at takeoff (Table 7.12), the corresponding values at takeoff in 105B were 0.178 ss at takeoff arising from touchdown (Table 7.11) and 0.127 ss arising from the lowest point (Table 7.13). These compare with the variability of the rotation potential at takeoff in the actual performances of 0.018 ss and 0.034 ss in 101B and 105B respectively, indicating that the variation obtained in the simulation outcome is much greater than variability in the actual performances, demonstrating that the diver made adjustment during the contact phase.

In 101B, the amount of variation arising from perturbation at lowest point 0.076 ss (Table 7.12) is comparable with the variability arising from touchdown conditions 0.085 ss (Table 7.10). Corresponding values in 105B 0.127 ss (Table 7.13) and 0.178 ss (Table 7.11) are similar. These indicate that most of the corrections have been made during the recoil phase. The peak variation in the hip and knee angles appeared in the second half of takeoff phase in both 101B and 105B. This means that hip and knee angles were varied during the recoil phase to make adjustment and reduce the amount of rotation potential variability at takeoff. The variability in the orientation angle decreased in this period in 101B, indicating that the hip and knee were varied to correct the body orientation. Conversely, the variability in the body orientation angle increased

during the recoil phase in 105B. This may suggest that more correction is required for more difficult dives.

The rotation potential variability at takeoff obtained from varying the initial conditions was much greater than the variability in the original matching simulation. This demonstrates that although the divers are consistent at the start and the end of each phase, their configurational changes are different each time they perform the same dive. This means the coordination during the takeoff phase is dependent on the initial conditions at touchdown and the maximum board depression. If the diver makes an error at touchdown there is about 0.2 s to notice the error at maximum board depression and 0.2 s to execute the correction. Most of the correction was made during the recoil phase and the peak variation in the hip and knee angles appeared in the second half of takeoff phase, suggesting that the diver noticed any error at the maximum board depression at about 0.2 s and varied his hip and knee angles during the recoil phase to correct the rotation potential at takeoff. This shows that the time of contact phase is enough to use feedback control which is supported by Schmidt (1975) that feedback control is involved in movement longer than 0.2 s.

It has been reported that the triggered reactions are faster than reaction time and there is no need for conscious processing and selection of a new movement program. If anything goes wrong during a movement, the performers just trigger off the reaction as if it was automatic. This would place the triggered reaction into the category of corrections for errors in performance (Schmidt and Lee, 1999). Since most of correction has been made during the recoil phase, the board depression phase may be feedforward and feedback correction is used in the second half of takeoff phase to correct body configuration leading to decrease the variability of the rotation potential at takeoff. This is supported by McNitt-Gray et al. (2001) who reported that feedforward control is used to stabilise the joints and satisfy the mechanical demand imposed on the lower extremity after contact during landing tasks.

Once the diver contacts the diving board at touchdown, the instant of the toe's contact on the board could be defined as sensory input for the neuro-musculo-skeletal system in which to respond during the contact phase and control the takeoff conditions by

feedback using initial touchdown conditions. Since the time of board depression is about 200 ms, it can be justified that the feedback from the initial touchdown conditions was used to correct the movement during the board recoil phase.

Kong (2005) found the average of 4.1% difference between the simulation and performance were similar to the averages of 2.7% and 3.1% difference between simulation and performance for 101B and 105B respectively (Table 7.4) found in this study. The execution of adjustment found in this study is also in agreement with Kong (2005) who demonstrated that a diver can perform the same dive with slightly different techniques and make corrections to compensate for any mistakes during the takeoff, flight, and/or entry.

In gymnastics, it is concluded that low variability is expected in critical aspects of movement and high variability is expected in less important aspects of movement. This can be understood that making adjustment is performed in the less important aspects of technique to ensure there is low variation in the most important aspects of movement (Hiley et al., 2013). Relatively high configurational variability in dive takeoff phase (Figure 7.2) indicates that correction is executed during the board contact to ensure that there is low variation at takeoff.

In the triggered reaction, the original goal is not necessarily to achieve the trajectory of the limbs with high precision, it is essential to achieve low variability in the most important aspects of the movement, indicating that the responsibility of the triggered reaction is correcting the error not only by changing the trajectory in performance but also in the selection of the program (Schmidt and Lee, 1999). If there is any variability in joint torques in the takeoff phase, the triggered reaction might have chosen a different trajectory to achieve the desired outcome. The diver might have used the triggered reaction for small perturbation at touchdown to bring the limb back on the original trajectory at board maximum depression, and a new trajectory is selected without reprogramming the movement to run off as a triggered reaction for a large perturbation at maximum board depression as the original goal is the rotation potential not the original trajectory. The preliminary position of the body and arms before the springboard begins its recoil phase is common to all takeoffs regardless of the rotation

or the number of somersaults but the difference between different dives begins before the recoil phase begins (O'Brien, 1992). The effect of variation on the rotation potential arising from perturbing the conditions at maximum board depression was comparable with the variability arising from the initial touchdown conditions, indicating that the main adjustment was made during the recoil phase. This may indicate that different trajectory without reprogramming the movement is selected to run off for required rotational movement during the recoil phase.

Although the divers should concentrate on the board tip throughout the approach, the focus of the eyes should be shifted to the far side of the pool in front prior to landing on the board and they must not watch the feet or the board tip in this period (O'Brien, 1992). It was reported that the maintenance of balance during the board depression and the body orientation at takeoff were the springboard divers' greatest concerns. In addition, divers acquired self-confidence faster in performing dives with lower degree of difficulty (Slobounov et al.,1997). The small adjustments during the board depression may be responsible for maintaining balance and the main correction during the recoil phase may be associated with the body orientation at takeoff for easier dives and angular momentum for more difficult dives.

The hip angle in takeoff phase was identified as the most important and invariable kinematic parameter in springboard diving (Slobounov et al.,1997). On the other hand, the consistency in the whole body angular orientation in takeoff determines the quality of the dive (Golden, 1981; Miller, 1974; Obrien, 1992). It was hypothesised that elite divers are able to determine and apply the best efficient strategies to successively freeze or free the biomechanical degrees of freedom in order to deliver the task demand (Slobounov et al.,1997).

In a normal jump, the arms swing upward as the legs extend to thrust the body into the air which is a natural jumping motion. In contrast, during all diving takeoffs the arms swing upward before the legs extend to begin the jump, making the arm timing so difficult to achieve due to unnatural jumping movement (O'Brien, 1992). Although the peak variability in the hip and knee in the second half of contact phase (Figure 7.2) are associated with the execution of adjustment, the peak variation in the arm angle during

the board depression may be due to maintaining balance due to unnatural jumping motion during board depression phase.

Removing the values greater than two standard deviations in the simulation analyses was to make the RMS values as small as possible to give the best chance to reduce the error and make the rotation potential at takeoff as close to the actual performance as possible.

Although the variability of orientation and arm angles decrease during the recoil phase in 101B, the corresponding values increase in 105B, the execution of more angular momentum may be responsible for greater variability in more difficult dives.

Figure 7.2 showed that the diver varied his hip and knee angles during the recoil phase to minimise the orientation variability at takeoff. The peak variation of orientation angle at takeoff in 105B might be due to the generation of angular momentum and the requirement to move the hands towards the knees for execution of $2\frac{1}{2}$ somersault whereas there is less angular momentum in 101B. The peak variation of the hip and knee appeared in the second half of contact phase in both 101B and 105B, indicating that there is greater variability during the recoil phase.

7.6. Summary

A subject-specific angle driven model of a diver and springboard (Kong, 2005) was applied to investigate whether the diver makes any correction during the contact phase in 1-m springboard forward pike dives. The mean score between the simulation and performance in forward pike and forward 2½ somersault pike dives obtained 2.7° and 3.1° respectively, indicating that the simulation model closely matched the performance. Two simulation analyses were carried out to determine the effect of varying the initial touchdown and initial recoil conditions on the rotation potential at takeoff. The variation obtained in the simulation outcome was much greater than variability in the actual performances, demonstrating that the diver made adjustment during the contact phase. The amount of variability arising from initial touchdown conditions was comparable with the variation arising from initial recoil conditions, indicating that the most of corrections have been made during the recoil phase. The results of the angle variability showed that hip and knee angles were varied during the recoil phase to make adjustment and reduce the amount of rotation potential at takeoff.

CHAPTER 8

VARIABILITY AND CONTROL IN FLIGHT PHASE

8.1. Introduction

8.1.1. Motor control

A competitive diver starts the hurdle by jumping from an active leg (drive leg) to achieve a hurdle height, contacts the diving board at the end of the hurdle flight using both feet to depress the board. During the board recoil phase, most of the potential energy is converted to the kinetic energy to generate velocity and angular momentum and project the diver from the end of the board into the air to achieve a dive height, giving time to perform the somersault rotation, coming out of pike and finally entry into the water (Figure 8.1). While the location of the centre of mass during the flight phase and velocity at takeoff are determined, the amount and type of rotation of the body is largely under the control of the performer since the somersault rotation is governed by the conservation of angular momentum in cases where the effects of air resistance can be neglected (Yeadon, 2015). Thus the balance between the moment of inertia and angular velocity is responsible for any change or adjustment in this phase.

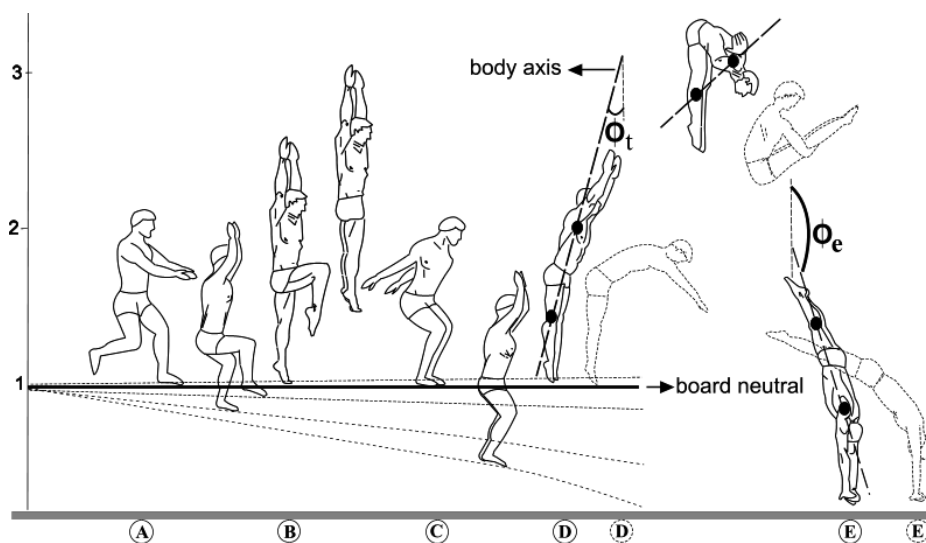


Figure 8.1. 1-m springboard forward pike (solid line) and forward 2½ somersault pike (dotted line) dives. Phases: hurdle contact (A-B) and hurdle flight (B-C), contact (C-D), flight (D-E). Φ_t = initial body orientation angle at takeoff at board neutral position, Φ_e = entry orientation angle.

Movements with longer duration (longer than reaction time) may be performed under feedback control (Klapp, 1975). According to the reaction time which is the time interval between the presentation of a stimulus and the beginning of the response, the typical minimum simple reaction time ranges from 0.13 s to 0.2 s, indicating that if there is any error in the initiation of movement, it takes about one reaction time to start correcting the movement (Schmidt & Lee, 1999). The average time of dive flight phase in forward pike and forward $2\frac{1}{2}$ somersault pike dives were 1.349 s and 1.221 s respectively (Table 5.3) indicate that the control strategy during the flight phase may be associated with feedforward and feedback correction.

Once the diver leaves the diving board at takeoff, the instant of losing contact with the board could be defined as sensory input for the neuro-musculo-skeletal system in which to respond during the flight phase, using feedback from initial takeoff conditions, to control the aerial movement in order to achieve required entry conditions. In the case of aerial movements, a single correction might have some effect on the other aspects of the movement at the end of the phase and the problem is aggravated by the inherent feedback system delay which can be up to 0.1 s to 0.2 s for long loop / triggered and voluntary responses. Any correction in the air therefore has to be made based on the state of the mechanical system at a previous time. As a consequence, there will be limits to the level of adjustment that can be made in the aerial phase (Latash, 1998; 2008; Yeadon and Hiley, 2014).

Both feedforward and feedback control strategies are used in acrobatic aerial movements, such that takeoff and the early part of flight phase use solely feedforward control (or preplanned) whereas feedback control is used to modify the planned feedforward movement in the later stages of the aerial phase in order to make configuration changes to control rotational movement (Yeadon and Hiley, 2014). Although, this might suggest that feedback from takeoff conditions is used to control the flight phase, there is no evidence reported whether there is any compensation for variation at takeoff and what kind of information processing (feedforward or feedback) is used to control the aerial movement in springboard diving.

During the aerial phase when the landing area is viewed late, there will be limited time to perform hip flexion or extension to adjust the somersault rate and therefore the

somersault orientation before landing, indicating that a single adjustment is likely to be made. On the other hand, however, asymmetrical arm movements using continual feedback throughout the flight phase may control the build-up of twist in an unstable non-twisting straight double somersault (Yeadon, Mikulcik, 1996).

The divers' s motor outcomes often seem very repeatable and stereotyped but, the existence of variation in their orientation and configuration is inevitable. The variability of the diver's movement may be viewed as an unwanted source of error which needs to be minimised or eliminated to improve accuracy. However, the diver's adjustment during the performance may appear as a deliberate source of variation to obtain an optimal output. It is hypothesised that divers make adjustment during the flight phase to cope with variation in the initial conditions and modify the joint coordination pattern in order to minimise the body orientation variability at entry into the water.

One crucial aspect for success in competitive springboard diving is to achieve the correct whole-body orientation at entry to the water. For elite divers, it is to be expected that they will be very consistent at takeoff so that the initial conditions of the flight phase will have low variability. The variability at entry will be dependent upon the variability of the initial conditions and the variability of the configuration changes during flight arising from planning and execution errors. Additionally, there is the possibility of making adjustments to body configuration during flight in order to achieve the correct entry orientation. The aim of this chapter is to investigate the extent to which configuration changes (technique) during the flight phase control the entry angle in 1-m springboard forward dives.

8.2. Method

An 11-segment computer simulation model of Yeadon et al. (1990b) was applied (Chapter 4, section 4.1.2) to investigate the effect of variability arising from initial takeoff conditions and configuration during the flight phase on orientation angle at entry in 1-m springboard forward pike and forward 2½ somersault pike dives. Body orientation and the time histories of the joint angles were obtained from video analysis (Chapter 3). Segmental inertia values were determined (Chapter 4) from

anthropometric measurements (Appendix 2) on a diver using Yeadon's (1990a) mathematical inertia model of the human body.

The angular momentum values for each movement as described in Yeadon (1990c) were calculated using the segmental inertia values and the time histories of the 3 orientation angles and 14 joint angles describing the relative movements of the 11 segments in the aerial phase. The angular momentum, initial vertical velocity and joint angle time histories in the flight phase were used as input into the model. The model output comprised the orientation angle time histories during the simulation of 15 forward pike (101B) and 15 forward 2½ somersault pike (105B) dives.

8.2.1. Normalizing the angular momentum

Angular momentum was normalised to give the number of straight somersault and rotations in revolutions. The angular momentum values were divided by the moment of inertia of the body about the lateral axis for a straight body configuration. This makes the unit of angular momentum as the number of straight somersaults per second (ss/s). Rotation potential (Chapter 3) gives the equivalent number of straight somersaults and this equals to the normalised angular momentum multiplied by the flight time.

Table 8.1 displays the flight time, the normalised and SI values of the angular momentum, rotation potential and entry orientation angle of 101B and 105B. For each movement, the initial angular momentum, the segmental inertia parameters, the initial values of the orientation angles, and the time histories of the configuration angles were input into the Yeadon's et al. (1990b) computer simulation model. The model calculated the time histories of the orientation angle during the simulation.

Table 8.1. Entry orientation angle, rotation potential, normalised and SI values of the angular momentum and flight time of 101B and 105B

Trial	Dive	Flight time (s)	Angular Momentum		Rotation potential	Entry
			Normalised (ss/s)*	SI (kg.m ² /s)		
1	101B	1.340	0.275	15.6	0.210	161.0
	105B	1.232	1.13	66.6	0.898	131.4
2	101B	1.352	0.291	17.1	0.230	165.3
	105B	1.232	1.188	69.8	0.941	135.7
3	101B	1.348	0.275	15.8	0.213	163.6
	105B	1.248	1.211	70.1	0.945	132.6
4	101B	1.356	0.315	17.5	0.235	167.6
	105B		N/A			
5	101B	1.364	0.340	18.9	0.254	166.6
	105B	1.232	1.181	70.0	0.944	136.3
6	101B	1.352	0.318	18.1	0.244	170.4
	105B	1.240	1.136	66.7	0.899	138.1
7	101B	1.344	0.293	16.0	0.215	166.6
	105B	1.208	1.182	71.1	0.958	127.9
8	101B	1.320	0.326	18.4	0.247	163.7
	105B	1.220	1.225	72.7	0.980	127.4
9	101B	1.356	0.300	16.2	0.218	164.8
	105B	1.228	1.207	71.3	0.961	129.4
10	101B	1.348	0.279	16.0	0.216	162.6
	105B	1.216	1.169	69.9	0.943	133.9
11	101B	1.360	0.296	16.5	0.222	164.3
	105B	1.200	1.164	70.8	0.955	123.3
12	101B	1.320	0.309	17.3	0.233	163.9
	105B	1.240	1.187	69.7	0.940	123.9
13	101B	1.356	0.285	16.2	0.218	163.6
	105B	1.208	1.178	70.9	0.956	132.7
14	101B	1.352	0.303	16.9	0.227	164.6
	105B	1.204	1.115	67.6	0.911	132.5
15	101B	1.368	0.315	17.1	0.230	167.0
	105B	1.212	1.174	70.3	0.948	129.6
16	105B	1.204	1.125	68.3	0.921	123.1
Mean	101B	1.349	0.301	16.9	0.23	165.0
	105B	1.222	1.172	69.7	0.94	130.5
SD	101B	0.014	0.019	1.1	0.01	2.3
	105B	0.015	0.033	1.7	0.02	4.7
Coefficient of variation (%)	101B	1.020	6.363	6.4	0.06	14
	105B	1.261	2.781	2.4	0.02	4

*ss/s = straight somersault per second

8.2.2. Tuning angular momentum

After running the simulation, the amounts of the angular momenta obtained from all dives were not satisfactory. Although the initial orientation angles of simulation and video data were exactly the same, there was a discrepancy in the final orientation angle. The angular momentum was adjusted slightly to produce the correct orientation angle at entry so that the simulation rotates the same amount as the performance.

The amount of change in rotation was calculated as a percentage to determine the extent to which the angular momentum needed to be adjusted for each dive. Table 8.1 shows the adjusted angular momentum for each dive with the corresponding value of entry angle in simulation of all dives.

8.2.3. Rotation potential

A linear and stepwise multiple regression analysis were carried out to investigate the relationship between the rotation potential, average hip angle, minimum hip angle and average arm angle in 101B and 105B.

Table 8.2 shows the mean and standard deviation of the rotation potential in 101B and 105B. The average rotation potential is 0.31 straight somersaults for 101B compared to 1.15 straight somersaults for 105B. Table 8.3 shows the average and minimum hip and arm angles in each dive in 101B and 105B.

Table 8.2. Mean and standard deviation of the rotation potential in straight somersault in 101B and 105B

Rotation potential (straight somersault)		
Dive	101B	105B
1	0.282	1.107
2	0.312	1.160
3	0.288	1.179
4	0.319	1.164
5	0.347	1.116
6	0.331	1.158
7	0.290	1.196
8	0.327	1.181
9	0.297	1.147
10	0.291	1.146
11	0.303	1.166
12	0.308	1.155
13	0.297	1.097
14	0.307	1.149
15	0.315	1.109
Mean	0.308	1.149
SD	0.018	0.029

Table 8.3. Mean and minimum hip and arm angles in 101B and 105B

	Hip				Arm			
	101B		105B		101B		105B	
	Mean	Minimum	Mean	Minimum	Mean	Minimum	Mean	Minimum
1	99.2	32.2	62.5	25.9	110.1	60.3	70.6	37.8
2	109.1	49.3	65.5	28.9	99.8	45.4	68.1	40.7
3	102.8	34.1	67.4	24.9	104.7	54.3	70.2	33.9
4	111.4	33.0	65.6	25.2	102.6	58.5	70.9	37.2
5	124.9	35.7	61.4	24.1	101.6	62.4	70.5	33.9
6	115.3	38.1	68.0	26.1	106.7	55.7	68.0	36.0
7	97.4	39.2	71.2	31.6	107.0	54.0	70.0	41.0
8	119.9	37.4	67.8	29.8	101.2	57.8	69.4	38.5
9	105.2	36.6	64.5	27.3	102.9	52.6	69.7	40.4
10	103.2	32.9	64.2	30.5	105.9	51.9	67.9	36.2
11	110.5	31.8	65.8	27.7	104.4	59.1	70.6	36.5
12	112.5	34.6	64.7	26.2	105.2	59.8	68.6	35.4
13	105.2	36.2	59.2	24.8	99.0	45.5	69.6	39.9
14	109.5	34.0	65.8	24.5	102.6	57.0	67.3	36.4
15	112.5	29.6	60.5	26.9	105.2	61.0	72.2	41.2
Mean	109.2	35.6	64.9	27.0	103.9	55.7	69.6	37.7
SD	7.5	4.6	3.1	2.3	3.0	5.2	1.4	2.5

8.2.4. Matching performances

The accuracy of the simulation output was evaluated by data obtained from the actual performance. The orientation angle time histories determined from the simulation were compared with the values obtained from the video analysis. Figures 8.2 and 8.3 show the orientation angle time histories of matching the aerial phase of 15 forward pike and 15 forward 2½ somersault pike dives respectively. These indicate how good the simulation model was in reproducing the diving performances.

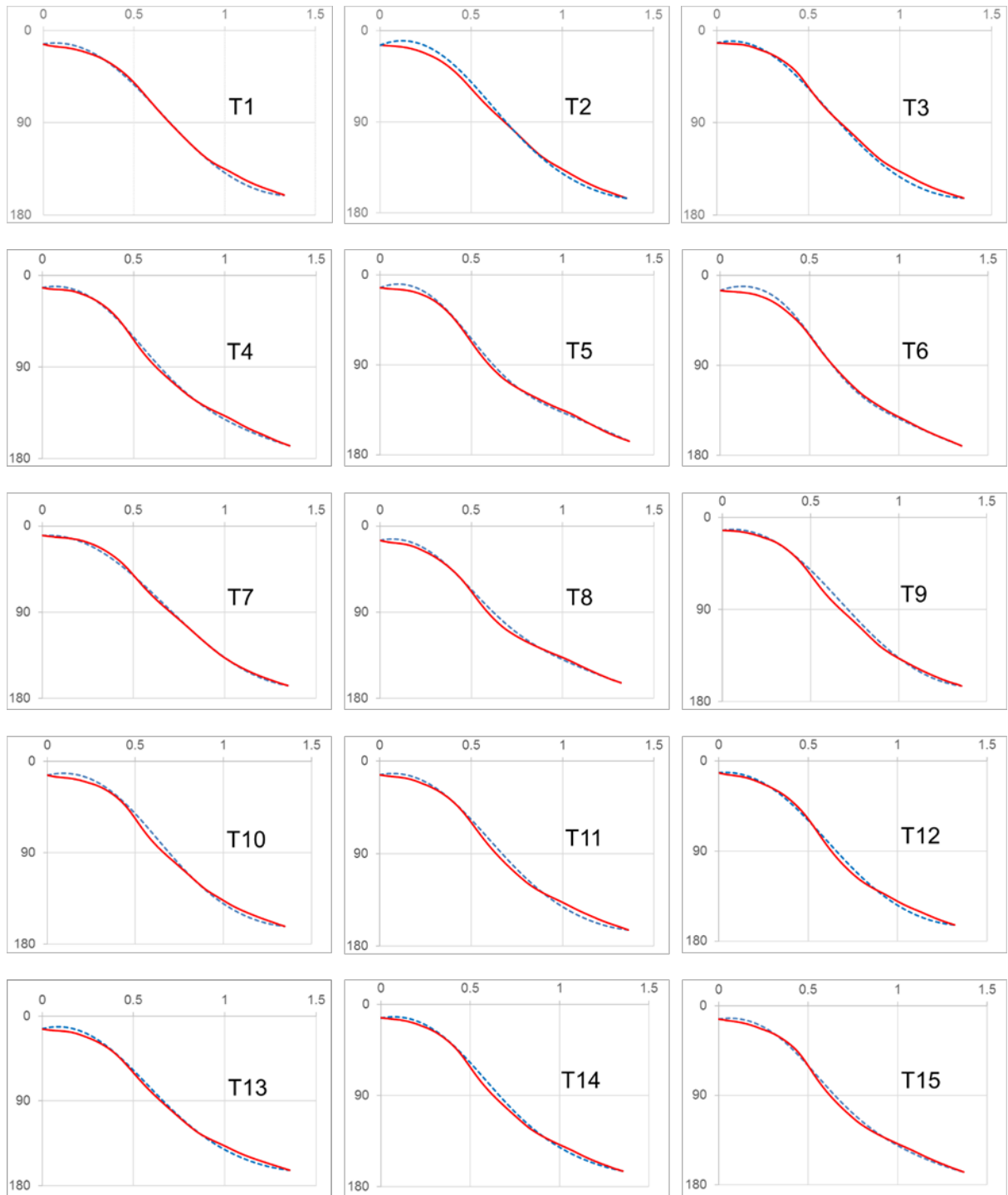


Figure 8.2. Matching simulation (solid line) and performance (dashed line) orientation angles (°) during the aerial phase of 15 forward pike dives.

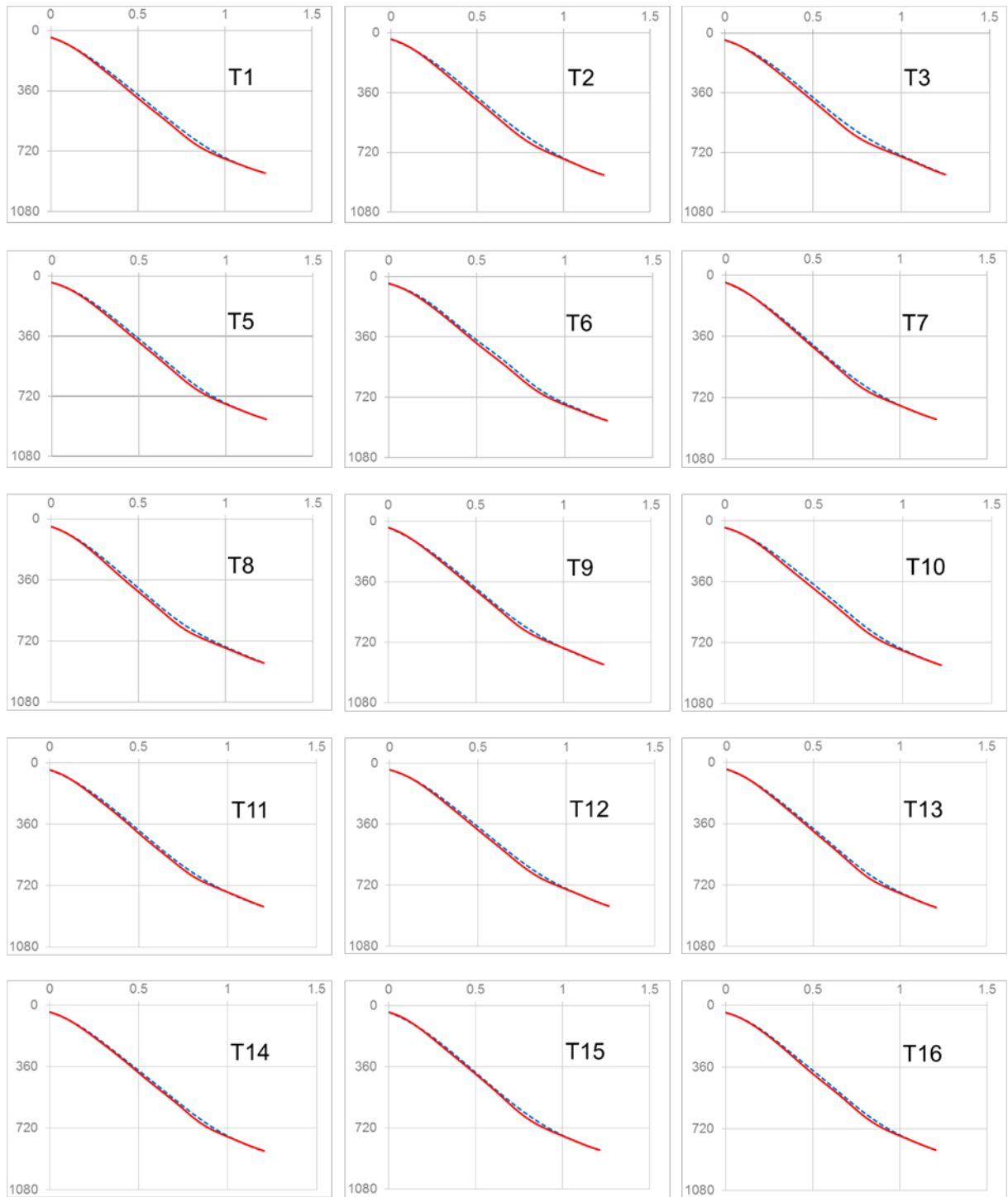


Figure 8.3. Matching simulation (solid line) and performance (dashed line) orientation angles (°) during the aerial phase of 15 forward 2½ somersault pike dives.

8.2.5. Accuracy of simulation

The orientation angles of simulation and video data were compared to determine root mean square differences (RMS). When the orientation angle time history of each dive was matched individually, the simulation matched the performance very well for 101B with the mean score of 3.1°, whereas the corresponding value is 13.5° in 105B (Table 8.4).

Table 8.4. Root mean square differences of the orientation angles between simulation and video data (S2 score)

RMS difference in orientation angle (°)		
Dive	101B	105B
1	2.64	15.2
2	4.83	16.4
3	3.24	17.9
4	2.33	7.7
5	2.87	14.0
6	3.31	15.1
7	1.80	11.1
8	2.99	15.4
9	3.78	12.1
10	3.92	17.1
11	3.65	12.3
12	3.36	13.2
13	2.81	12.2
14	3.45	11.1
15	2.81	10.9
Mean	3.18	13.5
SD	0.72	2.77

8.3. SIMULATION

In order to investigate the diver's movement during the flight phase, the simulation model of aerial phase developed by Yeadon et al. (1990b) was used. The simulation study was performed on all individual trials separately to investigate the extent to which the variability in the entry orientation angle arose from the variability of (a) initial conditions (angular momentum and flight time) (b) joint angles during the flight phase (technique). Therefore, two analyses were carried out to examine the variability and control in forward pike dives. In the first analysis, the variability in the entry angle arising from the variability of initial takeoff conditions was considered. In the second analysis, the variability in the entry angle arising from joint angle changes during the flight phase was investigated.

8.3.1. First Analysis

For each trial in the first analysis, flight time and angular momentum of all 15 dives were used as input into the model while retaining the joint angles time histories of that trial, thereby producing 15 simulations of each trial, giving 225 simulations in total. The variability of the entry angle from the 15 simulations indicates the variability arising from the variability in the vertical velocity and angular momentum at takeoff. Thus, the results of the first analysis highlight the variability of entry angle arising from the variability in initial conditions only. This is run to investigate how the entry angle varies if the diver performs a dive by changing the initial takeoff conditions without changing the body configuration (fixed technique) during the flight phase.

8.3.2. Second Analysis

Each trial in the second analysis was run, such that the time histories of joint angles of each of the 15 dives were used as input into the model while retaining the initial conditions, flight time and angular momentum of that trial, leading to 225 simulations in total. The variability of the entry angle from the 15 simulations indicates the variability arising from the variability in the body configuration during the flight phase. The results of the second analysis highlight the variability of the entry angle arising from configuration changes during the flight phase only. This is run to investigate how the

entry angle varies if the diver performs a dive by varying the joint angle time history without changing (fixed) the initial takeoff conditions.

8.3.3. Net change

The difference between the mean entry angle from 15 simulations and each individual simulated entry angle in the first analysis and second analysis was calculated to determine the net change in entry angle of each dive.

8.4. Results

8.4.1. Actual performance

Table 8.5. shows the mean and standard deviation of joint angles, body orientation, CM velocities, flight time, angular momenta and rotation potential in 101B and 105B. The standard deviations of orientation angles, flight times and angular momenta were all small showing that the diver was consistent from trial to trial. The mean orientation angles at takeoff and entry in 101B are 13° (SD=1.3°) and 165° (SD=2.3°) respectively, indicating that the body orientation was near vertical at the start and end of the flight phase in forward pike dives. The corresponding values in 105B were 40° (1.9°) and 131 (4.7°) respectively, showing that the body orientation was short of 2½ somersault in 105B. The average rotation potential is 0.31 straight somersaults for 101B compared to 1.15 straight somersaults for 105B.

Table 8.5. Mean and standard deviation of the angles at takeoff and entry, CM velocities, flight time, angular momentum and rotation potential during flight of 15 forward and 15 forward 2½ somersault dives

Variable	Dive	Takeoff	entry
Hip (°)	101B	168 (3.3)	180 (3.6)
	105B	110 (5.0)	142 (13)
Knee (°)	101B	179 (0.8)	180 (0.6)
	105B	179 (1.8)	177 (1.2)
Arm (°)	101B	157 (7.5)	149 (4.9)
	105B	101 (7.3)	138 (6.1)
Orientation (°)	101B	13 (1.3)	165 (2.3)
	105B	40 (1.9)	131 (4.7)
Vertical velocity (m/s)	101B	5.8 (0.09)	7.4 (0.06)
	105B	5.2 (0.08)	6.8 (0.08)
Horizontal Velocity (m/s)	101B	0.86 (0.06)	
	105B	1.1 (0.07)	
Flight Time (ms)	101B	1349 (13.8)	
	105B	1221 (15.4)	
Angular Momentum (kg.m ² /s)	101B	17 (1.0)	
	105B	70 (1.7)	
Rotation potential (ss)	101B	0.31 (0.018)	
	105B	1.15 (0.03)	

Figures 8.4 and 8.5 show hip and arm variability within joint angle time history and standard deviation of orientation angle during flight phase of 15 forward and 15 forward 2½ somersault pike dives respectively. The variation in the hip angle was small in the first half of flight and then became large in the second half in both 101B and 105B. For 101B the variation became small again at the end of the dive (Figure 8.4) but for 105B it remained large (Figure 8.5). The peak variation in the orientation angle occurred in mid-flight in both 101B and 105B (Figures 8.4 and 8.5).

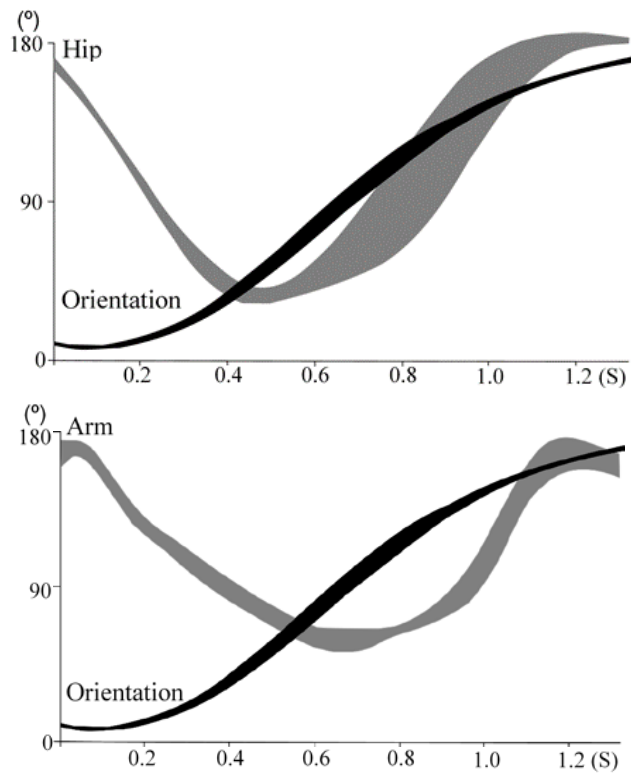


Figure 8.4. The angle envelope generated from 15 forward pike dives.

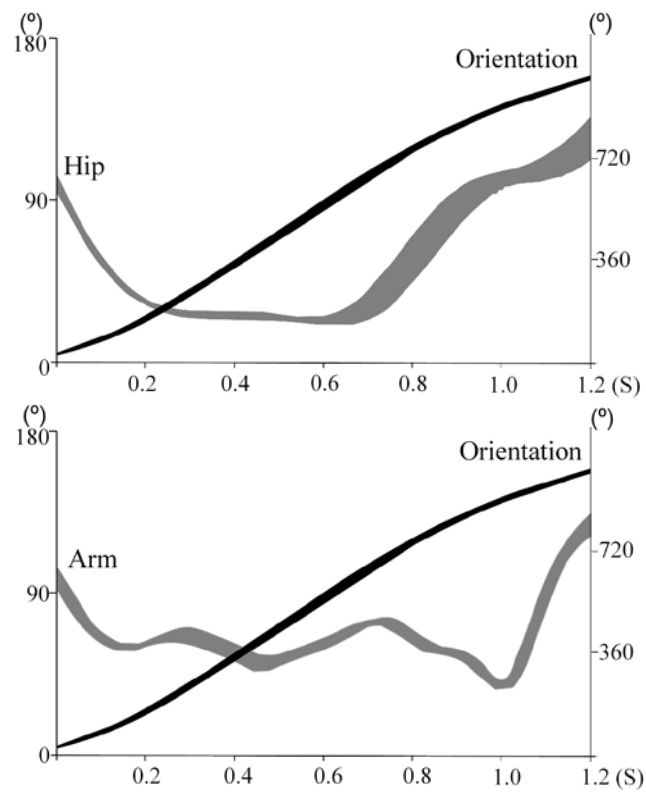


Figure 8.5. The angle envelope generated from 15 forward 2½ somersault pike dives.

8.4.2. Simulation outcome

8.4.2.1. Analysis 1

In the first analysis, in simulation 1, the model was run for 15 times each time using the flight time and angular momentum of one of the 15 dives while retaining the joint angle time history of dive 1. Each simulation output constitutes 15 orientation entry angles. Tables 8.6 and 8.7 show the orientation angles at entry and their corresponding video values obtained from the simulation 1 of 101B and 105B respectively. As it can be seen in Table 8.6, at the first run of the simulation 1, using the flight time 1.340 s and angular momentum 15.6 kg m²/s of dive 1, while retaining the joint angle time history of dive 1, the entry angle obtained 161.0°. This entry orientation angle is the same value of the performance as the model virtually replicates the dive 1. Similarly, in the second run using the flight time and angular momentum of dive 2, the entry angle obtained 176.4°, which is greater than the first run (161.0°). This is due to longer flight time and greater angular momentum, such that the simulation model produced more rotation at entry. The same scenario can be seen in Table 8.7 for 105B.

Table 8.6. Entry orientation angle in 101B obtained from Analysis 1

Simulation 1 of 101B				
Trial (dive)	Flight time (s)	Angular momentum (kg m ² /s)	Video value (°)	Orientation angle (°)
1	1.340	15.6	161.0	161.0
2	1.352	17.1	165.0	176.4
3	1.348	15.9	163.5	163.6
4	1.356	17.4	166.8	180.0
5	1.364	18.9	164.6	194.8
6	1.352	18.2	169.6	187.4
7	1.344	16.0	166.5	165.0
8	1.320	18.4	165.0	186.5
9	1.356	16.2	164.4	168.1
10	1.348	16.0	162.4	165.6
11	1.360	16.5	163.8	171.2
12	1.320	17.4	164.2	176.4
13	1.356	16.2	163.3	168.0
14	1.352	16.8	164.2	174.0
15	1.368	17.1	165.6	177.4
Mean	1.349	16.9	164.6	174.4
SD	0.014	0.987	2.1	9.8

Table 8.7. Entry orientation angle in 105B obtained in Analysis 1

Simulation 1 of 105B				
Trial (dive)	Flight time (s)	Angular momentum (kg m ² /s)	Video value (°)	Orientation angle (°)
1	1.232	66.6	131.4	131.4
2	1.232	69.8	131.4	169.7
3	1.248	70.1	135.8	177.9
5	1.232	70.0	131.4	172.6
6	1.240	66.7	133.7	134.7
7	1.208	71.1	124.4	177.3
8	1.220	72.7	93.5	163.3
9	1.228	71.3	130.3	186.9
10	1.216	69.9	126.8	166.3
11	1.200	70.8	93.5	141.8
12	1.240	69.7	133.7	171.2
13	1.208	70.9	124.4	174.9
14	1.204	67.6	123.1	134.1
15	1.212	70.3	125.6	169.4
16	1.204	68.3	123.1	142.7
Mean	1.2	69.7	124.1	160.9
SD	0.0	1.7	13.1	18.6

It is assumed that by increasing the flight time and angular momentum, while retaining the joint angle time history, more rotation in flight and subsequently a greater entry angle would result in the simulation outcome. However, by running the simulation on dives 8 and 12 in 101B, their entry angles were 186.5° and 176.4° respectively (Table 8.6). These angles are greater than the actual performance (161.0°) as their angular momenta are greater even though their flight times are shorter. The same results were obtained in Table 8.7 for dives 7, 8, 9, 10, 11, 13, 14, 15 and 16 in 105B.

Tables 8.8 and 8.9 show the orientation angles at entry into the water of 15 simulations, using the flight time and angular momentum of all 15 dives while retaining the joint angle time histories of each individual dive (Analysis 1) of 101B and 105B respectively. Since each simulation was run 15 times, the outcome is a total number of 225 simulations (entry orientation angles) in Analysis 1.

Table 8.8. Simulated entry orientation angle obtained from the simulations of 15 trials 101B while retaining joint angle time history the same as actual performance

Orientation angle at entry while retaining the joint angle time history of one particular dive																	
Dive	Time	AM*	Sim 1	Sim 2	Sim 3	Sim 4	Sim 5	Sim 6	Sim 7	Sim 8	Sim 9	Sim 10	Sim 11	Sim 12	Sim 13	Sim 14	Sim 15
1	1.340	0.211	161.0	150.9	160.8	149.6	137.1	145.4	162.5	141.4	157.7	158.0	154.4	149.1	156.7	152.3	151.4
2	1.352	0.231	176.4	165.3	176.6	164.2	150.6	159.6	178.2	154.9	173.1	173.3	169.3	163.5	171.9	167.0	166.1
3	1.348	0.214	163.6	153.3	163.6	152.0	139.4	147.8	165.1	143.7	160.3	160.6	156.9	151.5	159.3	154.7	153.9
4	1.356	0.235	180.0	168.6	180.3	167.6	153.7	162.8	181.7	158.0	176.6	177.0	172.7	166.8	175.4	170.4	169.5
5	1.364	0.255	194.8	182.5	195.0	181.5	166.6	176.4	196.7	170.9	191.3	191.8	187.0	180.6	190.0	184.6	183.5
6	1.352	0.245	187.4	175.5	187.8	174.7	160.2	170.4	189.4	164.6	184.0	184.3	179.9	173.9	182.9	177.7	176.6
7	1.344	0.216	165.0	154.6	165.0	153.4	140.7	149.1	166.6	144.9	161.7	162.0	158.3	152.9	160.7	156.1	155.3
8	1.320	0.248	186.5	174.7	187.1	173.8	159.5	168.5	188.8	163.7	183.3	183.6	179.0	173.5	182.0	177.1	175.7
9	1.356	0.219	168.1	157.5	168.2	156.2	143.3	151.9	169.6	147.6	164.8	165.1	161.2	155.6	163.7	159.0	158.1
10	1.348	0.216	165.6	155.2	165.6	153.9	141.2	149.7	167.2	145.5	162.3	162.6	158.8	153.4	161.2	156.6	155.8
11	1.360	0.223	171.2	160.5	171.3	159.2	146.0	154.9	172.7	150.3	167.9	168.4	164.3	158.5	166.8	162.0	161.1
12	1.320	0.234	176.4	165.3	176.8	164.2	150.6	159.4	178.5	154.9	173.2	173.6	169.3	163.9	172.0	167.4	166.1
13	1.356	0.219	168.0	157.5	168.1	156.2	143.2	151.9	169.5	147.5	164.7	165.1	161.2	155.5	163.6	158.9	158.1
14	1.352	0.227	174.0	163.0	174.1	161.8	148.4	157.2	175.6	152.7	170.6	170.9	166.9	161.2	169.5	164.6	163.7
15	1.368	0.231	177.4	166.4	177.1	165.0	151.3	160.5	178.9	155.7	174.0	174.4	170.2	164.1	172.8	168.0	167.0
Mean	1.349	0.228	174.4	163.4	174.5	162.2	148.8	157.7	176.1	153.1	171.0	171.4	167.3	161.6	169.9	165.1	164.1
SD	0.014	0.013	9.8	9.1	9.9	9.3	8.5	9.0	9.9	8.5	9.7	9.7	9.4	9.2	9.6	9.4	9.3

The diagonally bold numbers are the same as in the video values.

*Angular momentum (straight somersault).

Table 8.9. Simulated entry orientation angle obtained from the simulations of 15 trials 105B while retaining joint angle time history the same as actual performance

Orientation angle at entry while retaining joint angle time history of one particular dive																	
	Time	AM*	Sim 1	Sim 2	Sim 3	Sim 4	Sim 5	Sim 6	Sim 7	Sim 8	Sim 9	Sim 10	Sim 11	Sim 12	Sim 13	Sim 14	Sim 15
1	1.232	0.899	131.4	98.8	87.6	113.2	96.6	134.5	84.7	64.4	77.8	100.9	87.4	85.5	92.0	130.2	93.3
2	1.232	0.941	169.7	135.7	123.8	150.6	133.5	172.9	120.6	99.4	113.6	137.9	123.7	122.0	128.5	168.4	130.0
3	1.248	0.945	177.9	143.8	132.6	159.5	141.8	181.7	128.4	108.0	122.0	146.8	132.6	130.7	137.3	176.8	138.1
4	1.204	0.921	142.7	109.2	96.2	123.1	107.1	144.8	95.4	74.2	86.8	110.7	96.4	94.7	102.0	141.1	104.1
5	1.232	0.945	172.6	138.5	126.5	153.4	136.3	175.8	123.3	102.1	116.3	140.7	126.4	124.7	131.3	171.2	132.7
6	1.240	0.900	134.7	102.1	91.3	116.8	99.9	138.1	87.8	67.8	81.2	104.5	91.1	89.1	95.5	133.6	96.6
7	1.208	0.959	177.3	142.5	129.1	157.0	140.4	179.7	127.9	105.8	119.3	144.3	129.3	127.8	135.0	175.6	137.2
8	1.220	0.980	163.3	125.1	107.4	140.3	125.2	162.6	114.4	127.4	99.4	128.5	111.0	108.2	119.8	160.0	123.0
9	1.228	0.962	186.9	152.1	139.7	167.2	150.0	190.0	136.8	115.0	129.4	154.3	139.7	138.1	144.7	185.4	146.4
10	1.216	0.943	166.3	132.1	119.4	146.6	129.9	168.9	117.5	95.9	109.5	133.9	119.4	117.8	124.7	164.8	126.6
11	1.200	0.955	141.8	104.4	87.3	119.2	104.5	141.0	94.2	70.5	79.4	107.7	123.4	87.9	99.3	138.5	102.4
12	1.240	0.940	171.2	137.2	125.7	152.5	135.1	174.7	122.0	101.2	115.4	139.8	125.6	123.9	130.3	170.0	131.5
13	1.208	0.956	174.9	140.2	126.9	154.6	138.1	177.3	125.6	103.6	117.0	141.9	127.0	125.5	132.7	173.2	134.8
14	1.204	0.911	134.1	101.0	88.1	114.7	98.8	136.2	87.3	66.3	78.7	102.4	88.2	86.5	93.8	132.5	95.9
15	1.212	0.948	169.4	135.1	122.1	149.5	132.9	171.9	120.5	98.7	112.2	136.8	122.1	120.5	127.6	167.9	129.6
Mean	1.222	0.940	160.9	126.5	113.6	141.2	124.7	163.3	112.4	93.4	103.9	128.7	116.2	112.2	119.6	159.3	121.5
SD	0.015	0.023	18.6	18.2	18.6	18.5	18.1	19.0	17.4	19.7	18.1	18.2	17.1	18.4	17.9	18.7	17.8

The diagonally bold numbers are the same as in the video values.

*Angular momentum (straight somersault).

8.4.2.2. Analysis 2

In the second analysis in simulation 1, the model was run for 15 times using the joint angle time history of all 15 dives, while retaining the initial takeoff conditions, flight time and angular momentum of dive 1. Tables 8.10 and 8.11 show the outcomes of the simulation 1 for 101B and 105B respectively. In the first run of the simulation 1 for 101B, using the flight time 1.340 s and angular momentum 15.6 kg.m²/s of dive 1 while retaining the angle time history of dive 1, the entry angle obtained 161.0° (Table 8.10). This is the same value obtained in the recorded performance as the model virtually replicates the dive 1. Similarly, in the second run using the joint angle time history of dive 2, while retaining the flight time and angular momentum of dive 1, the entry angle obtained 150.9°. This is less than the value obtained in the first run (161.0°). Except for the 7th run, all other values of entry angle are smaller than the first run. This is due to relatively shorter flight time in dive 1 since it has been used for all 15 runs. Similar effect occurred for 105B (Table 8.11).

Table 8.10. The outcome of simulation 1 using joint angle time histories of 15 forward pike dives while retaining the flight time and angular momentum of dive 1

Simulation 1 of 101B				
Trial (dive)	Flight time (s)	Angular momentum (kg m ² /s)	Entry orientation angle (°)	
			Video value (°)	Orientation angle (°)
1	1.34	15.6	161.0	161.0
2	1.34	15.6	165.0	150.9
3	1.34	15.6	163.5	160.8
4	1.34	15.6	166.8	149.6
5	1.34	15.6	164.6	137.1
6	1.34	15.6	169.6	145.4
7	1.34	15.6	166.5	162.5
8	1.34	15.6	165.0	141.4
9	1.34	15.6	164.4	157.7
10	1.34	15.6	162.4	158.0
11	1.34	15.6	163.8	154.4
12	1.34	15.6	164.2	149.1
13	1.34	15.6	163.3	156.7
14	1.34	15.6	164.2	152.3
15	1.34	15.6	165.6	151.4
Mean			164.6	152.5
SD			2.1	7.3

Table 8.11. The outcome of simulation 1 using the joint angle time histories of 15 forward 2½ somersault pike dives while retaining the flight time and angular momentum of dive 1

Simulation 1 of 105B				
Trial (dive)	Flight time (s)	Angular momentum (kg m ² /s)	Entry orientation angle	
			video value (°)	Simulation output (°)
1	1.232	66.6	131.4	131.4
2	1.232	66.6	135.7	98.8
3	1.232	66.6	125.6	87.6
4	1.232	66.6	132.4	113.2
5	1.232	66.6	136.3	96.6
6	1.232	66.6	135.0	134.5
7	1.232	66.6	135.2	84.7
8	1.232	66.6	132.2	64.4
9	1.232	66.6	130.9	77.8
10	1.232	66.6	139.7	100.9
11	1.232	66.6	134.0	87.4
12	1.232	66.6	120.9	85.5
13	1.232	66.6	141.3	92.0
14	1.232	66.6	140.8	130.2
15	1.232	66.6	135.5	93.3
Mean			133.8	98.6
SD			5.4	20.5

Tables 8.12 and 8.13 show the orientation angles at entry into the water of 15 simulations using the joint angle time histories of all 15 dives while retaining the set of flight time and angular momentum of one particular dive. Since there are 15 simulations and there are 15 entry orientation angles in each simulation outcome, there is a number of 225 simulations (entry orientation angles) in Analysis 2.

Table 8.12. Simulated entry orientation angle obtained from the simulations of 15 trials of 101B while retaining the initial conditions the same as actual performance

Trials	Time	AM*	Orientation angle at entry while retaining flight time and angular momentum of one particular dive														
			Sim 1	Sim 2	Sim 3	Sim 4	Sim 5	Sim 6	Sim 7	Sim 8	Sim 9	Sim 10	Sim 11	Sim 12	Sim 13	Sim 14	Sim 15
1	1.340	0.211	161.0	176.4	163.6	180.0	194.8	187.4	165.0	186.5	168.1	165.6	171.2	176.4	168.0	174.0	177.4
2	1.352	0.231	150.9	165.3	153.3	168.6	182.5	175.5	154.6	174.7	157.5	155.2	160.5	165.3	157.5	163.0	166.4
3	1.348	0.214	160.8	176.6	163.6	180.3	195.0	187.8	165.0	187.1	168.2	165.6	171.3	176.8	168.1	174.1	177.1
4	1.356	0.235	149.6	164.2	152.0	167.6	181.5	174.7	153.4	173.8	156.2	153.9	159.2	164.2	156.2	161.8	165.0
5	1.364	0.255	137.1	150.6	139.4	153.7	166.6	160.2	140.7	159.5	143.3	141.2	146.0	150.6	143.2	148.4	151.3
6	1.352	0.245	145.4	159.6	147.8	162.8	176.4	170.4	149.1	168.5	151.9	149.7	154.9	159.4	151.9	157.2	160.5
7	1.344	0.216	162.5	178.2	165.1	181.7	196.7	189.4	166.6	188.8	169.6	167.2	172.7	178.5	169.5	175.6	178.9
8	1.320	0.248	141.4	154.9	143.7	158.0	170.9	164.6	144.9	163.7	147.6	145.5	150.3	154.9	147.5	152.7	155.7
9	1.356	0.219	157.7	173.1	160.3	176.6	191.3	184.0	161.7	183.3	164.8	162.3	167.9	173.2	164.7	170.6	174.0
10	1.348	0.216	158.0	173.3	160.6	177.0	191.8	184.3	162.0	183.6	165.1	162.6	168.4	173.6	165.1	170.9	174.4
11	1.360	0.223	154.4	169.3	156.9	172.7	187.0	179.9	158.3	179.0	161.2	158.8	164.3	169.3	161.2	166.9	170.2
12	1.320	0.234	149.1	163.5	151.5	166.8	180.6	173.9	152.9	161.9	155.6	153.4	158.5	163.9	155.5	161.2	164.1
13	1.356	0.219	156.7	171.9	159.3	175.4	190	182.9	160.7	182.0	163.7	161.2	166.8	172.0	163.6	169.5	172.8
14	1.352	0.227	152.3	167.0	154.7	170.4	184.6	177.7	156.1	177.1	159.0	156.6	162.0	167.4	158.9	164.6	168.0
15	1.368	0.231	151.4	166.1	153.9	169.5	183.5	176.6	155.3	175.7	158.1	155.8	161.1	166.1	158.1	163.7	167.0
Mean	1.349	0.228	152.6	167.3	155.0	170.7	184.9	178.0	156.4	176.3	159.3	157.0	162.3	167.4	159.3	164.9	168.2
SD	0.014	0.013	7.3	8.0	7.5	8.2	8.9	8.5	7.5	9.4	7.7	7.5	7.8	8.1	7.7	7.9	8.1

The diagonally bold numbers are the same as film values.

*Angular momentum (straight somersault).

Table 8.13. Simulated entry orientation angle obtained from the simulations of 15 trials of 105B while retaining the initial conditions the same as actual performance

Orientation angle at entry while retaining flight time and angular momentum of one particular dive																	
	Time	AM	Sim 1	Sim 2	Sim 3	Sim 4	Sim 5	Sim 6	Sim 7	Sim 8	Sim 9	Sim 10	Sim 11	Sim 12	Sim 13	Sim 14	Sim 15
1	1.232	0.899	131.4	169.7	177.9	142.7	172.6	134.7	177.3	200.7	186.9	166.3	171.5	171.2	174.9	134.1	169.4
2	1.232	0.941	98.8	135.7	143.8	109.2	138.5	102.1	142.5	165.3	152.1	132.1	136.8	137.2	140.2	101.0	135.1
3	1.248	0.945	87.6	123.8	132.6	96.2	126.5	91.3	129.1	152.2	139.7	119.4	123.0	125.7	126.9	88.1	122.1
4	1.204	0.921	113.2	150.6	159.5	123.1	153.4	116.8	157.0	180.3	167.2	146.6	151.0	152.5	154.6	114.7	149.5
5	1.232	0.945	96.6	133.5	141.8	107.1	136.3	99.9	140.4	163.2	150.0	129.9	134.7	135.1	138.1	98.8	132.9
6	1.240	0.900	134.5	172.9	181.7	144.8	175.8	138.1	179.7	203.6	190.0	168.9	173.5	174.7	177.3	136.2	171.9
7	1.208	0.959	84.7	120.6	128.4	95.4	123.3	87.8	127.9	149.8	136.8	117.5	122.4	122.0	125.6	87.3	120.5
8	1.220	0.980	64.4	99.4	108.0	74.2	102.1	67.8	105.8	127.4	115.0	95.9	100.4	101.2	103.6	66.3	98.7
9	1.228	0.962	77.8	113.6	122.0	86.8	116.3	81.2	119.3	141.9	129.4	109.5	113.4	115.4	117.0	78.7	112.2
10	1.216	0.943	100.9	137.9	146.8	110.7	140.7	104.5	144.3	167.3	154.3	133.9	138.4	139.8	141.9	102.4	136.8
11	1.200	0.955	87.4	123.7	132.6	96.4	126.4	91.1	129.3	152.2	139.7	119.4	123.3	125.6	127.0	88.2	122.1
12	1.240	0.940	85.5	122.0	130.7	94.7	124.7	89.1	127.8	150.8	138.1	117.8	121.7	123.9	125.5	86.5	120.5
13	1.208	0.956	92.0	128.5	137.3	102.0	131.3	95.5	135.0	157.6	144.7	124.7	129.4	130.3	132.7	93.8	127.6
14	1.204	0.911	130.2	168.4	176.8	141.1	171.2	133.6	175.6	199.1	185.4	164.8	169.7	170.0	173.2	132.5	167.9
15	1.212	0.948	93.3	130.0	138.1	104.1	132.7	96.6	137.2	159.5	146.4	126.6	131.6	131.5	134.8	95.9	129.6
Mean	1.222	0.940	98.6	135.4	143.9	108.6	138.1	102.0	141.9	164.7	151.7	131.6	136.1	137.1	139.6	100.3	134.5
SD	0.015	0.023	20.5	21.4	21.4	20.9	21.5	20.5	21.8	22.3	21.9	21.5	21.7	21.4	21.7	20.7	21.6

The diagonally bold numbers are the same as film values.

*Angular momentum (straight somersault).

8.4.2.3. Net change & RMS differences

Tables 8.14 and 8.15 show the net change and the RMS differences in entry angle between each individual simulated and mean values of 15 simulations in 101B and 105B respectively.

Table 8.14. Differences between each individual simulated and mean entry angle from 15 simulations and the net changes in analysis 1 and analysis 2 of 101B

Difference between simulated entry angle and actual performance (°)						
Trial	Actual entry	Analysis 1		Analysis 2		Difference (analysis 1&2)
		Entry	net change	Entry	net change	
sim1	161.0	174.4	13.4	152.6	-8.4	4.9
sim2	165.3	163.4	-1.9	167.3	2.0	0.1
sim3	163.6	174.5	10.9	155.0	-8.6	2.3
sim4	167.6	162.2	-5.4	170.7	3.1	-2.2
sim5	166.6	148.8	-17.8	184.9	18.3	0.5
sim6	170.4	157.7	-12.7	178.0	7.6	-5.1
sim7	166.6	176.1	9.5	156.4	-10.2	-0.7
sim8	163.7	153.1	-10.6	176.3	12.6	2.0
sim9	164.8	171.0	6.2	159.3	-5.5	0.8
sim10	162.6	171.4	8.8	157.0	-5.6	3.2
sim11	164.3	167.3	3.0	162.3	-2.0	1.0
sim12	163.9	161.6	-2.3	167.4	3.5	1.2
sim13	163.6	169.9	6.3	159.3	-4.3	2.0
sim14	164.6	165.1	0.5	164.9	0.3	0.8
sim15	167.0	164.1	-2.9	168.2	1.2	-1.7
Mean	165.0	165.4	0.3	165.3	0.3	0.6
SD	2.30	7.95	9.21	9.29	8.06	2.41
RMS			8.9		7.8	2.4

The simulation 1 in the analysis 1 in Table 8.14 indicates that when the model was run for 15 times, each time using the angular momentum and flight time of all 15 dives, while retaining the joint angle time history of dive 1 of 101B, the mean entry angle of these 15 simulations obtained 174.4° whereas the entry angle for the simulated dive 1 was 161.0°, which is the same value of the performance as it virtually replicates the dive 1. Thus, the effect of varying the angular momentum and flight time of dive 1 was to change the entry angle by 13.4° compared to the mean simulated entry angle (174.4°). Similarly, the simulation 1 in the analysis 2, indicates that when the model was run for 15 times, each time using the joint angle time history of all 15 dives while

retaining the initial takeoff conditions, the mean entry angle of the 15 simulations obtained 152.6° whereas the entry angle for the simulated dive 1 was 161.0°. This shows that the effect of varying the joint angle time history of the flight phase of dive 1 was to change the entry angle by -8.4° compare to the mean simulated entry angle (152.6°). Putting these two effects together gives a net change of 4.9° in the entry angle of dive 1 arising from the individual initial conditions and joint angle changes of dive 1. Repeating this process for all 15 dives gives an RMS of the combined net change of 2.4° compared with RMS values of 8.9° and 7.8° for the net change arising from initial conditions (analysis 1) and from joint angle time histories (analysis 2) respectively. This 2.4° compares with a standard deviation of the actual entry angle of 2.3°.

Table 8.15. Comparison between simulated orientation angle and actual entry angle and the net change between analysis 1 and analysis 2 of 105B

Difference between simulated entry angle and actual performance (°)						
Trial	Actual Entry	Analysis 1		Analysis 2		Difference (Analyses 1&2)
		Entry	net change	Entry	net change	
Sim1	131.4	160.9	29.5	98.6	-32.8	-3.3
Sim2	135.7	126.5	-9.2	135.4	-0.3	-9.5
Sim3	132.6	113.6	-19.0	143.9	11.3	-7.8
Sim5	136.3	124.7	-11.6	138.1	1.8	-9.8
Sim6	138.1	163.3	25.2	102.0	-36.1	-10.9
Sim7	127.9	112.4	-15.5	141.9	14.0	-1.5
Sim8	127.4	93.4	-34.0	164.7	37.3	3.3
Sim9	129.4	103.9	-25.5	151.7	22.3	-3.2
Sim10	133.9	128.7	-5.2	131.6	-2.3	-7.5
Sim11	123.4	116.2	-7.2	136.1	12.7	5.5
Sim12	123.9	112.2	-11.7	137.1	13.2	1.5
Sim13	132.7	119.6	-13.1	139.6	6.9	-6.2
Sim14	132.5	159.3	26.8	100.3	-32.2	-5.4
Sim15	129.6	121.5	-8.1	134.5	4.9	-3.3
Sim16	123.1	141.2	18.1	108.6	-14.5	3.6
Mean	130.5	126.5	-4.0	130.9	0.4	-3.6
SD	4.7	21.0	19.6	19.7	21.1	5.2
RMS		19.4		20.4		6.2

Similarly, repeating the same process for all 15 forward 2½ somersault pike dives (Table 8.15) gives an RMS of the combined net change of 6.2° compared with RMS

values of 19.4° and 20.4° for the net changes arising from initial conditions (analysis 1) and body configuration in flight (analysis 2) respectively. This 6.2° compares with a standard deviation of the actual entry angle of 4.7°.

8.4.3. Actual performance and simulation outcome

It was hypothesised that if there is a certain amount of variability in the initial conditions, by running the simulation with the same amount of variation at initial condition, the errors would be cumulative throughout the simulation to the end due to the process of numerical integration inherent in a simulation. This means the variability of the simulation outcome should be greater than the actual performance. Table 8.16 shows a comparison in the entry angle variability obtained in the actual performance and simulation outcome.

Table 8.16. Mean and standard deviation of orientation angle at initial takeoff conditions and entry in actual performance and simulation of 101B and 105B

Orientation angle		Simulation (°)					
		Actual performance (°)		Analysis 1		Analysis 2	
		101B	105B	101B	105B	101B	105B
Initial	Mean	13	40	13	40	13	40
	(SD)	(1.3)	(1.9)	(1.3)	(1.9)	(1.3)	(1.9)
Entry		165	131	165	126.5	165	131
		(2.3)	(4.7)	(9.2)	(19.6)	(8.1)	(21.1)

The average standard deviation of orientation angle at entry into the water obtained from the first analysis and second analysis of 101B were 9.2° and 8.1° respectively. The corresponding values in 105B were 19.6° and 21.1° respectively. This means that by varying the initial takeoff conditions while retaining the joint angles, the entry orientation angle varies by 9.2° and by varying the angles while retaining the initial takeoff conditions, the entry angle varies by 8.1° in 101B. In other words, the variation in the simulated orientation angle at entry arising from variation in the initial conditions, flight time and angular momentum in 101B was 9.2° compared to a variation of 2.3° in the recorded performances. The variation in the simulated orientation angle at entry arising from variation in the joint angle time histories was 8.1° in 101B compare to a

variation of 2.3° in the recorded performances. Similarly, in 105B the average standard deviation of orientation angle in actual performance was 4.7° and obtained from analysis 1 and analysis 2 were 19.6° and 21.1° respectively (Table 8.16). Since the actual variability either in 101B or 105B is much smaller than the average variation obtained in the simulation outcome, it can be used as an evidence that the diver used feedback correction. In fact, the low variability in the actual entry angle is due to the diver's adjustment.

The de-trended standard deviation time history of 101B and 105B depicted in Figure 8.6 shows how the joints angles change relative to the orientation angle. The de-trended graphs were produced by subtracting the average angle time history from each angle time history to produce the standard deviation time history away from the average time history.

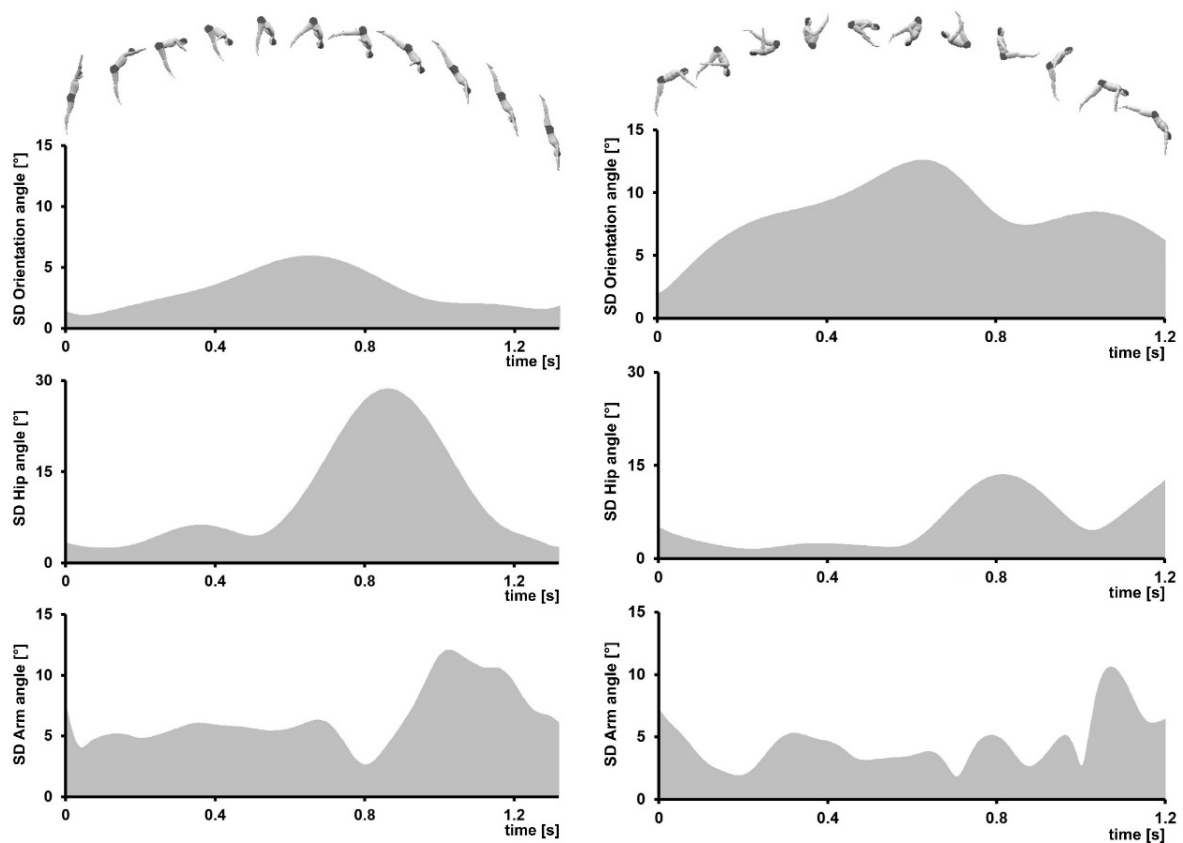


Figure 8.6. The de-trended standard deviation angle time history of 101B and 105B.

During the first half of flight the configurational changes were very similar with trial to trial differences in later flight. For both dives the variation in the orientation angle increased to a maximum in mid-flight around 0.6 s and then decreased to a low value at entry. The variation in the hip angle was small in the first half of flight rising to a maximum after around 0.8 s before starting to fall again. For 101B the hip variation became small again at the end of the dive but for 105B it became large again. The variation in the arm angle was small for the majority of the flight, rising to a maximum at 1.028 s for 101B and 1.072 s for 105B before falling to a low value at entry (Figures 8.4, 8.5 and 8.6).

As it can be seen in Figure 8.6, the variability of the hip and arm angles may depend on the variability of the orientation angle at previous time. It is expected that when there is over rotation or less rotation, the hip and arm are varied to make correction. The regression analysis can reveal the relationship between the joints angles and the orientation angle.

8.4.4. Regressions

When rotation potential was regressed against mean hip angle and mean arm angle, only the mean hip angle was included as a predictor in each of the two cases, with the mean hip angle increasing with the rotation potential (Table 8.17). For the pike dives the mean hip angle explained 91% of the variance in rotation potential ($p < .01$) and 89% ($p < .01$) for the $2\frac{1}{2}$ pike dives (Table 8.17). The configuration changes during flight were adjusted as a function of the takeoff conditions. When rotation potential was regressed against minimum hip angle and minimum arm angle, no predictors were included.

Table 8.17. Stepwise multiple regression analysis of rotation potential against average hip angle, minimum hip angle, average arm angle and minimum arm angle during the flight phase of 15 forward and 15 forward 2½ somersault pike dives

Model Summary ^b						
Dive	Model	R Square	Unstandardised		standardised	
			B		Beta	Sig.
101B	1	0.907 ^a	Constant	0.056		0.025
			Average hip angle	0.0023	.952	0.000
105B	1	0.893 ^a	Constant	0.572		0.000
			Average hip angle	0.009	.945	0.000
a. Predictors: (Constant), Average hip angle (°)						
b. Dependent Variable: Rotation potential (Straight somersault)						

The b-value 0.009 (Table 8.17) indicates that as the average hip angle increases by one unit, rotation potential increases by 0.009 units. The hip angle and rotation potential were measured in degree (°) and straight somersault (ss) respectively; thus, for every 1° increase in hip angle, an extra 0.009 straight somersault would be achieved in rotation potential.

The standard coefficient beta 0.945 (Table 8.17) indicates that as the average hip angle increases by 1 standard deviation (7.45°), the rotation potential increases by 0.945 standard deviations. The standard deviation for the rotation potential in 101B is 0.018 (ss) and so this constitutes a change of 0.017 (ss) in the rotation potential. Therefore, if the average hip angle increases an extra 7.45°, 0.017 extra straight somersault in rotation potential is expected. This interpretation is true only if the effect of the minimum hip angle, average arm angle and minimum arm angle are held constant. These results indicate that the hip movement is highly correlated to the rotation potential in both 101B and 105B.

Figures 8.7 and 8.8 show the linear regression of the rotation potential against hip angle at the time of maximum hip variation in 101B and 105B respectively. The hip angle at the time of maximum hip angle variation is significantly correlated to the rotation potential in both dives, indicating that the hip movement at the time of maximum variation is dependent on the amount of rotation potentiation generated at

takeoff. This means that if there is large rotation potential, the hip angle is extended to increase the moment of inertia and slow the rotation down. In the case of low rotation potential, the hip angle is decreased to speed the rotation up.

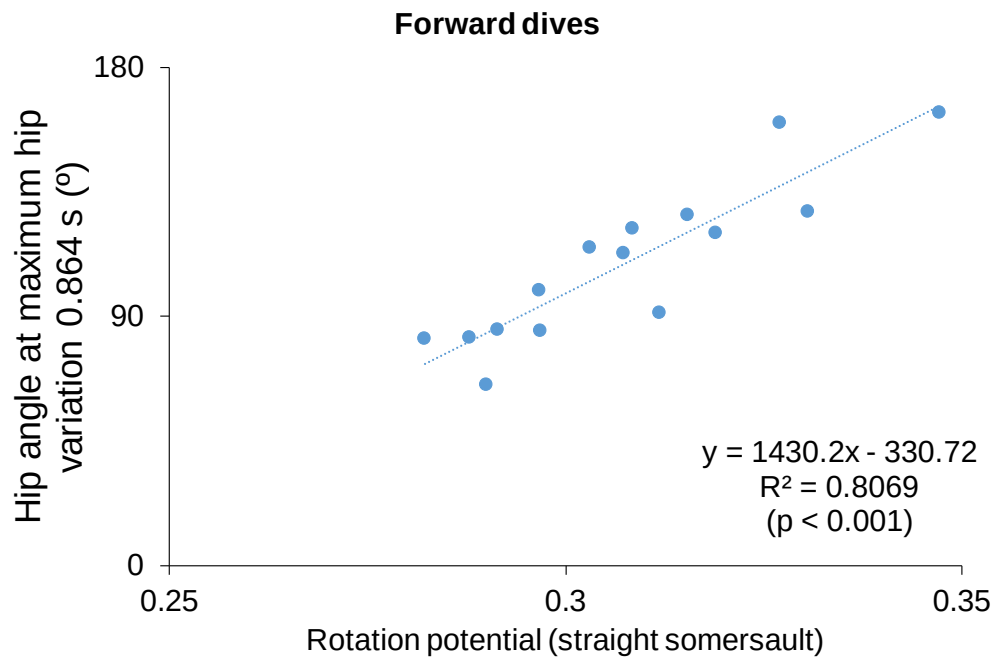


Figure 8.7. Regression of rotation potential against hip angle at 0.864 s (maximum hip variation) of 101B.

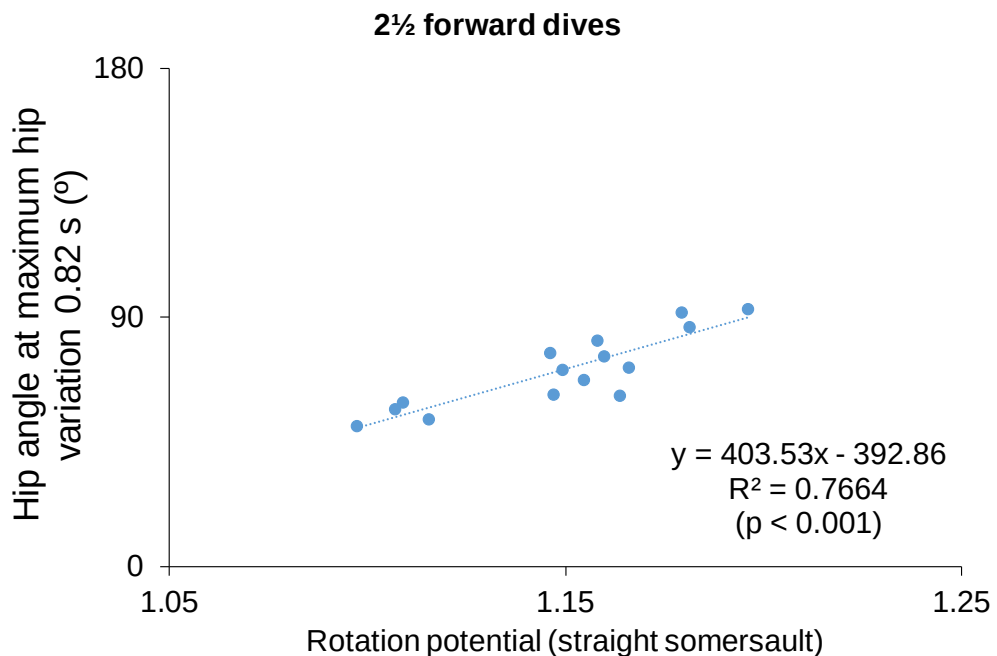


Figure 8.8. Regression of rotation potential against hip angle at 0.864 s (maximum hip variation) of 105B.

Somersault rotation in flight was controlled by hip angle adjustment and also arm angle adjustment in the case of the 2½ somersault dive. When the hip angle at the time of maximum hip variation was regressed against rotation potential positive relationships were obtained for both dives, with greater hip angle for greater rotation potential (Table 8.18). When the arm angle at the time of maximum arm variation was regressed against the orientation angle at the earlier time of maximum hip variation, there was no relationship for 101B but a positive relationship for 105B, with greater arm angle associated with greater orientation angle (Table 8.19). When the final hip angle of 105B was regressed against the orientation angle at the earlier time of maximum arm angle variation, a positive relationship was found, with greater final hip angle associated with greater orientation angle (Table 8.19). These indicates that there are three corrections in 105B. The hip angle at 0.82 makes the first adjustment and the second one is made by the arm angle at 1.072 s. Essentially, this means that if the diver is over rotated after varying his hip angle for the first correction at 0.82 s, he raises his arms later at 1.072 s to increase the moment of inertia and slow the rotation down. The third correction is made by the hip angle at entry based on how the orientation angle is at 1.072 s.

Table 8.18. Linear regression analysis of (a) arm angle at maximum arm variation against orientation angle at maximum hip variation and (b) hip angle at entry against orientation angle at maximum arm variation of 15 forward 2½ somersault pike dives

Dive	R ²	coefficients		p
		unstandardised	standardised	
101B	0.807	constant	-330.7	< 0.001
		rotation potential	1430.2	0.898 < 0.001
105B	0.766	constant	-392.9	< 0.001
		rotation potential	403.5	0.875 < 0.001
predictors: (constant), rotation potential (straight somersaults)				
dependent variable: hip angle (°) at max hip angle variation (101B at 0.864 s, 105B at 0.820 s)				

Table 8.19. Linear regression analysis of hip angle at maximum hip variation against rotation potential of 15 forward pike dives and 15 forward 2½ somersault pike dive

Analysis	R ²	coefficients		p
		unstandardised	standardised	
(a)	0.371	constant	-454.9	0.033
		orientation angle	0.812	0.609
(b)	0.453	constant	-672.6	0.017
		orientation angle	1.021	0.673
(a) predictors: (constant), orientation angle (°) at max hip variation (0.82 s)				
dependent variable: arm angle (°) at max arm variation (1.072 s)				
(b) predictors: (constant), orientation angle (°) at max arm variation (1.072 s)				
dependent variable: hip angle (°) at entry (1.2 s)				

8.5. DISCUSSION

The aim of this chapter was to investigate the extent to which technique in the aerial movement control the outcome in 1-m springboard forward pike dives. Two analyses were carried out to investigate whether configuration changes during the flight phase control the entry angle. In the first analysis, the variability in entry angle arising from variability in initial takeoff conditions were 9.2° and 19.6° for 101B and 105B respectively. In the second analysis, the variability in entry angle arising from variability in configuration changes during the aerial phase were 8.1° and 21.1° for 101B and 105B respectively (Table 8.16). These compare with the standard deviation of the actual entry orientation angle of 2.3° and 4.7° in 101B and 105B respectively (Table 8.5), showing that the average variation obtained in the simulation outcome was much larger than the variability in the performance outcome. This demonstrated that the diver made adjustment during the flight phase leading to reduce his variability in the entry orientation angle. In other words, the first simulation analysis showed that each flight phase in the recorded performances appeared to have been tailored to the corresponding takeoff parameters and the second simulation analysis showed that the variation of body configuration over 15 trials produced similar variability in entry angle to the first analysis (Table 8.16). This also indicates that the combination of takeoff parameters and body configuration in flight are linked.

The net change variability in the first and second analyses in 101B obtained 9.2° and 8.1° . The corresponding values in 105B obtained 19.6° and 21.1° , putting (comparing) these two together gives the RMS net changes in the orientation angle 2.4° and 6.2° in 101B and 105B respectively (Tables 8.14 and 8.15). Although the net changes in simulation 14 in 101B (Table 8.14) and simulation 2 in 105B (Table 8.15) have the same sign due to very small values, the net changes from analysis 1 and 2 genuinely operate in opposite direction (different sign), indicating the analysis 1 and 2 cancel each other out, giving smaller values of RMS net changes. These two values are comparable with the standard deviations in the orientation angle obtained in the actual performance of 2.3° and 4.7° respectively (Tables 8.14 and 8.15). Since there would be errors in planning, execution and motor sensory system in the initiation of correction (van Beers et al., 2004), it is likely that the diver's configuration changes in the flight

phase are performed to correct the errors arising from takeoff conditions to minimise the performance outcome variability. The RMS differences in 101B and 105B may suggest that the variability in body configuration in the flight phase (analysis 2) acts on the variability arising from initial takeoff conditions (analysis 1) compensating the errors to correct the movement and reduce the variability in body orientation angle. This may also suggest that the remaining variability in the entry orientation angle is the variability of planning and execution.

There is small variation in the hip and arms angles in the first half of flight phase in both 101B and 105B (Figures 8.4 and 8.5). The hip angle showed a large variation between 0.6 s and 1.1 s in 101B and in the last half of the flight phase in 105B (Figures 8.4 and 8.5). The arm angle also showed large variation between 0.8 s and 1.1 s in 101B and from 1.0 s to the end of the dive in 105B (Figures 8.4 and 8.5). The orientation angle increased gradually, and its peak variation appeared in the middle of the flight phase at around 0.6 s in both 101B and 105B, indicating the body orientation changed at the maximum dive height. The average hip angle during flight of each trial was able to explain around 90% of the variation in rotation potential, showing that the hip angle was the primary means of adjustment during the flight phase (Table 8.17). Since the minimum hip angle during flight was not significantly correlated with the rotation potential, it must have been the timing of the hip angle changes that was adjusted. The variation in the hip angle was small in the first half of flight rising to a maximum after around 0.8 s before starting to fall again (Figure 8.6). This indicates that feedforward control (fixed technique) was employed for the first part of flight with feedback control (adjustments) of the hip angle being used subsequently. The variation in the arm angle was small for the first three quarters of flight rising to a maximum late in flight before falling to a low value at entry (Figure 8.6), suggesting that the arms made a correction after the first hip angle adjustment. For the 2½ somersault dives the hip angle variability became large at entry (Figure 8.6), indicating that a final hip angle adjustment was made. This is in agreement with the finding in the study by Hiley & Yeadon (2016) that both feedforward and feedback are used in acrobatic aerial movements.

Since the stepwise multiple regression analysis (Table 8.17) showed that the average hip angle significantly accounts for variance in the rotation potential, this could be an evidence that hip angle is adjusted to be larger when there is more angular momentum. The larger variability in the hip angle in the second half of flight started at about 0.6 s where the diver comes out of pike (Figures 8.4, 8.5 and 8.6) indicating that the diver desired state to start making the correction is at the maximum dive height where the feedback from takeoff conditions is received (Latash, 1998). It is to be understood that the pike position is held for longer to rotate enough to compensate for low rotation potential, resulting in a lower average hip angle, and it is held for a shorter time to rotate less to compensate for larger rotation potential, resulting in a higher average hip angle. This is in agreement with Yeadon (2013) who reported that adjustments are made in performances with aerial movement to compensate for too much angular momentum. The initiation of adjustment to compensate for variation in takeoff found in the present study highlights the view that variability has a functional role in human movement proposed by Bartlett et al. (2007) and supported by Hiley and Yeadon (2016).

The primary means of controlling the somersault rotation in flight was a single hip angle adjustment, followed by subsequent arm and hip adjustments for the more complex $2\frac{1}{2}$ somersault dive. The hip angle at the time of maximum hip angle variation was positively correlated with rotation potential for both dives (Table 8.18), with greater hip angle for greater rotation potential, indicating that the hip angle variation constituted an adjustment based on the variation in rotation potential. When the rotation potential was greater so was the hip angle thereby increasing the moment of inertia to slow the somersault. For the $2\frac{1}{2}$ somersault dives the arm angle at the time of maximum arm variation was positively correlated with the orientation angle at the earlier time of maximum hip variation (Table 8.19), indicating that the arms made a second adjustment. If the orientation angle was large then so was the arm angle, again increasing the moment of inertia and slowing the somersault. The final hip angle of the $2\frac{1}{2}$ somersault dives was positively correlated with the orientation angle at the earlier time of maximum arm angle variation (Table 8.19), indicating that a final adjustment was made using the hip angle. If the orientation angle was low then the hip angle remained low so as to compensate.

In springboard diving, divers might be over rotated or perform less rotation at entry. The potential problem may possibly be associated with coaching. If a diver is over rotated or has less rotation at entry into the water, the error might come from takeoff conditions or flight phase. If coaches are not aware of such adjustments, they might focus on the wrong part of the dive. The performance outcome is dependent on the errors in takeoff conditions and the diver's ability to control the variability. This highlights the view that it is beneficial for coaches to be aware of such adjustments (Yeadon and Hiley, 2014).

It has been reported that the use of a single visual reference point during the flight phase assists the divers to determine the body position and contributes to the consistency of the dives (Gerald, 1983). Spotting skills enabled Greg Louganis to see the water in between each somersault by picking out a point on the surface of the pool (Louganis, 1995) providing more information to make correction and perform more accurately than other divers. The adjustment found in this study demonstrated that viewing technique could be a controlling component used in the feedback correction to reduce the performance outcome variability.

According to Fitts' (1954) model of human movement, the whole aerial phase might be considered as a combination of short and long movements, such that the early part of the flight phase is preprogrammed and feedback from previous stages could be used to make the final correction. On the other hand, according to the theory of time to collision (Lee, 1976), the diver might have used information about time to entry for controlling the outcome variable rather than information about distance, velocity or acceleration. This might suggest that although the diver has used feedback from previous stage such as information from takeoff initial conditions through sensory system to make the adjustments, information about time to entry might have played an important role to control the entry orientation angle.

Barris et al. (2014) demonstrated the functional role of movement variability as the adaptability of the divers under different training conditions referred as functional adaptive movement variability. Although the use of spatial coordinate has been used in traditional biomechanical studies to analyse a combination of individual and group

subjects performing under a variety conditions may lead to unrealistic sample variability and biasing statistical tests (Rewt, 1993; Gentner, 1987), the study of the consistency of technique performing under the same condition may reveal important information to understand the movement control process. The advantage of single subject or group case studies appears as an area of divergence but, it is imperative to carefully select each experimental design based upon research questions, hypotheses and envisioned conclusions of the research study in addition to considering visual and/or statistical support for the chosen research method (Smith, 2012). The advantage of group experimental designs is the ability to generalise the results to a similar population. However, they don't reflect the uniqueness of the performance characteristics of the individual such as technique since there are so many performance options (strategies) available for performing the same task in inter-subject experiments. The nature of the present study is to investigate technique which is involved in movement variability within subject. This highlights the single subject design as an appropriate research method to control for variation in anthropometry, strength and position in order to investigate the uniqueness of the performance characteristics of a single diver.

Within the flight phase there were periods of relative invariance and periods of increased variability (within the joint angle time histories). These periods of high variability may be considered to be examples of functional variability. The concept of functional variability is where the presence of more variability plays a functional role in stabilising other aspects of the movement or increasing the consistency of the movement outcome (Hamill, van Emmerik, Heiderscheit, & Li, 1999; van Emmerik, Hamill, and McDermott, 2005). For example, movement variability may be present as a strategy to reduce the occurrence of injury in repetitive cyclical tasks such as running (Bartlett et al., 2007; Hamill et al., 1999). However, Hamill et al. (1999) did state that a lack of variability was a symptom of pain (injury) in running, not necessarily the cause. Or increased variability may be associated with expertise and an indication of increased adaptability as Wilson et al. (2008) found for triple jumpers. Similarly, in diving, Barris, Farrow & Davids (2014) described the functional role of movement variability as the adaptability of springboard divers under different training conditions. It was found that by training with induced variability that divers indeed demonstrated

more variability in the hurdle phase but they also became more consistent in terms of reducing the number of dives aborted (balked). In other words, it appeared that the divers had learned to cope with the induced variability by making the appropriate adjustments. However, care should be taken when identifying increased movement variability and attributing it to higher levels of ability or adaptability, since it must be established how the observed variability is related to the success of the task, i.e. determine the mechanism and its function. In the present study there is a clear link between the increased variability, the initial conditions and the mechanics of correcting such errors.

There is evidence for the use of both feedforward control and feedback control in the flight phase. In both dives analysed the hip and arm angle variation remained low in the first part of the flight phase as the body adopted the pike position and subsequently the hip angle variation increased followed by the arm angle variation as the body straightened prior to entry. This indicates that feedforward control was used in the first part of flight while feedback control was used in the later phases to adjust the rotation for entry into the water. It is likely that the diver plans the aerial phase in advance (Miall & Wolpert, 1996; Wolpert, 1997; van Beers et al., 2004), including adopting the correct piked configuration, based on the best estimate of the initial conditions (i.e. expected linear and angular momentum). During the period from takeoff until before the point of maximum variation the diver evaluates the planned movement in relation to the actual values of the initial conditions or the current state (Miall & Wolpert, 1996; Wolpert, 1997). This evaluation most likely uses information relating to the orientation angle or angular velocity. If there is a discrepancy between the predicted movement and the current state, the pre-planned movements (feedforward control) are then adapted (feedback control) so that the diver either stays in the piked shape for longer or opens out earlier, leading to the increased recorded variation in hip angle.

The low variability in both the takeoff and water entry measures (Table 8.5) demonstrated that the diver's performances were very consistent. The consistency in the early stages of flight of the joint angle time histories suggests that the diver planned to use the same technique on each attempt (Figure 8.6) in a feedforward manner. However, the subsequent increase in variation in the joint angle time histories (Figure

8.6) indicates that the diver adjusted to the discrepancy between the pre-planned movement and the actual state of the movement (Wolpert, 1997). These adjustments appeared to have been very precise and specific to the discrepancy detected in the individual trials (Hiley & Yeadon, 2016). This was evident from the low variability achieved in the entry angle (Table 8.5). The variability seen in the diver's movements was therefore not random and performed a functional role in ensuring a consistent entry into the water. The adjustments were performed through feedback control, since the actions were related to events occurring earlier in the flight phase. The exact nature of the feedback on which the adjustments were based remains unknown. However, there is a strong indication that it is based on information regarding the angular momentum and time of flight of the dive (Table 8.17). In the forward pike dive the water can be kept in view for the whole of the flight phase and so visual information may contribute to the estimation of somersault rotation, angular velocity and flight time. In the forward $2\frac{1}{2}$ somersault pike the water can be viewed continuously from the extension from the pike to water entry and so visual information may be used to time the final arm and hip corrections based upon the amount of somersault rotation. While somersaulting in the piked position it is likely that the somersault rate is estimated from the otolith and semicircular canal information and that this governs the timing of the extension from the pike.

Divers performances are very precise and reproducible since low variability is recorded in the initial takeoff conditions and performance outcome (entry angle). It might be expected that they try to use the same technique each time they perform the same dive. However, in the forward pike dive the somersault rotation is adjusted by the timing of the hip extension from the pike. In the forward $2\frac{1}{2}$ pike dive the somersault rotation is first adjusted by timing the hip extension from the pike as a function of the estimated rotation potential arising from the vertical velocity and angular momentum at takeoff. Subsequently the timing of the raising of the arms is based upon the rotation angle at an earlier time and the final hip angle depends on an earlier estimate of the rotation angle. Divers make such correction with high precision, leading to low variability in performance outcome. Their deliberate compensation will produce variability in technique greater than the level of their precision variability. Although other divers might make adjustments somewhat differently the need for adjustment will remain if a

good performance is to be achieved since the orientation angle at entry is greatly affected by small variations at takeoff. Thus, the essential need for adjustment can justify the existence of execution variability in making configurational changes from trial to trial (Yeadon and Hiley, 2014) and supports the view that within each attempt there is also some variability in the technique used (Newell and Corcos, 1993; Wilson et al., 2008).

8.6. Summary

The aim of this chapter was to investigate the extent to which configuration changes (technique) during the flight phase control the entry angle in 1-m springboard forward dives. The angle-driven simulation model of aerial movement was used to determine the effects of initial conditions variability and flight phase configuration variability on outcome (orientation at entry) variability. The variation in the simulated orientation at entry arising from variability in the initial conditions was greater than the actual variation, demonstrating that the diver used feedback correction to make adjustments during flight to reduce the variability of his entry orientation angle. The early part of the flight phase uses feedforward control not only due to the small kinematic variation but also because the rate of change of the variability is nearly constant in both 101B and 105B. The larger variability in the last half of the aerial phase indicated that adjustments are made in flight from dive to dive by changing hip and arm movements to minimise the performance outcome (entry angle) variability. The hip angle has been adjusted to be larger when there is more angular momentum and the diver desired state to make the correction was at the maximum dive height where the feedback from the takeoff conditions is received.

CHAPTER 9

SUMMARY AND CONCLUSION

9.1. INTRODUCTION

In this final chapter, the main finding of the present study is reviewed to determine whether the objective of the research has been addressed through a biomechanical analysis of springboard diving using video data and applying simulation model. The research questions stated in the introduction (Chapter 1) are addressed. Finally, the conclusion of the study is provided.

9.2. SUMMARY OF MAIN FINDING

9.2.1. Kinematic variability

An experimental study was performed to investigate the movement variability and control in 1-m springboard diving. 15 forward dive and 15 forward 2½ somersault pike dives performed by an international diver were recorded using high speed video (250 Hz) and were digitised manually (Chapter 3). The kinematic variables were determined and the relationships between the key variables at different stages of the dives were determined. The standard deviation of joint and body angles showed that the diver was highly consistent in performing forward pike (101B) and forward 2½ somersault pike (105B) dives (Chapter 5). The movement pattern demonstrated the ability of the diver to reproduce the same pattern of movement repeatedly. Low variability at the entry orientation angle showed that the diver was consistent in performance outcome in both 101B and 105B.

The joint and body angles variability at the start and end of each phase along with the variation in the peak dive height and time of flight phases were quantified. The relationships between variables at hurdle landing, during board contact, at dive height and entry orientation angle were identified to determine factors influencing dive height and entry orientation angle (Chapter 5). It was found that the amount of hip extension from maximum hip flexion to the maximum hip extension during board contact phase determines the amount of additional energy that the diver put into the system to depress the board. This suggested that instead of using the amount of board

depression as a function of mechanical work done by the diver, the amount of hip extension during board contact can explain the extent to which the additional energy has been used to achieve a great dive height.

Larger hip extension during board contact increased the dive height in forward pike dives whereas the execution of hip flexion at the end of board contact in order to generate angular momentum degrades the dive height in forward 2½ somersault pike dives (Chapter 5). The execution of lower dive height was less observable in forward pike dives as there is less need for a large angular momentum. Less height in more difficult dives such as 2½ somersaults dives is unavoidable because greater somersault angular momentum is required.

9.2.2. Adjustment in foot placement

The variability of foot position during board contact phase at hurdle touchdown and dive takeoff (Chapter 6) showed that the diver made adjustment based on feedback control that corrected the foot position at dive takeoff slightly, providing post-response error information in each dive to learn the correct foot placement in the following dive. The diver varied his centre of mass horizontal velocity at hurdle takeoff to adjust the hurdle length and compensate for the variability of foot position at hurdle touchdown to control his distance on the board at hurdle landing, leading to land safely and as close to the board tip as possible in forward 2½ somersault pike dives.

9.2.3. Movement control in takeoff

A subject-specific angle driven model of a diver and springboard (Kong, 2005) was used to investigate whether there is any movement correction during the contact phase in 1-m springboard forward dives (Chapter 7). The RMS difference of 2.7° and 3.1° for forward pike and forward 2½ somersault pike dives respectively, demonstrated that the simulation model was closely matched with the performance. Two simulation analyses were carried out to determine the effect of varying the initial touchdown and initial recoil conditions on the rotation potential at takeoff. The variation obtained in the simulation outcome was much greater than variability in the actual performances,

demonstrating that the diver made adjustment during the contact phase. The amount of variability arising from initial touchdown conditions was comparable with the variation arising from initial recoil conditions, indicating that the most of corrections have been made during the recoil phase. The results of the angle variability in the actual performance showed that hip and knee angles were varied during the recoil phase to make adjustment and reduce the amount of rotation potential at takeoff.

9.2.4. Movement control in flight

An 11-segment simulation model of aerial movement (Yeadon et al., 1990b) was applied to investigate the extent to which technique in the aerial movement control the outcome in 1-m springboard forward pike dives (Chapter 8). The average root mean square (RMS) difference in orientation angle time history between simulation and performance obtained 3.1° , indicating that simulation closely matched performance. Two analyses were carried out to investigate the variability in the entry angle arising from the variability of initial conditions and the variability in entry angle arising from the joint angle changes during the flight phase. The average variation obtained in the simulation outcome was much larger than the variability in the performance outcome, demonstrating that the diver made adjustment during the flight phase leading to reduce his variability in the entry orientation angle.

The diver's configuration changes in the flight phase were performed to correct the errors arising from takeoff conditions to minimise the performance outcome variability. This suggested that the early part of the flight phase uses feedforward control not only due to the small variation but also because the rate of change of the movement variability was nearly constant in both 101B and 105B. The larger variability in the last half of the aerial phase also suggested that adjustments were made in flight from dive to dive by changing hip and arm movements to reduce the performance outcome (entry angle) variability.

The variation in the rotation potential was corrected during the flight phase (Chapter 8). The average hip angle significantly explained the variance in the rotation potential which means that the diver average hip angle increased by increasing the rotation

potential. This was found to be an evidence that hip angle has been adjusted to be larger when there is more angular momentum. The larger variability in the hip angle in the second half of flight started at the maximum dive height indicated that the diver desired state to provide the correction is at the peak dive height where the feedback from takeoff conditions is received (Latash, 1998) such that the pike position is held for longer to rotate enough to compensate for low rotation potential, resulting in a lower average hip angle, and it is held for a shorter time to rotate less to compensate for larger rotation potential, resulting in a higher average hip angle.

The springboard diver's performances often seem very precise and reproducible but, the existence of variation in their orientation and configuration is inevitable. There may be a potential problem arising from coaching. If a diver is over rotated or has less rotation at entry into the water, the error might come from takeoff conditions or flight phase. If coaches are unaware of such adjustments, they might focus on the wrong part of the dive. The performance outcome is dependent on the errors in takeoff conditions and the diver's ability to control the variability. It suggested that it is beneficial for coaches to be aware of such adjustments.

Elite divers use spotting skills to see the water in between each somersault by picking out a point on the surface of the pool to have more information for timing the aerial movement (Louganis, 1995). The adjustments found in this study (Chapter 8) were consistent with viewing technique being a controlling component used in the feedback correction to reduce the performance outcome variability.

The initiation of adjustment to compensate for variability arising from initial touchdown and takeoff conditions found in the present study (Chapter 7 and 8) highlights the view that variability has a functional role in human movement proposed by Bartlett et al. (2007) and supported by Hiley and Yeadon (2016).

Low variability recorded in the initial takeoff conditions and performance outcome (entry angle) in springboard diving performances showed that the divers are very precise and reproducible. When divers perform the same dive, it may be viewed that they use the same technique each time. There has been significant variation in body configuration during contact and flight phases due to adjustment for the small variation

in the initial conditions (Chapter 7 and 8). Divers complete the dives with low variability in performance outcome by making such correction with high precision. This indicates that each diver has a certain precision such that their deliberate compensation to make the adjustment will produce variability in technique greater than the level of their precision variability, suggesting that the essential need for adjustment can justify the existence of execution variability in making configurational changes from trial to trial.

9.3. RESEARCH QUESTIONS

1. How consistent is the diver from dive to dive? i.e. what is the variation of the time of phases, mass centre velocities and the joint angles at start and end of each phase in 1-m springboard forward dives?

The mean and standard deviations of the time of phases, mass centre velocities and the joint angles at touchdown, takeoff and entry (Chapter 5) showed that the diver was consistent dive after dive in both forward pike and forward $2\frac{1}{2}$ somersault pike dives.

2. Is there any compensation for variability in foot placement of the hurdle?

There is an adjustment made based on feedback control that corrected the hurdle foot position slightly, providing post-response error information in each dive to learn the correct foot placement in the following dive. In forward $2\frac{1}{2}$ somersault pike dives, the diver controlled his distance to the board tip at dive takeoff phase depending on how far his foot position was from the end of the board at hurdle contact. The diver varied his centre of mass horizontal velocity at hurdle contact to adjust the hurdle length and compensate for the variability of foot position at hurdle contact, leading to land safely and as close to the board tip as possible.

3. To what extent can variation at touchdown and maximum board depression affect the takeoff conditions?

The variability arising from the initial touchdown conditions in forward pike dives gave an RMS difference of rotation potential 0.085 straight somersault (ss) at takeoff and the variation arising from maximum board depression (lowest point) gave an RMS difference of 0.076 ss at takeoff, the corresponding values in forward $2\frac{1}{2}$ somersault

pike dives were 0.178 ss arising from initial touchdown conditions and 0.127 ss arising from initial lowest point. These compare with the variability of the rotation potential at takeoff in the actual performances of 0.018 ss and 0.034 ss in 101B and 105B respectively indicate that the variation obtained in the simulation outcome is significantly greater than variability in the actual performances, demonstrating that the diver made adjustment during the contact phase to reduce the variability of rotation potential.

4. Is there any adjustment in the flight phase of 1-m springboard forward dives?

The variability in entry orientation angle arising from variability in the initial takeoff conditions were 9.2° and 19.6° for 101B and 105B respectively. The variability in entry angle arising from variability in configuration changes during the aerial phase were 8.1° and 21.1° for 101B and 105B respectively. These compare with the standard deviation of the actual entry orientation angle of 2.3° and 4.7° in 101B and 105B respectively (Chapter 8) indicate that the average variation obtained in the simulation outcome was much larger than the variability in the performance outcome. This demonstrates that the diver made adjustment during the flight phase to reduce his variability in the entry orientation angle.

9.4. CONCLUSION

The aim of this research was to distinguish between execution variability and adjustment in springboard diving performances. The angle-driven simulation model of a diver and springboard was successfully applied to investigate the effect of perturbation of initial touchdown conditions and initial recoil conditions on the variability of the rotation potential at takeoff. The variability arising from initial conditions was significantly greater than the variation in the actual performance, indicating that the diver made adjustment during the board contact phase to control his velocity and angular momentum at takeoff. The angle-driven simulation model of the aerial movements was also successfully applied to investigate the variability in the performance outcome arising from initial takeoff conditions and arising from body configuration changes during the aerial movement. The variability obtained in the simulation outcome was considerably greater than the variation in the actual performance, demonstrating that the diver made adjustment during the flight phase. The diver varied his body configuration during the flight phase to compensate for the variation arising from initial takeoff conditions, and to control his body orientation angle at entry into the water. The adjustment found in the present study demonstrated that variability has a functional role in human movement.

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APPENDIX 1

INFORMED CONSENT FORM

Insert Name of Research Proposal

**INFORMED CONSENT
FORM**

(to be completed after Participant Information Sheet has been read)

The purpose and details of this study have been explained to me. I understand that this study is designed to further scientific knowledge and that all procedures have been approved by the Loughborough University Ethical Approvals (Human Participants) Sub-Committee.

I have read and understood the information sheet and this consent form. I have had an opportunity to ask questions about my participation. I understand that I am under no obligation to take part in the study.

I understand that I have the right to withdraw from this study at any stage for any reason, and that I will not be required to explain my reasons for withdrawing.

I understand that all the information I provide will be treated in strict confidence and will be kept anonymous and confidential to the researchers unless (under the statutory obligations of the agencies which the researchers are working with), it is judged that confidentiality will have to be breached for the safety of the participant or others.

I agree to participate in this study.

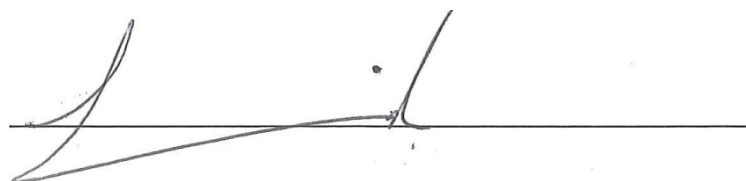
Your name

FREDDIE WOODWARD

Your signature



Signature of investigator



Date 21/03/2014

APPENDIX 2

ANTHROPOMETRIC MEASUREMENTS

Anthropometric Measurements for Segmental Inertia Parameters

Participant ID: Freddie Woodward

Age [years months]: 19 years 2m

Height [mm]: 1.790 meter

Weight [kg]: 69.7 kg

Date: 21/08/2014

All measurements in millimetres

Torso

	T0	T1	T2	T3	T4	T5	→	T6	T7	T8
Level	Hip	Umbilicus	Ribcage	Nipple	Shoulder	Neck		Nose	Ear	Top
Length	0					576	0			266
Perimeter	914	752	748	968		380			568	
Width	319	315	269	340	382					

Left Arm

	A0	A1	A2	A3	A4	→	A5	A6	A7
Level	Shoulder	Midarm	Elbow	Forearm	Wrist		Thumb	Knuckle	Nails
Length	0				536	0			191
Perimeter		258	241	255	159			180	
Width									

Right Arm

Level	Shoulder	Midarm	Elbow	Forearm	Wrist	→	Thumb	Knuckle	Nails
Length	0					0			
Perimeter		278	249	263	165			183	
Width									

Left Leg

	J0	J1	J2	J3	J4	J5	→	J6	J7	J8	J9
Level	Hip	Crotch	Midthigh	Knee	Calf	Ankle		Heel	Arch	Ball	Nails
Length	0					840	0				
Perimeter		553	504	380	358	210		312		220	
Foot length (heel-toe)			257								
Ankle height (from floor)			65								

Right Leg

Level	Hip	Crotch	Midthigh	Knee	Calf	Ankle	→	Heel	Arch	Ball	Nails
Length	0					831	0				
Perimeter		550	523	371	364	211		314		222	
Foot length (heel-toe)			259								
Ankle height (from floor)			64								

APPENDIX 3

INERTIA PARAMETERS

```

veni 14-segment model (2-segment foot)
  5.2406  0.0312  0.0312  0.0159 head
 28.7558  0.8706  1.0155  0.2966 trunk
  1.7918  0.0132  0.0132  0.0017 left upper arm
  1.4625  0.0196  0.0196  0.0009 left lower arm
  3.2543  0.1141  0.1140  0.0026 left straight arm
  2.1337  0.0169  0.0169  0.0023 right upper arm
  1.6126  0.0230  0.0229  0.0010 right lower arm
  3.7463  0.1407  0.1407  0.0033 right straight arm
  9.4395  0.1420  0.1421  0.0329 left thigh
  4.0243  0.0558  0.0558  0.0057 left shank
  0.8427  0.0025  0.0024  0.0006 left whole foot
  0.7238  0.0014  0.0012  0.0006 left ankle/ball
  4.8670  0.1325  0.1324  0.0063 left shank + foot
  0.1188  0.000049  0.000077  0.000049 left ball/toes
  9.5248  0.1391  0.1392  0.0337 right thigh
  4.0141  0.0539  0.0539  0.0057 right shank
  0.8577  0.0026  0.0025  0.0007 right foot
  0.7346  0.0014  0.0012  0.0006 right ankle/ball
  0.1231  0.000050  0.000080  0.000051 right ball/toes
  4.8719  0.1298  0.1297  0.0064 right shank + foot

```

SEGMENTAL INERTIA PARAMETER VALUES

UNITS: MASS IN KG
 DISTANCE IN METRES
 MOMENT OF INERTIA IN KG*M**2

FORMAT AND SEQUENCE OF DATA PRESENTATION
 SEGMENT NAME
 MASS, DISTANCE OF MASS CENTRE FROM PROXIMAL JOINT,
 SEGMENT LENGTH
 PRINCIPAL MOMENTS OF INERTIA

SUBJECT: frd2

HEAD H			
	5.241	0.134	0.266
	0.031	0.031	0.016
TRUNK PTC			
	28.756	0.305	0.576
	0.871	1.015	0.297
HEAD-TRUNK PTCH			
	33.996	0.368	0.842
	1.629	1.774	0.313
UPPER ARM 1A			
	1.792	0.129	0.285
	0.013	0.013	0.002
UPPER ARM 1B			
	2.134	0.132	0.297
	0.017	0.017	0.002

FOREARM 2A			
1.111	0.108	0.254	
0.006	0.006	0.001	
FOREARM 2B			
1.233	0.113	0.264	
0.007	0.007	0.001	
HAND 3A			
0.352	0.075	0.191	
0.001	0.001	0.000	
HAND 3B			
0.380	0.077	0.198	
0.001	0.001	0.000	
STRAIGHT ARM A			
3.254	0.272	0.730	
0.114	0.114	0.003	
STRAIGHT ARM B			
3.746	0.275	0.759	
0.141	0.141	0.003	
FOREARM + HAND 23A			
1.462	0.162	0.445	
0.020	0.020	0.001	
FOREARM + HAND 23B			
1.613	0.167	0.462	
0.023	0.023	0.001	
THIGH 1J			
9.439	0.179	0.413	
0.142	0.142	0.033	
THIGH 1K			
9.525	0.178	0.409	
0.139	0.139	0.034	
CALF 2J			
4.024	0.175	0.427	
0.056	0.056	0.006	
CALF 2K			
4.014	0.174	0.422	
0.054	0.054	0.006	
FOOT 3J			
0.843	0.074	0.200	
0.003	0.002	0.001	
FOOT 3K			
0.858	0.074	0.200	
0.003	0.002	0.001	
ankle/ball 4J			
0.724	0.059	0.138	
0.001	0.001	0.001	
ankle/ball 4K			
0.735	0.059	0.138	
0.001	0.001	0.001	
ball/toes 5J			
0.119	0.026	0.062	
0.000049	0.000077	0.000049	
ball/toes 5K			
0.123	0.026	0.062	
0.000050	0.000080	0.000051	
SHANK + FOOT 23J			
4.867	0.231	0.627	
0.133	0.132	0.006	
SHANK + FOOT 23K			
4.872	0.231	0.622	
0.130	0.130	0.006	

TOTAL MASS = 69.70 KG

DENSITY = 1.031

Appendix 4

Subject Profile

Name: FREDDIE WOODWARD

Coach: Tom Owens

Date of Birth: 23/06/1995

Sex: Male

Diving experience

1. How many years have you been diving?
 - 10 years
2. What level of competition are you currently at?
 - Senior international
3. What fulcrum number do you normally use for dives in the forward group?
 - 8½ for all forward group
4. What factors would you recognise as the key of good springboard takeoff?
 - Distance from the diving board
 - Height of the dive
 - Position
5. How would you describe your springboard takeoff technique?
 - Trying to push forward slightly during recoil phase so that I can gain a better height.
6. Is there any particular aspect of technique regarding the springboard takeoff that you would like to improve on?
 - The amount and the direction of pushing forward is tricky and I need to practise to find the best technique.

Diving achievement

1. British Championships

2014 - Ponds Forge, Sheffield

Event	Score	Medal/Position
1m Springboard	384.60	GOLD
3m Springboard	359.85	5th
3m Synchro	405.63	GOLD

2013 - Life Centre, Plymouth

Event	Score	Medal/Position
3m Springboard	359.80	4th

2012 - Ponds Forge, Sheffield

Event	Score	Medal/Position
3m Springboard	340.40	9th
3m Synchro	371.34	SILVER

2011 - Ponds Forge, Sheffield

Event	Score	Medal/Position
-------	-------	----------------

3m Synchro	333.03	4th
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2010 - Ponds Forge, Sheffield

Event	Score	Medal/Position
1m Springboard	264.40	8th
3m Springboard	277.35	12th
3m Synchro	299.16	4th

2008 - Aquatics Centre, Manchester

Event	Score	Medal/Position
3m Springboard	227.90	11th
10m Platform	235.10	11th
3m Synchro	282.06	BRONZE

2. National Cup

2014 - Southend Diving Centre, Southend

Event	Score	Medal/Position
-------	-------	----------------

1m Springboard	344.50	4th
3m Springboard	390.80	BRONZE
3m Synchro	354.00	4th

2012 - Southend Diving Centre, Southend

Event	Score	Medal/Position
3m Springboard	284.50	12th

2011 - Southend Diving Centre, Southend

Event	Score	Medal/Position
1m Springboard	257.80	12th
3m Springboard	272.00	14

3. Elite Junior Championships

2013 - John Charles Centre For Sport, Leeds

Event	Score	Medal/Position
Group A 1m Springboard	469.50	SILVER

Group A 3m Springboard	526.70	SILVER
14-18 Yrs 3m Synchro	278.10	GOLD

2012 - Life Centre, Plymouth

Event	Score	Medal/Position
Group A 1m Springboard	437.15	SILVER
Group A 3m Springboard	489.10	BRONZE

4. Commonwealth Games

2014 - Glasgow, Scotland

Event	Score	Medal/Position
1m Springboard	340.05	10th
3m Synchro	364.41	BRONZE

5. Grand Prix

2014 - Guanajuato, Mexico

Event	Score	Medal/Position
-------	-------	----------------

3m Springboard	402.10	7th
3m Synchro	347.10	4th

2014 - San Juan, Puerto Rico

Event	Score	Medal/Position
3m Springboard	399.60	4th
3m Synchro	346.35	4th

6. European Junior Championships

2013 - Poznan, Poland

Event	Score	Medal/Position
Group A 1m Springboard	453.55	9th
Group A 3m Springboard	533.55	7th
3m Synchro	309.00	SILVER

2012 - Graz, Austria

Event	Score	Medal/Position
-------	-------	----------------

Group A 1m Springboard	408.40	14th
3m Synchro	285.24	5th

2011 - Belgrade, Serbia

Event	Score	Medal/Position
Group A 3m Springboard	492.70	4th
3m Synchro	269.10	4th